CITY OF CADILLAC

MAHL/LOCAL LIMITS REPORT

PREPARED FOR:

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Project No. 854440

TABLE OF CONTENTS

1.0 BACKGROUND & PURPOSE

The City of Cadillac wastewater system currently provides wastewater collection and treatment service within the City limits and to portions of the townships of Haring, Selma, Cherry Grove, and Clam Lake. The City of Cadillac has an active Industrial Pretreatment Program (IPP) due to the robust commercial and industrial activity in the area. The program procedures and local limits were adopted in 1994. The Michigan Department of Environment, Great Lakes, and Energy (EGLE) has requested the City update the program.

The first step in updating the IPP for EGLE approval was to evaluate the Maximum Allowable Headworks Loadings (MAHLs) for Pollutants of Concern. MAHLs were evaluated for select compatible and incompatible (e.g. metals) pollutants in order to determine appropriate Local Limits to include in an updated Sewer Use Ordinance and updated industrial discharge permits. At the time of this MAHL evaluation the City had 15 permitted industrial facilities, as follows:

- AAR Cadillac Manufacturing (Haynes St.)
- AAR Cadillac Manufacturing (6th Ave.)
- ARVCO Container Corp.
- AKWEL
- Avon Protection
- Borg Warner of Cadillac
- Cadillac Castings, Inc. (CCI)
- Cadillac Renewable Energy (CRE)
- Fiamm Technologies, Inc.
- Hutchinson Antivibration Systems
- Michigan Rubber Products, Inc.
- Munson Healthcare – Cadillac
- Piranha Hose, Inc.
- Rec Boat Holdings – Trailer Division
- Spencer Plastics

As part of the IPP update, the City was able to reclassify Avon Protection, Munson Healthcare, Spencer Plastics, and Rec Boat Holdings as non-significant industrial users who do not require industrial discharge permits.

The City currently has Local Limits for the following metals: Arsenic, Cadmium, Chromium, Copper, Total Cyanide, Mercury, Molybdenum, Nickel, Silver, Lead, Zinc; and the following organics: 1,2-Dichlorobenzene, Chloroform, Ethylbenzene, Lindane, PCBs, Phenol, Styrene, Toluene, and Xylene. It is recommended that local limits also be evaluated for 5-Day Biochemical Oxygen Demand (BOD_5), Total Suspended Solids (TSS), Ammonia (NH_3-N), and Total Phosphorus (TP) due to NPDES permit limitations for these parameters, and for selenium based on biosolids land application requirements.

This report discusses the approaches used, data, and calculations used for the MAHL Evaluation and Local Limits recommendations for the City of Cadillac.

The MAHL evaluation was done in accordance with EGLE and US EPA guidelines. Detailed descriptions and rationale are provided where procedures or methods varied from those standard procedures outlined in the guidance.

2.0 DATA COLLECTION

Historic WWTP and industrial user (IU) data was used in conjunction with data collected specifically for this MAHL evaluation in accordance with the EGLE-approved sampling plan. WWTP influent, primary effluent, final effluent and representative domestic/background wastewater samples were collected in October 2022 to provide a basis for the MAHL evaluation.

The average non-compatible pollutant concentrations (mg/L) for WWTP influent, primary effluent, final effluent, and domestic/background wastewater samples are shown in Table 1. It should be noted that secondary sludge is co-settled with primary sludge, resulting in higher primary effluent concentrations than facilities that do not co-settle.

The flow weighted average for the background concentrations use respective average concentrations and domestic flowrates from two data sets. The average domestic sample concentration from the dataset one is multiple by the respective domestic flowrate. The same is done for dataset two. The sum of this is divided by the sum of domestic flowrate one and domestic flowrate two. See the following equation:

X1 = Average domestic sample concentration #1 (refer to Appendix A)

X2 = Average domestic sample concentration #2 (refer to Appendix A)

Q1 = Domestic sample flowrate #1, 49.5 gpm

Q2 = Domestic sample flowrate #2, 500 gpm

$$Flow\ Weighted\ Average = \frac{(X1 * Q1) + (X2 * Q2)}{(\mathrm{Q1} + \mathrm{Q2})}$$

Table 1 – Non-Compatibles - Data Collection Average Results

PARAMETER	FLOW WEIGHTED BACKGROUND AVERAGE (mg/L)	WWTP INFLUENT (mg/L)	PRIMARY EFFLUENT (mg/L)	FINAL EFFLUENT (mg/L)
Arsenic	0.0012	0.0023	0.0049	0.0005
Cadmium	0.0001	0.0002	0.0004	0.0001
Chromium	0.0016	0.0056	0.0198	0.001
Copper	0.087	0.066	0.1485	0.0058
Cyanide, T	0.0025	0.0025	0.0025	0.0173
Lead	0.0032	0.0025	0.0041	0.0005
Mercury	0.00012	0.0001	0.00031	0.0001
Molybdenum	0.0009	0.0064	0.0079	0.0053
Nickel	0.0027	0.003	0.007	0.0036
Selenium	0.0007	0.0009	0.0013	0.0007
Silver	0.0004	0.0003	0.0005	0.0003
Zinc	0.143	0.1036	0.203	0.015

*Note that analytical results below the method detection limit were taken as half of the detection limit when calculating the average concentration for these parameters.

Samples for specific organic pollutants in the background and at the WWTP were also collected in October 2022, and summarized in Table 2.

Table 2 – Organic Pollutants - Data Collection Average Results

PARAMETER	FLOW WEIGHTED BACKGROUND AVERAGE (mg/L)	WWTP INFLUENT (mg/L)	PRIMARY EFFLUENT (mg/L)	FINAL EFFLUENT (mg/L)
PCBs	<0.00011	<0.00012	<0.00011	<0.0001
Chloroform	<0.0048	<0.0048	<0.0048	<0.0048
1,2-Dichlorobenzene	<0.005	<0.005	<0.005	<0.005
Ethylbenzene	<0.005	<0.005	<0.005	<0.005
Styrene	<0.005	<0.005	<0.005	<0.005
Toluene	0.003	<0.005	0.071	<0.005
Xylene	<0.010	<0.010	<0.010	<0.010
Total Phenols	0.109	0.055	0.157	<0.20
Lindane	<0.000053	<0.000062	<0.000053	<0.00051

*Note that analytical results below the method detection limit were taken as half of the detection limit when calculating the average concentration for these parameters. If all data was non-detect for a particular parameter, it is shown as < highest RL in the table.

Discharge samples were collected from each of the currently permitted industrial user discharges and analyzed for the organic pollutants to help determine whether these parameters need to continue to have local limits. In general, phenols were the only parameter with detections in the IU discharge.

Per EGLE's request, the City also reviewed SDS sheets for each of the permitted industrial users to determine whether there is potential for any of the organic pollutants to be in the permitted IU discharge prior to treatment. A careful review of all SDS sheets for each facility, including process chemicals, lubricants, boiler additives, and cleaning products were reviewed. From the SDS sheets, it was determined that there is potential for several of the organic pollutants to be present in the raw wastewater from several of the facilities including: AKWEL (phenols), AAR Haynes Street facility (ethylbenzene, xylene, toluene, phenols), Borg Warner (phenols), CRE (ethylbenzene, toluene, xylene), and Hutchinson (ethylbenzene, xylene, toluene, phenols). The City collected a wastewater sample in April 2024 from each of these facilities upstream of any pretreatment process to determine whether these compounds were present. ARVCO did not provide SDS sheets as requested by the City; therefore, the City collected a raw wastewater sample and had it analyzed for all six of the organic pollutants. The results of the raw IU wastewater sampling event are included in Appendix A.

Total phenol was the only organic parameter present in the raw IU wastewater above detection at any of the IUs with a potential to discharge the current locally regulated limits. Therefore, total phenols is the only organic parameter recommended in the local limits evaluation.

WWTP operating data for 2021-2022 were also included in the MAHL evaluation for compatibles pollutants. The average value for the WWTP influent, primary effluent and final effluent for 2021-2022 is included in Table 3. Table 3 also presents the flow-weighted average concentration for the domestic/background samples based on the October 2022 MAHL sampling.

Table 3 – Compatibles - Data Collection Average Results

PARAMETER	FLOW WEIGHTED BACKGROUND SAMPLES (mg/L)	WWTP INFLUENT (mg/L)	PRIMARY EFFLUENT (mg/L)	FINAL EFFLUENT (mg/L)
BOD_5	227	270	313	7
TSS	325	252	293	7.5
Phosphorus	6.3	4.2	4.6	0.22
Ammonia	33	24	24	0.6

The full analytical results are presented in Appendix A for reference.

3.0 Non-Compatible Pollutants – Methods

The non-compatible pollutant MAHL results were determined using the EGLE-developed spreadsheet model, customized by Fleis & VandenBrink Engineering, Inc. (F&V) for use at the City of Cadillac.

The following parameters are key user inputs in the non-compatibles MAHL spreadsheet:

- Influent flows, including: domestic/background, Significant Industrial Users and other non-domestic flows
- Domestic/background, industrial, and WWTP influent wastewater characteristics
- Biosolids flows and percent solids
- Treatment plant % removals

Table 4 summarizes the key flow and miscellaneous variables used in the Cadillac MAHL.

Table 4 – Variables and Values Used for Non-Compatible Pollutant MAHL

VARIABLE	VALUE	UNITS
Q_{dom}, Domestic/Background Flow (est.)	0.863	MGD
Q_{ind}, Total Industrial (SIU/CIU) Flow	0.857	MGD
Q_{POTW}, Total Avg WWTP Influent Flow	1.72	MGD
Q_{Max}, Maximum Design WWTP Influent Flow	4.5	MGD
Q_{dig}, Sludge Flow to Digester	0.07	MGD
Q_{Sludge}, Sludge Flow to Disposal	0.0025	MGD
Sludge (to disposal) % Solids	3.0	%

Table 5 summarizes the percent removals used and the source reference for the value selected to be used in the calculations. Site specific percent removals were used where applicable and appropriate. If insufficient data (i.e. influent and effluent were both less than detectable) was available to determine a site-specific percent removal, an appropriate literature value (e.g. US EPA Local Limits Development Guidance appendices) was used.

Table 5 – Treatment Plant % Removals

PARAMETER	PRIMARY % REMOVAL*	OVERALL % REMOVAL	SOURCE (primary, overall)
Arsenic	-112%	78%	Site Specific, Site Specific
Cadmium	-101%	52%	Site Specific, Site Specific
Chromium, Total	-256%	82%	Site Specific, Site Specific
Copper	-125%	91%	Site Specific, Site Specific
Cyanide	0%	69%	Site Specific, Literature
Lead	-67%	80%	Site Specific, Site Specific
Mercury	-217%	60%	Site Specific, Literature
Molybdenum	-24%	17%	Site Specific, Site Specific
Nickel	-100%	42%	Site Specific, Literature
Selenium	-44%	24%	Site Specific, Site Specific
Silver	-87%	75%	Site Specific, Literature
Zinc	-96%	86%	Site Specific, Site Specific
Phenols	-189%	8%	Site Specific, Site Specific

*Site specific data indicate negative removal efficiencies for the primary effluent because secondary sludge is co-settled

The allowable headworks loading for each parameter was calculated for the following criteria:

- 2023 NPDES Effluent Limits (arsenic, cadmium, available cyanide, mercury, silver)
- Secondary treatment inhibition
- Nitrification treatment inhibition
- Anaerobic Digester Inhibition
- Water quality based on chronic toxicity pass-through
- Water quality based on acute toxicity pass-through
- Biosolids (sludge) quality for land application (Class A)

3.1 NON-COMPATIBLE POLLUTANTS – LOADING CALCULATIONS

The formula and a table summarizing the Allowable Headworks Loading (AHL) results for each criterion are described below.

3.2 ALLOWABLE LOADING BASED ON NPDES PERMIT LIMITS

$$L_{NPDES} = \frac{Q_{POTW} * 8.34 * C_{NPDES}}{1 - R_{Avg}}$$

Where:

L_{NPDES} = Loading based on NPDES permit limits (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

C_{NPDES} = NPDES permit limit (mg/L)

R_{Avg} = Removal % for the WWTP (expressed as a decimal)

Table 6 – MAHLs based on NPDES Permit Limits

PARAMETER	C_{NPDES} (mg/L)	MAHL (lbs./day)
Arsenic	0.016	1.06
Cadmium	0.004	0.123
Cyanide, Available	0.006	0.273
Mercury	0.000002	0.0001
Silver	0.0014	0.0803

Note that Beryllium and Benzidine effluent limits were also added to the recent NPDES permit; however, these parameters were not sampled as part of the MAHL study, which began prior to issuance of the permit in April 2023.

3.3 Allowable Loading Based on Secondary Biological Treatment Inhibition

$$L_{INHIB,Sec} = \frac{Q_{POTW} * 8.34 * C_{INHIB,Sec}}{1 - R_{PRIM}}$$

Where:

$L_{INHIB,\ Sec}$ = Loading based on biological secondary inhibition (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

$C_{INHIB,\ Sec}$ = Concentration that inhibits biological treatment of the secondary process (mg/L), literature values from *EPA Local Limits Development Guidance, Appendix G* and *EPA Guidance Manual for Preventing Interference at POTW (Table 2-1)*

R_{PRIM} = Removal % across the primary process (expressed as a decimal)

Table 7 – MAHLs based on Secondary Treatment Inhibition

PARAMETER	C_{INHIB} (mg/L)	MAHL (lb/day)
Arsenic	0.1	0.67
Cadmium	1.0	7.14
Chromium, Total	1.0	4.02
Copper	1.0	6.37
Cyanide, Total	0.1	1.43
Lead	1.0	8.60
Mercury	0.1	0.45
Molybdenum	NA	NA
Nickel	1.0	7.17
Selenium	NA	NA
Silver	0.25	1.92
Zinc	5.0	36.6
Phenols	50.0	248

3.4 Allowable Loading Based on Nitrification Inhibition

$$L_{INHIB,Nit} = \frac{Q_{POTW} * 8.34 * C_{INHIB,Nit}}{1 - R_{Prim}}$$

Where:

$L_{INHIB,\,Nit}$ = Loading based on nitrification inhibition (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

$C_{INHIB,\,Nit}$ = Concentration that inhibits nitrification (mg/L), literature values from *EPA Local Limits Development Guidance, Appendix G* and *EPA Guidance Manual for Preventing Interference at POTW (Table 2-1)*

R_{PRIM} = Removal % across the primary process (expressed as a decimal)

Table 8 – MAHLs based on Nitrification Treatment Inhibition

PARAMETER	C_{INHIB} (mg/L)	MAHL (lb/day)
Arsenic	1.5	10.2
Cadmium	5.2	37.1
Chromium, Total	1.0	4.02
Copper	0.48	3.06
Cyanide, Total	0.5	7.17
Lead	0.5	4.30
Mercury	NA	NA
Molybdenum	NA	NA
Nickel	0.5	3.58
Selenium	NA	NA
Silver	0.25	1.9
Zinc	0.5	3.66
Phenols	4.0	19.8

3.5 Allowable Loading Based on Anaerobic Digester Inhibition

$$L_{INHIB,Dig} = \frac{Q_{Dig} * 8.34 * C_{INHIB,Dig}}{R_{Avg} * F}$$

Where:

$L_{INHIB,\,Dig}$ = Loading based on digester inhibition (lbs/day)

Q_{Dig} = Total sludge flow to the digester (MGD)

$C_{INHIB,\,Dig}$ = Concentration that inhibits anaerobic digestion (mg/L), literature values from *EPA Local Limits Development Guidance, Appendix G* and *EPA Guidance Manual for Preventing Interference at POTW (Table 2-1)*

R_{Avg} = Removal % across the WWTP (expressed as a decimal)

F = Fraction of overall removal due to sorption, set equal to 1.0 to arrive at the most conservative AHL

Table 9 – MAHLs based on Anaerobic Digester Inhibition

PARAMETER	C_{INHIB} (mg/L)	MAHL (lb/day)
Arsenic	1.0	0.74
Cadmium	1.0	1.11
Chromium, Total	5.0	3.56
Copper	40	25.6
Cyanide	30	25.4
Lead	50	36.7
Mercury	NA	NA
Molybdenum	NA	NA
Nickel	2.0	2.78
Selenium	NA	NA
Silver	13	10.1
Zinc	40	27.3
Phenols	NA	NA

3.6 Allowable Loading Based on Water Quality (Chronic Toxicity Pass-Through)

Chronic Loading Formula:

$$L_{Chronic} = Q_{POTW} * 8.34 * \frac{WQBEL_C/1000}{(1-R_{Avg})}$$

Where:

$L_{Chronic}$ = Loading based on chronic toxicity pass through (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

R_{avg} = average overall removal fraction for the treatment plant (fraction)

$WQBEL_C$ = Water quality based effluent limit for chronic toxicity protection (µg/L), provided by EGLE for Cadillac, refer to Interoffice Memo dated 4/19/2022, included in Appendix B for reference.

Table 10 – MAHLs based on Chronic Toxicity Pass-through

PARAMETER	$WQBEL_C$ (mg/L)	MAHL (lb/day)
Arsenic	0.016	1.06
Cadmium	0.004	0.12
Chromium, Total	0.094	7.48
Copper	0.011	1.8
Cyanide, Available	0.006	0.27
Lead	0.094	6.6
Mercury	0.0000013	0.00004
Molybdenum	3.6	62.4
Nickel	0.047	1.16
Selenium	0.006	0.11
Silver	0.001	0.08
Zinc	0.20	20.9
Phenols	0.51	7.9

3.7 Allowable Loading Based on Water Quality (Acute Toxicity Pass-Through)

Acute Toxicity Loading Formula:

$$L_{Acute} = Q_{POTW} * 8.34 * \frac{WQBEL_A/1000}{(1-R_{Avg})}$$

Where:

L_{Acute} = Loading based on acute toxicity pass through (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

R_{avg} = average overall removal fraction for the treatment plant (expressed as a fraction)

$WQBEL_A$ = Water quality-based effluent limit for acute toxicity protection (µg/L), provided by EGLE for Cadillac, refer to Interoffice Memo dated 4/19/2022, included in Appendix B for reference.

Table 11 – MAHLs based on Acute Toxicity Pass-through

PARAMETER	$WQBEL_A$ (mg/L)	MAHL (lb/day)
Arsenic	0.68	45.2
Cadmium	0.012	0.36
Chromium, Total	1.3	103
Copper	0.029	4.74
Cyanide	0.044	2.03
Lead	0.8	58
Mercury	NA	NA
Molybdenum	58	1,005
Nickel	0.8	19
Selenium	0.12	2.27
Silver	0.022	1.26
Zinc	0.36	36
Phenols	6.8	106

3.8 Allowable Loading Based on Biosolids Quality

$$L_{Sludge} = Q_{Sludge} * \frac{TSS_{Sludge}}{100} * 8.34 * \frac{C_{Sludge}}{R_{avg} * F_{Sorp}}$$

Where:

L_{Sludge} = Loading based on biosolids quality (lbs/day)

Q_{Sludge} = Average daily sludge generation rate (MGD)

TSS_{Sludge} = Average sludge % solids

C_{Sludge} = Target standard for biosolids quality (mg/kg) – use Part 24 “Clean” or “Ceiling” limit as applicable. Part 24 Table 3 values for biosolids were used for this evaluation because Cadillac prefers to land apply its biosolids as a Class A product

R_{avg} = average overall removal fraction for the treatment plant (expressed as a fraction)

F_{Sorp} = fraction of removal that occurs by sorption to biosolids (this was assumed to be 1.0 for all parameters)

Table 12 – MAHLs based on Biosolids Quality for Land Application of Biosolids

PARAMETER	C_{Sludge} (mg/kg)	MAHL (lb/day)
Arsenic	41	0.032
Cadmium	39	0.046
Chromium, Total	NA	NA
Copper	1500	1.014
Cyanide	NA	NA
Lead	300	0.232
Mercury	17	0.017
Molybdenum	75	0.268
Nickel	420	0.616
Selenium	100	0.256
Silver	NA	NA
Zinc	2800	2.018
Phenols	NA	NA

The most restrictive loading was chosen as the controlling MAHL and safety factors were then applied to determine the MAIL as shown in the following table.

Table 13 – Controlling MAHL, Criteria, MAIL, and Safety Factor

PARAMETER	Controlling Criteria	MAHL (lb/day)	Domestic Loading (lb/day)	Safety Factor (S.F.)	MAIL after SF (lb/day)
Arsenic	Biosolids	0.032	0.008	10%	**0.021**
Cadmium	Biosolids	0.046	0.001	10%	**0.040**
Chromium, Total	Digestion Inhibition	3.56	0.012	10%	**3.19**
Copper	Biosolids	1.01	0.627	10%	**0.290**
Cyanide, Available	Chronic Toxicity/NPDES	0.273	0.068	10%	**0.211**
Lead	Biosolids	0.232	0.023	10%	**0.187**
Mercury	Chronic Toxicity/NPDES	0.00005	0.0009	10%	**-**
Molybdenum	Biosolids	0.268	0.007	10%	**0.235**
Nickel	Biosolids	0.617	0.020	10%	**0.536**
Selenium	Chronic Toxicity	0.108	0.005	10%	**0.092**
Silver	Chronic Toxicity.NPDES	0.080	0.0028	10%	**0.070**
Zinc	Biosolids	2.02	1.031	10%	**0.793**
Phenols	Chronic Toxicity	7.98	0.784	10%	**6.40**

*Safety Factor as defined in Section 6.2.3 of US EPA's *Local Limits Guidance*

4.0 NON-COMPATIBLE POLLUTANTS – ALLOCATION & LOCAL LIMIT DEVELOPMENT

After the MAHL was determined for each of the non-compatible parameters, the Maximum Allowable Industrial Loading (MAIL) was determined by subtracting out the domestic background portion of the total loading.

The City has chosen to utilize a uniform allocation of the available MAIL. The available MAIL was converted into a calculated local limit by dividing by the current total industrial flow of 0.8506 MGD and the conversion factor 8.34. The calculated Local Limits for the non-compatible pollutants are summarized and compared with the City's current local limit in Table 14 below.

Table 14 –Standard Local Limits – Non-compatible Pollutants

POLLUTANT	Current Local Limit (Daily Max) (µg/L)	Calculated Local Limit (Daily Max) (µg/L)	Proposed Local Limit (Daily Max) (µg/L)
Arsenic	26	3	3
Cadmium	2	5.6	2
Chromium	369	447	369
Copper	99	40	40
Cyanide	4 (total)	30 (available)	4 (available)
Lead	100	26	26
Mercury	ND*	-	ND*
Molybdenum	118	33	33
Nickel	185	75	75
Selenium	-	13	13
Silver	3	10	3
Zinc	835	110	110
Phenols	1236	895	895

* Monitoring for mercury shall be in accordance with the following test methods: The discharge of mercury at or above the quantification level of 0.2 ug/l shall represent an exceedance of the local limit. Mercury sampling procedures, preservation and handling, and analytical protocol for compliance monitoring shall be in accordance with U.S. EPA Method 245.1, unless screening using Method 1631 is required by the City. The quantification level shall be 0.2 ug/l for Method 245.1 or 0.5 nanograms/liter (ng/l) for Method 1631, unless higher levels are appropriate due to sampling matrix interference.

It should be noted that a typical method for determining the Local Limit was not feasible for mercury because the observed domestic loading exceeds the calculated MAHL. Mercury will be assigned a non-detectable limit in accordance with the State of Michigan and City of Cadillac's pollutant minimization program for mercury.

5.0 COMPATIBLE POLLUTANTS – METHODS

MAHL evaluations for 5-day biochemical oxygen demand, total suspended solids, phosphorus, and ammonia were also completed.

Table 3 in the Data Collection section above summarizes the compatible pollutant sampling results for WWTP influent, primary effluent, final effluent, and domestic/background sources.

The MAHLs for the compatible pollutants were determined based on the treatment plant capacity or NPDES limitations, whichever loading was lowest. The allowable headworks loading based on the NPDES permit limit was calculated in accordance with the EPA Local Limits Development Guidance using the following equation:

NPDES Limit Formula: $L_{NPDES} = \frac{8.34 * C_{NPDES} * Q_{POTW}}{(1 - Ravg)}$

Where:

L_{NPDES} = Loading based on NPDES permit limit (lbs/day)

C_{NPDES} = NPDES permit limit

Q_{POTW} = Total flow to the WWTP (MGD)

R_{avg} = average overall removal fraction for the treatment plant (fraction)

5.1 BIOCHEMICAL OXYGEN DEMAND MAHL AND LOCAL LIMIT

The City's 1975 O&M Manual Basis of Design indicates that the WWTP is designed to treat an average BOD_5 loading of 3,300 lb BOD_5/day. Fine bubble aeration was added to the plant in 1996, resulting in a calculated BOD_5 capacity of 4,748 lb/day based the size of the aeration tanks and the *10 States Standards* loading of 40 lbs BOD_5 per 1000 ft^3 of aeration.

The estimated 1996 design basis AHL was compared to seasonal NPDES Limit formula calculations based on available performance data for the City. The calculations are included in Appendix A and summarized below.

Table 15 – NPDES Permit Based AHLs for BOD_5

Permit Basis	Season	C_{NPDES}	% Removal	L_{NPDES}
Monthly Average	June-September	7 mg/L	97.6	4,184 lb/d
Monthly Average	October-November	11 mg/L	97.5	6,312 lb/d
Monthly Average	December-March	23 mg/L	97.2	11,783 lb/d
Monthly Average	April-May	17 mg/L	96.8	7,621 lb/d

The monthly average June-September NPDES permit limit is the controlling criteria for the MAHL. The controlling AHL and MAHL is 4,184 lb/day.

The MAIL for BOD_5 was calculated as the MAHL (4,184 lb/day) minus the current average domestic/background loading (1,634 lb/day) and reserve loading (20 lb/day).

The calculated MAIL for BOD_5 is 2,530 lb day. The MAIL was distributed uniformly to all industrial users, resulting in a calculated BOD local of 354 mg/L. The proposed local limit is 350 mg/L.

5.2 Total Suspended Solids MAHL and Local Limit

The MAHL for TSS was determined similarly to the method described for the BOD_5 MAHL. The 1972 WWTP design capacity for TSS was 4,200 lb/day; however, additional primary and secondary clarification has been added, and tertiary treatment has also been upgraded. No TSS design basis information for the current treatment facility is available, therefore, it was necessary to calculate a theoretical TSS capacity to compare to the NPDES limit formula.

There are currently four 45-ft diameter secondary clarifiers. Peak solids capacity was calculated using *10 States Standards* peak solids loading rate of 40 lb/day/ft^2. With one secondary clarifier out of surface, the peak solids loading rate is 190,852 lb/day, of which approximately, 112,600 lbs/day is used by the RAS flow, leaving 78,200 lb/day for influent solids loading.

The MAHL and local limits are based on the most restrictive AHL which is the monthly average from June to September.

Table 16 – NPDES Permit Based AHLs for TSS

Permit Basis		Season	C_{NPDES}	% Removal	L_{NPDES}
Monthly Average		June-September	20 mg/L	97.0	9,563 lb/d
Monthly Average		October-May	30 mg/L	96.3	11,631 lb/d

The monthly average from June to September is the controlling criteria for the MAHL. After applying a 10% safety factor, the MAHL is 8,607 lb/day.

The MAIL for TSS was calculated as the MAHL (8,607 lb/day) minus the current average domestic loading (2,338lb/d) and reserve loading (100 lb/day).

The calculated MAIL for TSS is 6,168 lb day. As with BOD, a standard local limit was developed for TSS for all non-domestic users, by dividing the MAIL by the industrial flow. This yields an allowable concentration of 863 mg/L. The proposed standard local limit for TSS is 850 mg/L.

5.3 Ammonia MAHL and Local Limit

The MAHL for Ammonia nitrogen was determined similarly to the method described for the BOD_5 and TSS MAHL calculations. The 1972 WWTP O&M manual lists the max day ammonia design basis as 751 lb/day.

This design capacity was compared to the seasonal NPDES permit limits for ammonia. The spreadsheet calculations are attached for reference.

Table 17 – NPDES Permit Based AHLs for Ammonia

Permit Basis	Season	C_{NPDES}	% Removal	L_{NPDES}
Monthly Average	June-September	0.9 mg/L	98.5	861 lb/d
Monthly Average	October-November	1.6 mg/L	97.7	998 lb/d
Monthly Average	December-March	3.7 mg/L	96.3	1,434 lb/d
Monthly Average	April-May	4.8 mg/L	97.0	2,295 lb/d

The 1972 WWTP O&M manual WWTP design basis for ammonia is the controlling criteria for the MAHL of 751 lb/day. A 20% safety factor was chosen for ammonia based on input from City WWTP staff. After applying the 20% safety factor, the MAHL is 600 lb ammonia/day.

The MAIL for ammonia was calculated as the MAHL (600 lb/day) minus the current average domestic loading (238 lb/day) and a reserve loading of 60 lbs of ammonia. The calculated MAIL for ammonia is 303 lb/day.

As with BOD and TSS, a standard local limit was developed for all industrial users. The maximum concentration allowable for this limit was calculated by dividing the MAIL by the industrial flow rate. This yields an allowable concentration of 42 mg/L. The proposed standard local limit for ammonia is 40 mg/L.

5.4 Phosphorus MAHL and Local Limit

The MAHL for phosphorus was determined similarly to the method described for the other compatible pollutant MAHLs. The 1972 design basis report indicates the WWTP has a phosphorus treatment capacity of 667.2 lb/day.

The design capacity was compared to the NPDES permit based AHLs, summarized in Table 18.

Table 18 – NPDES Permit Based AHLs for Phosphorus

Permit Basis	Season	C_{NPDES}	% Removal	L_{NPDES}
Monthly Average	May-March	0.5 mg/L	94.5	130 lb/d
Monthly Average	April	0.65 mg/L	95.2	194 lb/d

The controlling AHL for phosphorus is the May-March monthly average permit limit of 0.5 mg/L. After applying a 5% safety factor, the MAIL for phosphorus was calculated as the MAHL (130 lb/day) minus the current average domestic loading (46 lb/day). The calculated MAIL for phosphorus is 78 lb/day, which was divided uniformly to industrial users. The calculated local limit for phosphorus is 11 mg/L, which is proposed.

5.5 Compatible Pollutant MAHL & Local Limits Summary

The average MAHL, domestic loading, and MAILs for the compatible pollutants are summarized in Table 19, along with the proposed daily maximum local limit based on uniform allocation of the MAIL.

Table 19 – Compatible MAHL, MAIL, and Safety Factor

PARAMETER	Safety Factor	MAHL (lb/day)	Background Loading (lb/day)	Reserve Loading (lb/day)	MAIL (lb/day)	Proposed Daily Max (mg/L)
BOD_5	-	4,184	1,634	20	2,712	350
TSS	10%	8,607	2,338	40	6,228	850
Ammonia	20%	600	238	60	303	40
Phosphorus	5%	130	46	-	78	11

Appendix A

City of Cadillac
Maximum Allowable Headworks Loading Evaluation
Table A1 - Flows and Data Summary

Q_{POTW}	1.72	MGD
Q_{Ind}	0.857	MGD
$Q_{AAR\ Facility\ +\ Rinse\ Water}$	0.298	MGD
$Q_{AAR\ 6th\ St}$	0.006	MGD
Q_{ARVCO}	0.005	MGD
Q_{AVON}	0.004	MGD
$Q_{Akwell\ 5th\ St}$	0.066	MGD
$Q_{Akwell\ 7th\ St}$	0.178	MGD
$Q_{Borg\ Warner}$	0.017	MGD
$Q_{Cadillac\ Castings}$	0.097	MGD, est.
$Q_{Cadillac\ Renewable\ En}$	0.065	MGD
Q_{Fiamm}	0.002	MGD
$Q_{Hutchinson}$	0.008	MGD
$Q_{Michigan\ Rubber}$	0.005	MGD
Q_{Munson}	0.027	MGD
$Q_{Piranha}$	0.077	MGD
$Q_{Rec\ Boat}$	0.0003	MGD
$Q_{Spencer\ Plastics}$	0.0006	MGD
$Q_{domestic/commercial}$	0.863	MGD
$Q_{max,\ design}$	4.50	MGD
$Q_{max,\ observed\ 2021-20}$	2.85	MGD
Q_{dig}	0.070	MGD
Q_{sludge}	0.0025	MGD
Sludge % TS	3%	
95% exceedence	2.6	cfs
Clam River	2	MGD

Sampling Results

Parameter	Background #1 (mg/L)	Background #2 (mg/L)	Flow-Weighted Avg Bkg (mg/L)	Influent (mg/L)	Primary Effluent (mg/L)	Final Effluent (mg/L)
Arsenic	0.0024	0.0010	0.0012	0.0023	0.0049	0.0005
Cadmium	0.0006	0.0001	0.0001	0.00021	0.0004	0.0001
Chromium	0.0082	0.0010	0.0016	0.0056	0.0198	0.0010
Copper	0.2648	0.0696	0.0871	0.0659667	0.1485	0.0058
Cyanide, Total	0.0030	0.0025	0.0025	0.0025	0.0025	0.0173
Cyanide, Availabl	0.006	0.005	0.005	0.003	0.005	0.0016
Lead	0.0278	0.0007	0.0032	0.0025	0.0041	0.0005
Mercury	0.00034	0.00010	0.00012	0.0001	0.00031667	0.0001
Molybdenum	0.00263	0.00077	0.00093	0.0063833	0.0079	0.0053
Nickel	0.0094	0.0021	0.0027	0.003	0.007	0.0036
Selenium	0.0012	0.0006	0.0007	0.0009	0.0013	0.0007
Silver	0.0018	0.0003	0.0004	0.0003	0.0005	0.0003
Zinc	0.6062	0.0974	0.1432	0.1036	0.2029	0.0150
cBOD5	482	202	227	270	309	6.8
TSS	1599	199	325	250	267	5.8
Ammonia	34	33	33	26	4.3	3.6
Phosphorus	15	6	6.3	4.3	0.34	0.22
Phenol	0.043	0.116	0.110	0.055	0.16	0.05

TABLE A2
Local Limits Determination Based on NPDES Monthly Effluent Limits

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE									MAXIMUM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Average Removal Efficiency (%) (Ravg)	Nonvolatile Removal Fraction (%)	NPDES Monthly Limit (mg/l) (Ccrit)		Domestic and Commercial Conc. (mg/l) (Cdom)	Flow (MGD) (Qdom)		Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8568	1.720	78.4	100	0.01600		0.0012	0.8632		1.0630	0.0083			0.9484	0.1328	10
Cadmium	0.8568	1.720	52.4	100	0.00410		0.0001	0.8632		0.1235	0.0011			0.1100	0.0154	10
Chromium	0.8568	1.720	82.0	100	-		0.0016	0.8632		-	0.0119			-	-	10
Copper	0.8568	1.720	91.2	100			0.0871	0.8632		-	0.6271			-	-	25
Cyanide, Available	0.8568	1.720	69.0	100	0.006		0.0048	0.8632		0.2729	0.0345			0.2111	0.0296	10
Lead	0.8568	1.720	79.6	100	-		0.0032	0.8632		-	0.0227			-	-	10
Mercury	0.8568	1.720	60.0	100	0.000002		0.0001	0.8632		0.0001	0.0009			-0.0008	-0.0001	10
Molybdenum	0.8568	1.720	17.2	100	-		0.0009	0.8632		-	0.0067			-	-	10
Nickel	0.8568	1.720	42.0	100	-		0.0027	0.8632		-	0.0196			-	-	10
Selenium	0.8568	1.720	24.1	100			0.0007	0.8632		-	0.0047			-	-	10
Silver	0.8568	1.720	75.0	100	0.0014		0.0004	0.8632		0.0803	0.0028			0.0695	0.0097	10
Zinc	0.8568	1.720	85.6	100	-		0.1432	0.8632		-	1.0309			-	-	10
Phenol	0.8568	1.720	8.4	100	-		0.1097	0.8632		-	0.7896			-	-	10
Beryllium	0.8568	1.720	0.0	100	0.004					0.0588						20

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.
(Q_{potw}) POTW's average influent flow in MGD.
(R_{avg}) Average removal efficiency across POTW as percent.
(C_{crit}) NPDES monthly maximum permit limit for a particular pollutant in mg/l.
(Q_{dom}) Domestic/commercial background flow in MGD.
(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.
(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).
(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).
(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.
(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.
$F_{(nonvol)}$ Fraction of overall POTW removal not due to volatilization as a percent
(SF) Safety factor as a percent.
8.337 Unit conversion factor

$$L_{hw} = \frac{8.337 * C_{crit} * Q_{potw}}{(1 - R_{potw}) * F_{(nonvol)}}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.337}$$

TABLE A3

Local Limits Determination Based on Secondary Treatment Inhibition Level

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE						MAXIMUM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Removal Efficiency (%) (Rprim)	Sec. Treatment Inhibition Level (mg/l) (Ccrit)	Domestic and Commercial Conc. (mg/l) (Cdom)	Domestic and Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8568	1.720	-112	0.1	0.0012	0.8632	0.678	0.008			0.602	0.08	10
Cadmium	0.8568	1.720	-101	1	0.0001	0.8632	7.141	0.001			6.426	0.90	10
Chromium	0.8568	1.720	-256	1	0.0016	0.8632	4.023	0.012			3.609	0.51	10
Copper	0.8568	1.720	-125	1	0.087	0.8632	6.369	0.627			4.149	0.58	25
Cyanide, Total	0.8568	1.720	0	0.1	0.005	0.8632	1.434	0.035			1.256	0.18	10
Lead	0.8568	1.720	-67	1	0.0032	0.8632	8.604	0.023			7.721	1.08	10
Mercury	0.8568	1.720	-217	0.1	0.0001	0.8632	0.453	0.001			0.407	0.06	10
Molybdenum	0.8568	1.720	-24		0.0009	0.8632	-	0.007			-	-	10
Nickel	0.8568	1.720	-100	1	0.0027	0.8632	7.170	0.020			6.433	0.90	10
Selenium	0.8568	1.720	-44		0.0007	0.8632	-	0.005			-	-	10
Silver	0.8568	1.720	-87	0.25	0.0004	0.8632	1.920	0.003			1.726	0.24	10
Zinc	0.8568	1.720	-96	5	0.143	0.8632	36.615	1.031			31.9	4.47	10
Phenol	0.8568	1.720	-189	50	0.110	0.8632	248.048	0.790			222.5	31.1	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.

(Q_{potw}) POTW's average influent flow in MGD.

(R_{prim}) Removal efficiency across across primary treatment as percent.

(C_{crit}) Activated sludge threshold inhibition level, mg/l. Source: US EPA Local Limits Development Guidance, Appendix G.

(Q_{dom}) Domestic/commercial background flow in MGD.

(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.

(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).

(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).

(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.

(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.

(SF) Safety factor as a percent.

8.337 Unit conversion factor

$$L_{hw} = \frac{8.337 * C_{crit} * Q_{potw}}{1 - R_{prim}}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.337}$$

TABLE A4

Local Limits Determination Based on Nitrification Inhibition Level

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE						MAXIMUM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Removal Efficiency (%) (Rprim)	Nitrification Inhibition Level (mg/l) (Ccrit)	Domestic and Commercial Conc. (mg/l) (Cdom)	Domestic and Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8568	1.720	-112	1.5	0.0012	0.8632	10.169	0.008			9.14	1.28	10
Cadmium	0.8568	1.720	-101	5.2	0.0001	0.8632	37.136	0.001			33.42	4.68	10
Chromium	0.8568	1.720	-256	1	0.002	0.8632	4.023	0.012			3.61	0.51	10
Copper	0.8568	1.720	-125	0.48	0.0871	0.8632	3.057	0.627			2.12	0.30	10
Cyanide, Total	0.8568	1.720	0	0.5	0.0048	0.8632	7.170	0.035			6.42	0.90	10
Lead	0.8568	1.720	-67	0.5	0.0032	0.8632	4.302	0.023			3.85	0.54	10
Mercury	0.8568	1.720	-217		0.0001	0.8632	-	0.001			-	-	10
Molybdenum	0.8568	1.720	-24		0.0009	0.8632	-	0.007			-	-	10
Nickel	0.8568	1.720	-100	0.5	0.0027	0.8632	3.585	0.020			3.21	0.45	10
Selenium	0.8568	1.720	-44		0.0007	0.8632	-	0.005			-	-	10
Silver	0.8568	1.720	-87	0.25	0.0004	0.8632	1.920	0.003			1.73	0.24	10
Zinc	0.8568	1.720	-96	0.5	0.143	0.8632	3.661	1.031			2.26	0.32	10
Phenol	0.8568	1.720	-189	4.0	0.110	0.8632	19.844	0.790			17.07	2.39	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.
(Q_{potw}) POTW's average influent flow in MGD.
(R_{prim}) Removal efficiency across primary treatment as percent.
(C_{crit}) Nitrification threshold inhibition level, mg/l.
(Q_{dom}) Domestic/commercial background flow in MGD.
(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.
(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).
(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).
(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.
(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.
(SF) Safety factor as a percent.
8.337 Unit conversion factor

$$L_{hw} = \frac{8.337 * C_{crit} * Q_{potw}}{1 - R_{sec}}$$

$$L_{ind} = L_{hw} * (1\text{-SF}/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.337}$$

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TABLE A5

Local Limits Determination Based on Anaerobic Digester Inhibition Level

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE								MAXIMUM LOADING						
	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Sludge Flow to Digester (MGD) (Qdig)	Sorption Removal Fraction (%)	Removal Efficiency (%) (Rpotw)	Anaerobic Digester Inhibition Level (mg/l) (Ccrit)	Domestic and Commercial Conc. (mg/l) (Cdom)	Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8568	1.720	0.07	100	78.4	1	0.0012	0.8632	0.744	0.0083			0.66145	0.09	10
Cadmium	0.8568	1.720	0.07	100	52.4	1	0.0001	0.8632	1.114	0.0011			1.00164	0.14	10
Chromium	0.8568	1.720	0.07	100	82.0	5	0.0016	0.8632	3.56	0.0119			3.19147	0.45	10
Copper	0.8568	1.720	0.07	100	91.2	40	0.0871	0.8632	25.59	0.6271			22.40102	3.14	10
Cyanide, Total	0.8568	1.720	0.07	100	69.0	30	0.005	0.8632	25.373	0.0345			22.80161	3.19	10
Lead	0.8568	1.720	0.07	100	79.6	50	0.0032	0.8632	36.7	0.0227			32.97255	4.62	10
Mercury	0.8568	1.720	0.07	100	60.0		0.0001	0.8632	-	0.0009			-	-	10
Molybdenum	0.8568	1.720	0.07	100	17.2		0.0009	0.8632	-	0.0067			-	-	10
Nickel	0.8568	1.720	0.07	100	42.0	2	0.0027	0.8632	2.78	0.0196			2.48149	0.35	10
Selenium	0.8568	1.720	0.07	100	24.1		0.0007	0.8632	-	0.0047			-	-	10
Silver	0.8568	1.720	0.07	100	75.0	13	0.0004	0.8632	10.12	0.0028			9.10123	1.27	10
Zinc	0.8568	1.720	0.07	100	85.6	40	0.1432	0.8632	27.3	1.0309			23.52134	3.29	10
Phenol	0.8568	1.720	0.07	100	8.4		0.110	0.8632	-	0.7896			-	-	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.
(Q_{potw}) POTW's average influent flow in MGD.
(Q_{dig}) Sludge flow to digester in MGD.
(R_{potw}) Removal efficiency across POTW as percent.
(C_{crit}) Anaerobic digester threshold inhibition level in mg/l.
(Q_{dom}) Domestic/commercial background flow in MGD.
(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.
(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).
(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).
(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.
(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.
$F_{(sorp)}$ Fraction of overall removal due to sorption as a percent
(SF) Safety factor as a percent.
8.337 Unit conversion factor

$$L_{hw} = \frac{8.337 * C_{crit} * Q_{dig}}{R_{potw} * F_{sorp}}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * \ 8.337}$$

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TABLE A6

Local Limits Determination Based on Chronic Water Quality Standards

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE									MAXIMIM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	Avg POTW Flow (MGD) (Qpotw)	20yr Avg POTW Flow (MGD) (Qmax)	River 95% Exceedence Flow (MGD)	Nonvolatile Removal Fraction (%)	Average Removal Efficiency (%) (Ravg)	Chronic WQBEL (mg/l) (Ccrit)	Domestic and Commercial Conc. (mg/l) (Cdom)	Domestic and Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Available Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8568	1.720	4.5	1.7	100	78.4	0.016	0.0012	0.8632	1.06	0.008			0.948	0.1	10
Cadmium	0.8568	1.720	4.5	1.7	100	52.4	0.004	0.0001	0.8632	0.12	0.001			0.110	0.02	10
Chromium	0.8568	1.720	4.5	1.7	100	82.0	0.094	0.0016	0.8632	7.48	0.012			6.721	1	10
Copper	0.8568	1.720	4.5	1.7	100	91.2	0.011	0.0871	0.8632	1.80	0.627			0.722	0.1	25
Cyanide, Available	0.8568	1.720	4.5	1.7	100	69.0	0.006	0.005	0.8632	0.27	0.035			0.211	0.03	10
Lead	0.8568	1.720	4.5	1.7	100	79.6	0.094	0.0032	0.8632	6.60	0.023			5.922	0.8	10
Mercury	0.8568	1.720	4.5	1.7	100	60.0	0.0000013	0.0001	0.8632	0.00005	0.0009			-0.001	-0.0001	10
Molybdenum	0.8568	1.720	4.5	1.7	100	17.2	3.600	0.0009	0.8632	62.37	0.007			56.127	8	10
Nickel	0.8568	1.720	4.5	1.7	100	42.0	0.047	0.0027	0.8632	1.16	0.020			1.026	0	10
Selenium	0.8568	1.720	4.5	1.7	100	24.1	0.006	0.0007	0.8632	0.11	0.005			0.092	0.01	10
Silver	0.8568	1.720	4.5	1.7	100	75.0	0.0014	0.0004	0.8632	0.080	0.003			0.070	0.01	10
Zinc	0.8568	1.720	4.5	1.7	100	85.6	0.21	0.1432	0.8632	20.87	1.031			17.750	2.5	10
Phenol	0.8568	1.720	4.5	1.7	100	8.4	0.51	0.110	0.8632	7.98	0.790			6.392	0.9	10

(Q_{ind})	Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.
(Q_{potw})	POTW's average influent flow in MGD.
(Q_{95ex})	MDEQ Designated 95% exceedence flow for receiving stream (MGD)
(Ravg)	Average removal efficiency across POTW as percent.
(C_{crit})	State chronic water quality standard for a particular pollutant in mg/l. Reference: EGLE Interoffice Memo 11/25/2019
(Q_{dom})	Domestic/commercial background flow in MGD.
(C_{dom})	Domestic/commercial background concentration for a particular pollutant in mg/l.
(L_{hw})	Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).
(L_{dom})	Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).
(L_{ind})	Maximum allowable industrial loading to the POTW in pounds per day.
(C_{ind})	Industrial allowable local limit for a given pollutant in mg/l.
$F(_{nonvol})$	Fraction of overall POTW removal not due to volatilization as a percent
(SF)	Safety factor as a percent.
(Q_{max})	Maximim permitted POTW flow
8.337	Unit conversion factor

$$C_{crit} = \frac{C_{std} * (Q_{max} + 25\%*Q_{95ex})}{Q_{max}}$$

$$L_{hw} = \frac{8.337 * Q_{potw} * C_{crit}}{(1 - R_{avg} * F_{nonvol})}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * \ 8.337}$$

TABLE A7

Local Limits Determination Based on Acute Water Quality Standards

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE									MAXIMIM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	Avg POTW Flow (MGD) (Qpotw)	Max POTW Flow (MGD) (Qmax)	River 95% Exceedence Flow (MGD)	Nonvolatile Removal Fraction (%)	Average Removal Efficiency (%) (Ravg)	Acute WQBEL (mg/l) (Ccrit)	Domestic & Commercial Conc. (mg/l) (Cdom)	Domestic & Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8568	1.720	4.5	1.7	100	78.4	0.680	0.0012	0.8632	45.18	0.0083			40.65	6	10
Cadmium	0.8568	1.720	4.5	1.7	100	52.4	0.012	0.00015	0.8632	0.361	0.0011			0.32	0.05	10
Chromium	0.8568	1.720	4.5	1.7	100	82.0	1.3	0.002	0.8632	103	0.0119			93.10	13	10
Copper	0.8568	1.720	4.5	1.7	100	91.2	0.029	0.0871	0.8632	4.74	0.6271			3.64	0.51	10
Cyanide, Available	0.8568	1.720	4.5	1.7	100	69.0	0.044	0.005	0.8632	2.035	0.0345			1.80	0.25	10
Lead	0.8568	1.720	4.5	1.7	100	79.6	0.8	0.0032	0.8632	58	0.0227			51.83	7.3	10
Mercury	0.8568	1.720	4.5	1.7	100	60.0	-	0.0001	0.8632	-	0.0009			-	-	10
Molybdenum	0.8568	1.720	4.5	1.7	100	17.2	58	0.0009	0.8632	1005	0.0067			904.37	127	10
Nickel	0.8568	1.720	4.5	1.7	100	42.0	0.8	0.0027	0.8632	19	0.0196			16.89	2.4	10
Selenium	0.8568	1.720	4.5	1.7	100	24.1	0.120	0.00065	0.8632	2.266	0.0047			2.04	0.28	10
Silver	0.8568	1.720	4.5	1.7	100	75.0	0.022	0.00039	0.8632	1.262	0.0028			1.13	0.16	10
Zinc	0.8568	1.720	4.5	1.7	100	85.6	0.360	0.143	0.8632	36	1.0309			31.17	4.4	10
Phenol	0.8568	1.720	4.5	1.7	100	8.4	6.800	0.110	0.8632	106	0.7896			94.97	13.3	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.
(Q_{potw}) POTW's average influent flow in MGD.
(Q_{95ex}) MDEQ Designated 95% exceedence flow for receiving stream (MGD)
(Ravg) Average removal efficiency across POTW as percent.
(C_{crit}) State acute water quality standard for a particular pollutant in mg/l. Reference: EGLE Interoffice Memo 11/25/2019
(Q_{dom}) Domestic/commercial background flow in MGD.
(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.
(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).
(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).
(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.
(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.
$F_{(nonvol)}$ Fraction of overall POTW removal not due to volatilization as a percent
(SF) Safety factor as a percent.
(Q_{max}) Maximim permitted POTW flow
8.337 Unit conversion factor

$$C_{crit} = C_{std}$$

$$L_{hw} = \frac{8.337 * Q_{potw} * C_{crit}}{(1 - R_{avg} * F_{nonvol})}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.337}$$

TABLE A8

Local Limits Determination Based on USEPA 503 Sludge Regulations

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE										MAXIMUM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Sludge Flow (MGD) (Qsldg)	Sorption Removal Fraction (%)	Percent Solids (%) (PS)	Average Digester Removal (%)	Average Removal Efficiency (%) (Ravg)	503 Sludge Criteria (mg/kg) (Cslcrit)	Domestic and Commercial Conc. (mg/l) (Cdom)	Domestic and Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8568	1.720	0.0025	100	3.0	0	78.4	41	0.0012	0.8632	0.0322	0.0083			0.021	0.003	10
Cadmium	0.8568	1.720	0.0025	100	3.0	0	52.4	39	0.0001	0.8632	0.0459	0.0011			0.040	0.006	10
Chromium	0.8568	1.720	0.0025	100	3.0	0	82.0		0.002	0.8632	-	0.0119			-	-	10
Copper	0.8568	1.720	0.0025	100	3.0	0	91.2	1500	0.087	0.8632	1.0140	0.6271			0.285	0.04	10
Cyanide, Total	0.8568	1.720	0.0025	100	3.0	0	69.0		0.005	0.8632	-	0.0345			-	-	10
Lead	0.8568	1.720	0.0025	100	3.0	0	79.6	300	0.0032	0.8632	0.2325	0.0227			0.186	0.03	10
Mercury	0.8568	1.720	0.0025	100	3.0	0	60.0	17	0.0001	0.8632	0.0175	0.0009			0.015	0.002	10
Molybdenum	0.8568	1.720	0.0025	100	3.0	0	17.2	75	0.0009	0.8632	0.2684	0.0067			0.235	0.033	10
Nickel	0.8568	1.720	0.0025	100	3.0	0	42.0	420	0.0027	0.8632	0.6167	0.0196			0.535	0.075	10
Selenium	0.8568	1.720	0.0025	100	3.0	0	24.1	100	0.0007	0.8632	0.2562	0.0047			0.226	0.032	10
Silver	0.8568	1.720	0.0025	100	3.0	0	75.0		0.0004	0.8632	-	0.0028			-	-	10
Zinc	0.8568	1.720	0.0025	100	3.0	0	85.6	2800	0.1432	0.8632	2.0180	1.0309			0.785	0.110	10
Phenol	0.8568	1.720	0.0025	100	3.0	0	8.4		0.1097	0.8632	-	0.7896			-	-	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.

(Q_{potw}) POTW's average influent flow in MGD.

(Q_{sldg}) Sludge flow to disposal in MGD.

(PS) Percent solids of sludge to disposal.

(R_{max}) Maximum removal efficiency across POTW as a percent.

(C_{slcrit}) 503 sludge criteria in mg/kg dry sludge.

(Q_{dom}) Domestic/commercial background flow in MGD.

(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.

(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).

(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).

(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.

(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.

$F_{(sorp)}$ Fraction of overall removal due to sorption as a percent

$R_{(dig)}$ Average removal fraction of digester as a percent

(SF) Safety factor as a percent.

8.337 Unit conversion factor

$$L_{hw} = \frac{8.337 * C_{slcrit} * (PS/100) * Q_{sldg}}{R_{max} * F_{sorp} * (1 - R_{dig})}$$

$$L_{ind} = L_{hw} * (1-SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.337}$$

CITY OF CADILLAC

TABLE A9
Summary of Determined Limits

				Arsenic	Cadmium	Chromium	Copper	Cyanide	Lead	Mercury	Molybdenum	Nickel	Selenium	Silver	Zinc	Phenol
NPDES Monthly Limits	Allowable Headworks Load	(lbs/day)	(Lhw)	1.063	0.123	-	-	0.273	-	0.0000717	-	-	-	0.080	-	-
	Domestic/Commercial	(lbs/day)	(Ldom)	0.008	0.001	0.012	0.627	0.035	0.023	0.0009	0.007	0.020	0.005	0.0028	1.031	0.790
	Allowable Industrial Load	(lbs/day)	(Lind)	0.948	0.110	-	-	0.211	-	-0.00081	-	-	-	0.070	-	-
	Local Limit (Cind)	(mg/l)	(Cind)	0.133	0.02	-	-	0.03	-	-0.0001	-	-	-	0.01	-	-
Secondary Treatment Inhibition	Allowable Headworks Load	(lbs/day)	(Lhw)	0.678	7.141	4.023	6.369	1.434	8.604	0.45283	-	7.170	-	1.920	36.615	248.0
	Domestic/Commercial	(lbs/day)	(Ldom)	0.008	0.001	0.012	0.627	0.035	0.023	0.0008774	0.007	0.020	0.005	0.0028	1.031	0.790
	Allowable Industrial Load	(lbs/day)	(Lind)	0.602	6.426	3.609	4.149	1.256	7.721	0.40667	-	6.433	-	1.726	31.922	222.5
	Local Limit (Cind)	(mg/l)	(Cind)	0.084	0.900	0.505	0.581	0.176	1.081	0.05693	-	0.901	-	0.242	4.469	31.1
Nitrification Inhibition	Allowable Headworks Load	(lbs/day)	(Lhw)	10.169	37.136	4.023	3.057	7.170	4.302	-	-	3.585	-	1.920	3.661	19.8
	Domestic/Commercial	(lbs/day)	(Ldom)	0.008	0.001	0.012	0.627	0.035	0.023	0.00088	0.007	0.020	0.005	0.003	1.031	0.790
	Allowable Industrial Load	(lbs/day)	(Lind)	9.144	33.421	3.609	2.124	6.418	3.849	-	-	3.207	-	1.726	2.264	17.1
	Local Limit (Cind)	(mg/l)	(Cind)	1.280	4.679	0.505	0.297	0.899	0.539	-	-	0.449	-	0.242	0.242	2.4
Digester Inhibition	Allowable Headworks Load	(lbs/day)	(Lhw)	0.744	1.114	3.559	25.587	25.373	36.661	-	-	2.779	-	10.116	27.280	-
	Domestic/Commercial	(lbs/day)	(Ldom)	0.008	0.001	0.012	0.627	0.035	0.023	0.0009	0.0067	0.020	0.005	0.003	1.031	0.790
	Allowable Industrial Load	(lbs/day)	(Lind)	0.661	1.002	3.191	22.401	22.802	32.973	-	-	2.481	-	9.101	23.521	-
	Local Limit (Cind)	(mg/l)	(Cind)	0.093	0.140	0.447	3.136	3.192	4.616	-	-	0.347	-	1.274	3.293	-
Chronic Water Quality Standards	Allowable Headworks Load	(lbs/day)	(Lhw)	1.063	0.123	7.481	1.799	0.273	6.605	0.00005	62.371	1.162	0.108	0.0803	20.868	7.98
	Domestic/Commercial	(lbs/day)	(Ldom)	0.008	0.001	0.012	0.627	0.035	0.023	0.00088	0.007	0.020	0.005	0.0028	1.031	0.790
	Allowable Industrial Load	(lbs/day)	(Lind)	0.948	0.110	6.721	0.722	0.211	5.922	-0.00084	56.127	1.026	0.092	0.06950	17.750	6.39
	Local Limit (Cind)	(mg/l)	(Cind)	0.133	0.015	0.941	0.101	0.030	0.829	-0.00012	7.8578	0.144	0.013	0.0097	2.485	0.89
Acute Water Quality Standards	Allowable Headworks Load	(lbs/day)	(Lhw)	45.179	0.361	103.461	4.743	2.035	57.617	-	1004.860	18.790	2.266	1.262	35.773	106.4
	Domestic/Commercial	(lbs/day)	(Ldom)	0.008	0.001	0.012	0.627	0.035	0.023	0.0008774	0.007	0.020	0.005	0.0028	1.031	0.790
	Allowable Industrial Load	(lbs/day)	(Lind)	40.653	0.324	93.103	3.642	1.797	51.832	-	904.368	16.891	2.035	1.133	31.165	95.0
	Local Limit (Cind)	(mg/l)	(Cind)	5.691	0.045	13.034	0.510	0.252	7.257	-	126.6123	2.365	0.285	0.159	4.363	13.3
USEPA 503 Sludge Regulations	Allowable Headworks Load	(lbs/day)	(Lhw)	0.032	0.046	-	1.014	-	0.232	0.01747	0.268	0.617	0.256	-	2.018	-
	Domestic/Commercial	(lbs/day)	(Ldom)	0.008	0.001	0.012	0.627	0.035	0.023	0.0008774	0.007	0.020	0.005	0.0028	1.031	0.790
	Allowable Industrial Load	(lbs/day)	(Lind)	0.021	0.040	-	0.285	-	0.186	0.01485	0.235	0.535	0.226	-	0.785	-
	Local Limit (Cind)	(mg/l)	(Cind)	0.003	0.006	-	0.040	-	0.026	0.00208	0.0329	0.075	0.032	-	0.110	-

CITY OF CADILLAC

TABLE A10
Allowable Headworks Load

			Arsenic	Cadmium	Chromium	Copper	Cyanide	Lead	Mercury	Molybdenum	Nickel	Selenium	Silver	Zinc	Phenol
NPDES Monthly Limit ML	(lbs/day)	(Lhw)	**1.063**	0.123	-	-	0.273	-	0.0001	-	-	-	0.080	-	-
Secondary Treatment Inhibition SI	(lbs/day)	(Lhw)	0.68	7.14	4.02	6.369	1.434	8.604	0.453	-	7.2	-	1.920	36.61	248
Nitrification Inhibition NI	(lbs/day)	(Lhw)	10.17	37.14	4.02	3.057	7.170	4.302	-	-	3.6	-	1.920	3.66	19.84
Anaerobic Digester Inhibition DI	(lbs/day)	(Lhw)	0.744	1.114	**3.559**	25.587	25.373	36.661	-	-	2.779	-	10.116	27.28	-
Chronic Water Quality Standards C	(lbs/day)	(Lhw)	1.06	0.123	7.48	1.799	**0.273**	6.605	**0.00005**	62	1.162	**0.108**	**0.080**	20.9	**7.98**
Acute Water Quality Standards A	(lbs/day)	(Lhw)	45.18	0.361	103.5	4.743	2.035	57.617	-	1005	19	2.266	1.262	35.77	106
USEPA 503 Sludge Regulations BS	(lbs/day)	(Lhw)	0.032	**0.046**	-	**1.014**	-	**0.232**	0.017	**0.268**	**0.617**	0.256	-	**2.02**	-
MAHL	(lbs/day)		0.032	0.046	3.559	1.014	0.273	0.232	0.00005	0.268	0.617	0.108	0.080	2.018	7.980
Basis			ML	BS	DI	BS	C	BS	C	BS	BS	C/ML	C	BS	C

TABLE A11
Allowable Industrial Load

			Arsenic	Cadmium	Chromium	Copper	Cyanide	Lead	Mercury	Molybdenum	Nickel	Selenium	Silver	Zinc	Phenol
NPDES Monthly Limit ML	(lbs/day)	(Lhw)	0.948	0.110	-	-	**0.211**	-	**-0.001**	-	-	-	**0.070**	-	-
Secondary Treatment Inhibition SI	(lbs/day)	(Lind)	0.602	6.426	3.609	4.149	1.256	7.721	0.407	-	6.433	-	1.726	31.922	222.5
Nitrification Inhibition NI	(lbs/day)	(Lind)	9.144	33.421	3.609	2.124	6.418	3.849	-	-	3.207	-	1.726	2.264	17.1
Anaerobic Digester Inhibition DI	(lbs/day)	(Lind)	0.661	1.002	**3.191**	22.401	22.802	32.973	-	-	2.481	-	9.101	23.521	-
Chronic Water Quality Standards C	(lbs/day)	(Lind)	0.948	0.110	6.721	0.722	**0.211**	5.922	**-0.001**	56.127	1.026	**0.092**	**0.070**	17.750	**6.39**
Acute Water Quality Standards A	(lbs/day)	(Lind)	40.653	0.324	93.103	3.642	1.797	51.832	-	904.368	16.891	2.035	1.133	31.165	95.0
USEPA 503 Sludge Regulations BS	(lbs/day)	(Lind)	**0.021**	**0.040**	-	**0.285**	-	**0.186**	0.015	**0.235**	**0.535**	0.226	-	**0.785**	-
MAIL	(lbs/day)		0.021	0.040	3.191	0.285	0.211	0.186	-0.001	0.235	0.535	0.092	0.070	0.785	6.39
Basis			BS	BS	DI	BS	ML/C	BS	ML/C	BS	BS	C	ML/C	BS	C

TABLE A12
Calculated Standard Local Limits - Uniform Allocation

		Arsenic	Cadmium	Chromium	Copper	Cyanide	Lead	Mercury†	Molybdenum	Nickel	Selenium	Silver	Zinc	Phenol
Daily Maximum	(mg/L)	0.003	0.006	0.45	0.040	0.03	0.026	0.000	0.033	0.075	0.013	0.010	0.110	0.895
	(ug/L)	3	6	447	40	30	26	0	33	75	13	10	110	895
Basis		BS	BS	DI	BS	ML/C	BS	ML/C	BS	BS	C	ML/C	BS	C

†Mercury Local Limit will be Non-Detect

TABLE A13 C_{crit} VALUES

Highlighted Values Were Used In This Local Limits Determination

The following tables are taken from EPA's Local Limits Development Guidance Appendixes, and Guidance Manual for Preventing Interference at PO

	Appendix R tables				Appendix V
	Removal Efficiencies				Domestic
	Primary	Activated sludge	Trickling Fltr	Tertiary	Level
Ag	20%	75%	66%	62%	0.019
As		45%			0.007
Cd*	15%	67%	68%	50%	0.008
CN	27%	69%	59%	66%	0.082
Cr	27%	82%	55%	72%	0.034
Cu	22%	86%	55%	85%	0.061
Hg	10%	60%	50%	67%	0.002
Mo					
Ni	14%	42%	29%	17%	0.047
Pb	57%	61%	55%	52%	0.058
Se		33%			
Zn	27%	79%	67%	78%	0.231
Phenol					

	Appendix G and Table 2-1 tables			
	Inhibition Levels			
	Activated sludge	Trickling Fltr	Nitrification	Digestion
Ag	0.25		0.25	13
As	0.1		1.5	1
Cd*	1		5.2	1
CN	0.1	30	0.5	30
Cr	1		1	5
Cu	1		0.48	40
Hg	0.1			
Mo				
Ni	1		0.5	2
Pb	1		0.5	50
Se				
Zn	5		0.5	40
Phenol	50		4	

Site Specific Data from Cadillac WWTP Sampling		
WWTF % Removals		
	Primary	Total Plant
Ag	-87%	0%
As	-112%	78%
Cd	-101%	52%
CN	0%	-590%
Cr	-256%	82%
Cu	-125%	91%
Hg	-217%	0%
Mo	-24%	17%
Ni	-100%	-8%
Pb	-67%	80%
Se	-44%	24%
Zn	-96%	86%
Phenol	-189%	8%

	Part 503 / 24 Land Application of Sludge			
	"Table 3"	Ceiling Concentration	Cumulative Pollutant Loading Rates	Annual Pollutant Loading Rate
As	41	75	37	1.8
Cd*	39	85	35	1.7
Cu*	1500	4300	1338	67
Hg	17	57	15	0.76
Mo	-	75	-	-
Ni*	420	420	375	19
Pb*	300	840	268	13
Se	100	100	89	4.5
Zn*	2800	7500	2498	125

City of Cadillac WWTP
Compatibles MAHL Calculations

Flows Summary

$Q_{City\ domestic,\ est.}$	0.86 MGD
$Q_{industrial}$	0.857 MGD
$Q_{POTW,\ Total}$	1.72 MGD

TABLE A14 - BOD MAHL CALCULATIONS

<u>Allowable BOD Loading - NPDES Permit Basis</u>

L_{NPDES}

$C_{NPDES,\ Monthly\ avg\ BOD5\ (Jun-Sept)}$ =	7 mg/L
$R_{Annual\ Avg,\ BOD5\ (Jun-Sept)}$ =	97.6%
Q_{POTW} =	1.72 MGD
$L_{NPDES,\ BOD5\ (Monthly\ Avg,\ Jun-Sept)}$ =	4,184 lb/day
$C_{NPDES,\ monthly\ avg\ BOD5\ (Oct-Nov)}$ =	11 mg/L
$R_{Avg,\ BOD5\ (Oct-Nov)}$ =	97.5%
Q_{POTW} =	1.72 MGD
$L_{NPDES,\ BOD5\ (Monthly\ Avg,\ Oct-Nov)}$ =	6,312 lb/day
$C_{NPDES,\ monthly\ avg\ BOD5\ (Dec-Mar)}$ =	23 mg/L
$R_{Avg,\ BOD5\ (Dec-Mar)}$ =	97.2%
Q_{POTW} =	1.72 MGD
$L_{NPDES,\ BOD5\ (Monthly\ Avg,\ Dec-Mar)}$ =	11,783 lb/day
$C_{NPDES,\ monthly\ avg\ BOD5\ (Apr-May)}$ =	17 mg/L
$R_{Avg,\ BOD5\ (Apr-May)}$ =	96.8%
Q_{POTW} =	1.72 MGD
$L_{NPDES,\ BOD5\ (Monthly\ Avg,\ Apr-May)}$ =	7,621 lb/day

Allowable BOD Loading - Design Capacity Basis

1975 O&M Manual - Average WWTP BOD Capacity, prior to fine bubble additior	3,300	lb BOD/day
Design basis check after 1996 improvements:		
Aeration Volume	330,000	gals
	118,705	cf
BOD loading (per 10 States Standards)	40	lbs/1000 cf
	4,748	lb BOD/day to secondary treatment
Maximum Allowable Headworks Loading =	**4,184**	**lb/day based on NPDES Monthly Avg (Jun-Sept)**

Determine Domestic Loading & MAIL

Average domestic BOD concentration 227 mg/L

Total Maximum Allowable Headworks Loading	4184	lb BOD5/day
Domestic Loading	1634	lb BOD5/day
Reserve Loading	20	lb BOD5/day
MAIL BOD	2530	lb BOD5/day

Uniform Allocation of BOD MAIL to Industrial Users	354	mg/L
Proposed Local Limit	**350**	**mg/L**

TABLE A15 - TSS MAHL CALCULATIONS
<u>TSS Loading - NPDES Permit Basis</u>

$$L_{NPDES} = \frac{8.34 * C_{NPDES} * Q_{POTW}}{1 - R_{Avg}}$$

$C_{NPDES,\ Monthly\ avg\ TSS\ (Jun\text{-}Sept)}$ =	20	mg/L
$R_{Avg,\ TSS\ (June\text{-}Sept)}$ =	97.0%	
Q_{POTW} =	1.72	MGD
$L_{NPDES,\ TSS\ (Jun\text{-}Sept)}$ =	9,563	lb TSS/day
$C_{NPDES,\ monthly\ avg\ TSS\ (Oct\text{-}May)}$ =	30	mg/L
$R_{Avg,\ TSS\ (Oct\text{-}May)}$ =	96.3%	
Q_{POTW} =	1.72	MGD
$L_{NPDES,\ TSS\ (Monthly\ Avg,\ Oct\text{-}May)}$ =	11,631	lb TSS/day

<u>TSS Capacity - Design Capacity Basis</u>

1972 Design Basis - Average TSS Capacity prior to adding additional primary and secondary clarification and upgrading tertiary treatment	4200	lb TSS/day; not applicable today

Secondary Clarifier Peak Solids Capacity Check:

10 states standards peak solids loading rate =	40	lb/day/ sf
No. of secondary clarifiers	4	
Diameter, each	45	ft
Surface area, each	1590	sf
Total Secondary Clarifier Surface area with largest out of service	4771	sf
Peak Solids loading allowed	190,852	lb TSS/day
$Q_{POTW\ Max}$ =	4.50	MGD
Target MLSS	3,000	mg/L
Loading used by RAS/MLSS	112,590	lb TSS/day at Max Flow.
Remaining TSS Capacity	78,262	lb TSS/day
Maximum Allowable Headworks Loading TSS =	**9,563**	**lb/day based on Monthly Average June-Sept.**

<u>Determine Domestic Loading & MAIL</u>

Average domestic TSS concentration	325	mg/L

Total Maximum Allowable Headworks Loading (with 10% SF)	8607	lb TSS/day
Domestic Loading	2338	lb TSS/day
Reserve Loading	100	lb TSS/day
MAIL TSS	6168	lb TSS/day

Uniform Allocation of TSS MAIL to Industrial Users	863	mg/L
Proposed TSS Local Limit	**850**	**mg/L**

TABLE A16 - AMMONIA NITROGEN MAHL CALCULATIONS

Ammonia Loading - NPDES Permit Basis

$$L_{NPDES} = \frac{8.34 * C_{NPDES} * Q_{POT}}{1 - R_{Avg}}$$

$C_{NPDES, Monthly Avg Ammonia (Jun-Sept)}$ =	0.9	mg/L
$R_{Avg, Ammonia (Jun-Sept)}$ =	98.5%	
$Q_{POTW, Avg}$ =	1.72	MGD
$L_{NPDES, Ammonia (Jun-Sept)}$ =	861	lb NH3/day
$C_{NPDES, Monthly Avg Ammonia (Oct-Nov)}$ =	1.6	mg/L
$R_{Avg, Ammonia (Oct-Nov)}$ =	97.7%	
$Q_{POTW, Avg}$ =	1.72	MGD
$L_{NPDES, Ammonia (Oct-Nov)}$ =	998	lb NH3/day
$C_{NPDES, Monthly Avg Ammonia (Dec-Mar)}$ =	3.7	mg/L
$R_{Avg, Ammonia (Dec-Mar)}$ =	96.3%	
$Q_{POTW, Avg}$ =	1.72	MGD
$L_{NPDES, Ammonia (Dec-Mar)}$ =	1434	lb NH3/day
$C_{NPDES, Monthly Avg Ammonia (Apr-May)}$ =	4.8	mg/L
$R_{Avg, Ammonia (Apr-May)}$ =	97.0%	
$Q_{POTW, Avg}$ =	1.72	MGD
$L_{NPDES, Ammonia (Apr-May)}$ =	2295	lb NH3/day

Ammonia Design Capacity Analysis

1972 O&M Manual WWTP design basis for ammonia	751	lb/day, max day
Ammonia Maximum Allowable Headworks Loading =	**751**	**1972 O&M Manual WWTP design basis for Ammonia**

Determine Domestic Loading & MAIL

Average domestic Ammonia concentration 33 mg/L

Total Maximum Allowable Headworks Loading w 20% SF	600	lb NH3/day
Domestic Loading	238	lb NH3/day
Reserve Loading	60	lb NH3/day
MAIL Ammonia	303	lb NH3/day

Uniform Allocation of Ammonia MAIL to Industrial Users	42	mg/L
Proposed Local Limit	**40**	**mg/L**

TABLE A17 - TOTAL PHOSPHORUS MAHL CALCULATIONS

Total Phosphorus Loading - NPDES Permit Basis

$$L_{NPDES} = \frac{8.34 * C_{NPDES} * Q_{POTW}}{1 - R_{Avg}}$$

$C_{NPDES,\ Monthly\ Avg\ (May-March)}$ =	0.5	mg/L
$R_{Avg,\ Total\ Phosphorus\ (May-March)}$ =	94.5%	
Q_{POTW} =	1.72	MGD
$L_{NPDES,\ Total\ Phosphorus}$ =	130	lb TP/day
$C_{NPDES,\ Monthly\ Avg\ (Apr)}$ =	0.65	mg/L
$R_{Avg,\ Total\ Phosphorus\ (Apr)}$ =	95.2%	
Q_{POTW} =	1.72	MGD
$L_{NPDES,\ Total\ Phosphorus}$ =	194	lb TP/day

Allowable Phosphorus Loading - Design Capacity Basis

1972 O&M Manual Design Basis for Phosphorus	667.2	lb TP/day
Maximum Allowable Headworks Loading for Total Phosphorus =	**130**	lbs TP/day, based on NPDES

Determine Domestic Loading & MAIL

Average domestic Total Phosphorus concentration	6.34	mg/L

Total Maximum Allowable Headworks Loading w/ 5% SF	124	lb TP/day
Domestic Loading	46	lb TP/day
Reserve Loading	0	lb TP/day
Total Phosphorus MAIL	78	lb TP/day

Uniform Allocation of Phosphorus MAIL to Industrial Users	11.0	mg/L
Proposed Phosphorus Local Limit	**11**	**mg/L**

Cadillac Castings Inc
Categorical Limit Calculations
Table A18

Regulated under 40 CFR Part 464.36

Processes include:

Dust Collection Scrubbing Operations (c), primarily ductile iron castings

Categorical Limits:

Pollutant	Max Day	Monthly Avg
	lbs/ billion SCF of air scrubbed	
Copper	0.218	0.12
Lead	0.398	0.195
Zinc	0.736	0.278
Total Phenols	0.646	0.225
TTO	2.04	0.664
FOG	22.5	7.51

Dust Scrubber Flow	52,000	scfm, each
Number of scrubbers	2	
Total flow, SCF/day	149,760,000	
Total flow, billion SCF/day	0.150	

Categorical Limits for CCI, based on scrubber flow:

Pollutant	**Max Day**	**Monthly Avg**
	lbs/ day	
Copper	**0.03**	**0.02**
Lead	**0.06**	**0.03**
Zinc	**0.11**	**0.04**
Total Phenols	**0.10**	**0.03**
TTO	**0.31**	**0.10**
FOG	**3.37**	**1.12**

Melting Furnace Scrubbing Operations (f), primarily ductile iron

Categorical Limits:

Pollutant	Max Day	Monthly Avg
	lbs/ billion SCF of air scrubbed	
Copper	1.02	0.561
Lead	1.86	0.911
Zinc	3.44	1.3
Total Phenols	3.01	1.05
TTO	8.34	2.73
FOG	105	35

Melt Furnace Scrubber Flow	30,000
Number of scrubbers	1
Total flow, SCF/day	43,200,000
Total flow, billion SCF/day	0.043

Categorical Limits for CCI, based on scrubber flow:

Pollutant	**Max Day**	**Monthly Avg**
	lbs/ day	
Copper	**0.15**	**0.08**
Lead	**0.28**	**0.14**
Zinc	**0.52**	**0.19**
Total Phenols	**0.45**	**0.16**
TTO	**1.25**	**0.41**
FOG	**15.72**	**5.24**

WWTP Influent

8389800	Grab Sample Parameters		Metals (Total)										
Collection	**Cyanide, Total**	**Cyanide, Available**	**Mercury**	**Arsenic**	**Cadmium**	**Chromium**	**Copper**	**Lead**	**Molybdenum**	**Nickel**	**Selenium**	**Silver**	**Zinc**
Date	**mg/L**	**µg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0050	<10.0	<0.0002	0.0027	0.00038	0.008	0.0757	0.002	0.0047	0.0036	0.0018	<0.0005	0.122
10/13/2022	<0.0050	3	<0.0002	0.0022	0.00022	0.0049	0.0646	0.0029	0.0077	0.0032	<0.001	<0.0005	0.107
10/14/2022	<0.0050	2.3	<0.0002	0.0021	0.00025	0.0054	0.0928	0.0045	0.0098	0.0037	0.0016	<0.0005	0.127
10/21/2022	<0.0050	3.3	<0.0002	0.0025	0.00021	0.0062	0.0626	0.0029	0.0078	0.0033	<0.001	<0.0005	0.114
10/22/2022	<0.0050	2.6	<0.0002	0.0023	<0.0002	0.005	0.0531	0.0013	0.0031	0.0032	<0.001	<0.0005	0.079
10/25/2022	<0.0050	2.5	<0.0002	0.0021	<0.0002	0.0038	0.047	0.0011	0.0052	0.0028	<0.001	<0.0005	0.073
Average	0.0025	0.0031	0.0001	0.0023	0.0002	0.0056	0.0660	0.0025	0.0064	0.0033	0.0009	######	0.104

8389800	Compatible Pollutants				Organic Pollutants									
Collection	**NH_3-N**	**BOD_5**	**TSS**	**Phosphorus**	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	19.7	203	162	4.5	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.144	<0.00051
10/13/2022	20.6	205	162	3.9	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000048
10/14/2022	23.4	258	180	4.5	<0.00012	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000062
10/21/2022	21	190	156	3.9	<0.00011	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000053
10/22/2022	24	228	63	4.4	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000050
10/25/2022	24.6	283	70	3	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.000050
Average	22	228	132	4.0	0.00005	0.0024	0.0025	0.0025	0.0025	0.0025	0.005	0.0546	0.00003

WWTP Primary Effluent

	Grab Sample Parameters		Metals (Total)											Compatible Pollutants				
Collection	**Cyanide, Total**	**Cyanide, Available**	**Mercury**	**Arsenic**	**Cadmium**	**Chromium**	**Copper**	**Lead**	**Molybdenum**	**Nickel**	**Selenium**	**Silver**	**Zinc**	**NH_3-N**	**BOD_5**	**TSS**	**Phosphorus**	**PCB**
Date	**mg/L**	**µg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0050	3.7	<0.0002	0.0042	0.00047	0.0169	0.13	0.0031	0.0064	0.0059	0.0016	0.0006	0.179	22.4	243	248	6.3	<0.0001
10/13/2022	<0.0050	5.1	<0.0002	0.004	0.00031	0.0161	0.114	0.0033	0.0096	0.0056	0.0011	<0.00050	0.154	22.1	218	232	5.2	<0.000096
10/14/2022	<0.0050	5.6	<0.0002	0.0056	0.00054	0.0254	0.208	0.0063	0.0101	0.0076	0.0022	0.0005	0.265	23.9	268	332	8.8	<0.000099
10/21/2022	<0.0050	5.3	0.0014	0.006	0.0006	0.0282	0.199	0.0053	0.0096	0.0084	0.0011	0.0006	0.285	22.8	175	428	13.2	<0.00011
10/22/2022	<0.0050	6.3	<0.0002	0.0033	<0.0002	0.0098	0.0622	0.0018	0.0046	0.0045	<0.001	<0.0005	0.0872	24.1	180	70	4.6	<0.000049
10/25/2022	<0.0050	4.4	<0.0002	0.0063	0.00051	0.0223	0.178	0.0047	0.0072	0.0076	0.0013	0.0005	0.247	26.4	253	380	11.6	<0.000099
Average	0.0025	0.005	0.00032	0.005	0.0004	0.0198	0.149	0.0041	0.0079	0.0066	0.0013	0.0005	0.203	24	223	282	8	0.00004
Prim Eff % ren	0.0%	-63%	-217%	-112%	-101%	-256%	-125%	-67%	-24%	-100%	-44%	-87%	-96%	-6%	2%	-113%	-105%	19%

Organic Pollutants							
Chloroform	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
mg/L	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
<0.0048	<0.005	<0.005	<0.005	0.118	<0.010	0.136	<0.00051
<0.0048	<0.005	<0.005	<0.005	0.103	<0.010	0.184	<0.000048
<0.0048	<0.005	<0.005	<0.005	0.127	<0.010	<0.0667	<0.000050
<0.0048	<0.005	<0.005	<0.005	0.0346	<0.010	0.178	<0.000053
<0.0048	<0.005	<0.005	<0.005	0.0218	<0.010	<0.100	<0.000005
<0.0048	<0.005	<0.005	<0.005	0.0199	<0.010	0.365	<0.000005
0.0024	0.0025	0.0025	0.0025	0.071	0.005	0.1577	0.00003
0%	0%	0%	0%	-2729%	0%	-189%	0%

WWTP Effluent

	Grab Sample Parameters		Metals (Total)											Compatible Pollutants				Organic Pollutants								
Collection	**Cyanide, Total**	**Cyanide, Available**	**Mercury**	**Arsenic**	**Cadmium**	**Chromium**	**Copper**	**Lead**	**Molybdenum**	**Nickel**	**Selenium**	**Silver**	**Zinc**	**NH_3-N**	**BOD_5**	**TSS**	**Phosphorus**	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**µg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0050	<0.002	<0.0002	<0.001	<0.0002	<0.002	0.0064	<0.001	0.0042	0.0036	0.001	<0.0005	0.0132	<0.10	7.5	<8.3	0.22	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00051
10/13/2022	0.091	<0.002	<0.0002	<0.001	<0.0002	<0.002	0.006	<0.001	0.0055	0.0039	<0.001	<0.0005	0.0159	<0.10	9.5	12.5	0.59	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000048
10/14/2022	<0.0050	0.0027	<0.0002	<0.001	<0.0002	<0.002	0.0067	<0.001	0.0064	0.0036	0.0011	<0.0005	0.0164	0.23	11.2	14	0.42	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.00005
10/21/2022	<0.0050	<0.002	<0.0002	<0.001	<0.0002	<0.002	0.0052	<0.001	0.0057	0.0033	<0.001	<0.0005	0.0138	<0.10	5.8	8.6	0.41	<0.000099	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00005
10/22/2022	<0.0050	0.0028	<0.0002	<0.001	<0.0002	<0.002	0.0051	<0.001	0.0063	0.0035	<0.001	<0.0005	0.015	<0.10	7.5	7	0.35	<0.000097	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.000049
10/25/2022	<0.0050	<0.002	<0.0002	<0.001	<0.0002	<0.002	0.0053	<0.001	0.0036	0.0035	<0.001	<0.0005	0.0154	<0.10	4.6	6.5	0.4	<0.000099	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.200	<0.0000050
Average	**0.01725**	**0.0016**	**0.0001**	**0.0005**	**0.0001**	**0.0010**	**0.0058**	**0.0005**	**0.0053**	**0.0036**	**0.0007**	**0.00025**	**0.015**	**0.08**	**8**	**9**	**0.40**	**0.000049**	**0.0024**	**0.0025**	**0.0025**	**0.0025**	**0.0025**	**0.005**	**0.0500**	**0.00006**

IU Sampling at Discharge Location

Akwel 5th St

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**hylbenzei**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/3/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.02	<0.00051
10/11/2022	<0.000097	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.000049
10/20/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000049
10/26/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.02	<0.000049

Akwel

Collection Date	Organic Pollutants								
	PCB mg/L	**Chloroform** mg/L	**1,2-Dichloro benzene** mg/L	**Ethylbenzene** mg/L	**Styrene** mg/L	**Toluene** mg/L	**Xylene** mg/L	**Phenolics** mg/L	**Lindane** mg/L
10/3/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00051
10/11/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	0.0051	<0.010	<0.100	<0.00048
10/20/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	0.0262	<0.010	<0.0667	<0.000049
10/26/2022	<0.000095	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000048

Borg Warner

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/3/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.0209	<0.00051
10/11/2022	<0.000097	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.00049
10/20/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000048
10/26/2022	<0.000095	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000048

AAR 6th St

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/4/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00052
10/12/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00048
10/20/2022	<0.000098	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	<0.100	<0.000049
10/24/2022	<0.00010	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000050

AAR Facility

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/4/2022	<0.0001	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	<0.0667	<0.00052
10/14/2022	<0.00012	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.067	<0.000062
10/17/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.0255	<0.000049
10/27/2022	<0.000096	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	<0.200	<0.000048

AAR Rinse Tank

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/4/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.00052
10/12/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.00048
10/17/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000050
10/28/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000049

CCI

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/4/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.0993	<0.00052
10/10/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.573	<0.000049
10/19/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.278	<0.000049
10/28/2022	<0.000098	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.233	<0.000049

CRE

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000052
10/13/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000050
10/21/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000050
10/24/2022	<0.00012	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000059

FIAMM

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.151	<0.00052
10/10/2022	<0.000099	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	1.57	<0.00050
10/21/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.108	<0.000049
10/27/2022	<0.000096	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.332	<0.000048

HUTCHINSON

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.377	<0.00052
10/10/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	1.7	<0.00050
10/21/2022	<0.000097	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.099	<0.000048
10/27/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.689	<0.000048

MI RUBBER

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.0703	<0.00052
10/10/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00050
10/21/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000050
10/27/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.200	<0.000048

MUNSON

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/6/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.457	<0.00051
10/14/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.071	<0.00005
10/18/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.234	<0.000049
10/24/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00005

SPENCER PLASTICS

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/6/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.679	<0.00051
10/14/2022	<0.00014	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.02	<0.000071
10/18/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.0362	<0.000048
10/24/2022	<0.00011	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000056

ARVCO

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/6/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.35	<0.00051
10/12/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.64	<0.00048
10/18/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000049
10/28/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000049

AVON PROTECTION

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.689	<0.000051
10/12/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.306	<0.000048
10/18/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.203	<0.000049
10/28/2022	<0.000098	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.474	<0.000049

PIRANHA HOSE

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/7/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000050
10/13/2022	<0.0001	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.715	<0.000050
10/19/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000048
10/25/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.00005

REC BOATS - TRAILER

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/7/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.329	<0.000050
10/13/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000050
10/19/2022	<0.000098	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.361	<0.000049
10/25/2022	<0.00011	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.27	<0.000053

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

May 07, 2024

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: IPP - MAHL
Pace Project No.: 50371395

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on April 24, 2024. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

CERTIFICATIONS

Project:	IPP - MAHL
Pace Project No.:	50371395

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268
Illinois Accreditation #: 200074
Indiana Drinking Water Laboratory #: C-49-06
Kansas/TNI Certification #: E-10177
Kentucky UST Agency Interest #: 80226
Kentucky WW Laboratory ID #: 98019
Michigan Drinking Water Laboratory #9050
Ohio VAP Certified Laboratory #: CL0065
Oklahoma Laboratory #: 9204
Texas Certification #: T104704355
Washington Dept of Ecology #: C1081
Wisconsin Laboratory #: 999788130
USDA Foreign Soil Permit #: 525-23-13-23119
USDA Compliance Agreement #: IN-SL-22-001

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
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SAMPLE SUMMARY

Project: IPP - MAHL
Pace Project No.: 50371395

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50371395001	**AKWEL 5th St.**	Water	04/23/24 13:45	04/24/24 11:20
50371395002	**CRE**	Water	04/23/24 12:55	04/24/24 11:20
50371395003	**BORG WARNER**	Water	04/23/24 13:25	04/24/24 11:20
50371395004	**AAR - RT**	Water	04/23/24 12:05	04/24/24 11:20
50371395005	**ARVCO**	Water	04/23/24 11:45	04/24/24 11:20

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Grand Rapids, MI 49512
(616)975-4500

SAMPLE ANALYTE COUNT

Project: IPP - MAHL
Pace Project No.: 50371395

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50371395001	**AKWEL 5th St.**	EPA 420.4	ZM	1	PASI-I
50371395002	**CRE**	EPA 608.3	KAV	2	PASI-I
		EPA 624.1	DAP	9	PASI-I
50371395003	**BORG WARNER**	EPA 420.4	ZM	1	PASI-I
50371395004	**AAR - RT**	EPA 624.1	DAP	9	PASI-I
		EPA 420.4	ZM	1	PASI-I
50371395005	**ARVCO**	EPA 608.3	KAV	2	PASI-I
		EPA 624.1	DAP	9	PASI-I
		EPA 420.4	ZM	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

Pace Analytical Services, LLC
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Grand Rapids, MI 49512
(616)975-4500

ANALYTICAL RESULTS

Project: IPP - MAHL
Pace Project No.: 50371395

Sample: AKWEL 5th St. **Lab ID: 50371395001** Collected: 04/23/24 13:45 Received: 04/24/24 11:20 Matrix: Water

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
420.4 Phenolics, Total	Analytical Method: EPA 420.4 Preparation Method: EPA 420.4 Pace Analytical Services - Indianapolis							
Phenolics, Total Recoverable	**405**	ug/L	200	1	05/06/24 11:30	05/07/24 11:53	64743-03-9	

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ANALYTICAL RESULTS

Project: IPP - MAHL
Pace Project No.: 50371395

Sample: CRE **Lab ID: 50371395002** Collected: 04/23/24 12:55 Received: 04/24/24 11:20 Matrix: Water

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
608.3 Pesticides	Analytical Method: EPA 608.3 Preparation Method: EPA 608.3							
	Pace Analytical Services - Indianapolis							
gamma-BHC (Lindane)	**<0.048**	ug/L	0.048	1	04/30/24 12:25	04/30/24 21:36	58-89-9	
Surrogates								
Decachlorobiphenyl (S)	70	%.	1-133	1	04/30/24 12:25	04/30/24 21:36	2051-24-3	
624.1 Volatile Organics	Analytical Method: EPA 624.1							
	Pace Analytical Services - Indianapolis							
Chloroform	**<4.8**	ug/L	4.8	1		04/26/24 18:17	67-66-3	
1,2-Dichlorobenzene	**<5.0**	ug/L	5.0	1		04/26/24 18:17	95-50-1	
Ethylbenzene	**<5.0**	ug/L	5.0	1		04/26/24 18:17	100-41-4	
Styrene	**<5.0**	ug/L	5.0	1		04/26/24 18:17	100-42-5	N2
Toluene	**<5.0**	ug/L	5.0	1		04/26/24 18:17	108-88-3	
Xylene (Total)	**<10.0**	ug/L	10.0	1		04/26/24 18:17	1330-20-7	
Surrogates								
Dibromofluoromethane (S)	95	%.	91-114	1		04/26/24 18:17	1868-53-7	
4-Bromofluorobenzene (S)	98	%.	85-120	1		04/26/24 18:17	460-00-4	
Toluene-d8 (S)	97	%.	85-117	1		04/26/24 18:17	2037-26-5	

Pace Analytical Services, LLC
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Grand Rapids, MI 49512
(616)975-4500

ANALYTICAL RESULTS

Project: IPP - MAHL
Pace Project No.: 50371395

Sample: BORG WARNER **Lab ID: 50371395003** Collected: 04/23/24 13:25 Received: 04/24/24 11:20 Matrix: Water

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
420.4 Phenolics, Total	Analytical Method: EPA 420.4 Preparation Method: EPA 420.4 Pace Analytical Services - Indianapolis							
Phenolics, Total Recoverable	**373**	ug/L	40.0	1	05/06/24 11:30	05/07/24 11:57	64743-03-9	

REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

ANALYTICAL RESULTS

Project: IPP - MAHL
Pace Project No.: 50371395

Sample: AAR - RT **Lab ID: 50371395004** Collected: 04/23/24 12:05 Received: 04/24/24 11:20 Matrix: Water

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
624.1 Volatile Organics	Analytical Method: EPA 624.1 Pace Analytical Services - Indianapolis							
Chloroform	**<4.8**	ug/L	4.8	1		04/26/24 19:13	67-66-3	
1,2-Dichlorobenzene	**<5.0**	ug/L	5.0	1		04/26/24 19:13	95-50-1	
Ethylbenzene	**<5.0**	ug/L	5.0	1		04/26/24 19:13	100-41-4	
Styrene	**<5.0**	ug/L	5.0	1		04/26/24 19:13	100-42-5	N2
Toluene	**<5.0**	ug/L	5.0	1		04/26/24 19:13	108-88-3	
Xylene (Total)	**<10.0**	ug/L	10.0	1		04/26/24 19:13	1330-20-7	
Surrogates								
Dibromofluoromethane (S)	94	%.	91-114	1		04/26/24 19:13	1868-53-7	
4-Bromofluorobenzene (S)	98	%.	85-120	1		04/26/24 19:13	460-00-4	
Toluene-d8 (S)	96	%.	85-117	1		04/26/24 19:13	2037-26-5	
420.4 Phenolics, Total	Analytical Method: EPA 420.4 Preparation Method: EPA 420.4 Pace Analytical Services - Indianapolis							
Phenolics, Total Recoverable	**<20.0**	ug/L	20.0	1	05/06/24 11:30	05/07/24 11:58	64743-03-9	

Pace Analytical Services, LLC
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Grand Rapids, MI 49512
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ANALYTICAL RESULTS

Project: IPP - MAHL
Pace Project No.: 50371395

Sample: ARVCO **Lab ID: 50371395005** Collected: 04/23/24 11:45 Received: 04/24/24 11:20 Matrix: Water

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
608.3 Pesticides	Analytical Method: EPA 608.3 Preparation Method: EPA 608.3							
	Pace Analytical Services - Indianapolis							
gamma-BHC (Lindane)	**<0.99**	ug/L	0.99	20	04/30/24 12:25	04/30/24 22:40	58-89-9	ED
Surrogates								
Decachlorobiphenyl (S)	0	%.	1-133	20	04/30/24 12:25	04/30/24 22:40	2051-24-3	S4
624.1 Volatile Organics	Analytical Method: EPA 624.1							
	Pace Analytical Services - Indianapolis							
Chloroform	**<240**	ug/L	240	50		04/29/24 12:32	67-66-3	
1,2-Dichlorobenzene	**<250**	ug/L	250	50		04/29/24 12:32	95-50-1	
Ethylbenzene	**<250**	ug/L	250	50		04/29/24 12:32	100-41-4	
Styrene	**<250**	ug/L	250	50		04/29/24 12:32	100-42-5	N2
Toluene	**<250**	ug/L	250	50		04/29/24 12:32	108-88-3	
Xylene (Total)	**<500**	ug/L	500	50		04/29/24 12:32	1330-20-7	
Surrogates								
Dibromofluoromethane (S)	96	%.	91-114	50		04/29/24 12:32	1868-53-7	D3,F1
4-Bromofluorobenzene (S)	98	%.	85-120	50		04/29/24 12:32	460-00-4	
Toluene-d8 (S)	98	%.	85-117	50		04/29/24 12:32	2037-26-5	
420.4 Phenolics, Total	Analytical Method: EPA 420.4 Preparation Method: EPA 420.4							
	Pace Analytical Services - Indianapolis							
Phenolics, Total Recoverable	**<2000**	ug/L	2000	1	05/06/24 11:30	05/07/24 11:59	64743-03-9	D3

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
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QUALITY CONTROL DATA

Project: IPP - MAHL
Pace Project No.: 50371395

QC Batch: 786885 | Analysis Method: EPA 624.1
QC Batch Method: EPA 624.1 | Analysis Description: 624.1 MSV
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50371395002, 50371395004

METHOD BLANK: 3599694 Matrix: Water

Associated Lab Samples: 50371395002, 50371395004

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,2-Dichlorobenzene	ug/L	<5.0	5.0	04/26/24 11:17	
Chloroform	ug/L	<4.8	4.8	04/26/24 11:17	
Ethylbenzene	ug/L	<5.0	5.0	04/26/24 11:17	
Styrene	ug/L	<5.0	5.0	04/26/24 11:17	N2
Toluene	ug/L	<5.0	5.0	04/26/24 11:17	
Xylene (Total)	ug/L	<10.0	10.0	04/26/24 11:17	
4-Bromofluorobenzene (S)	%.	96	85-120	04/26/24 11:17	
Dibromofluoromethane (S)	%.	98	91-114	04/26/24 11:17	
Toluene-d8 (S)	%.	97	85-117	04/26/24 11:17	

LABORATORY CONTROL SAMPLE: 3599695

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,2-Dichlorobenzene	ug/L	20	21.3	106	65-135	
Chloroform	ug/L	20	19.8	99	70-135	
Ethylbenzene	ug/L	20	21.4	107	60-140	
Styrene	ug/L	20	22.4	112	77-124	N2
Toluene	ug/L	20	20.2	101	70-130	
Xylene (Total)	ug/L	60	62.3	104	76-119	
4-Bromofluorobenzene (S)	%.			98	85-120	
Dibromofluoromethane (S)	%.			99	91-114	
Toluene-d8 (S)	%.			97	85-117	

MATRIX SPIKE SAMPLE: 3599697

Parameter	Units	50371395004 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,2-Dichlorobenzene	ug/L	<5.0	20	22.4	112	18-190	
Chloroform	ug/L	<4.8	20	22.7	109	51-138	
Ethylbenzene	ug/L	<5.0	20	23.7	118	37-162	
Styrene	ug/L	<5.0	20	24.0	120	59-169	N2
Toluene	ug/L	<5.0	20	25.2	114	47-150	
Xylene (Total)	ug/L	<10.0	60	68.9	115	53-165	
4-Bromofluorobenzene (S)	%.				99	85-120	
Dibromofluoromethane (S)	%.				95	91-114	
Toluene-d8 (S)	%.				98	85-117	

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: IPP - MAHL
Pace Project No.: 50371395

SAMPLE DUPLICATE: 3599696

Parameter	Units	50371395002 Result	Dup Result	RPD	Max RPD	Qualifiers
1,2-Dichlorobenzene	ug/L	<5.0	<5.0		57	
Chloroform	ug/L	<4.8	<4.8		54	
Ethylbenzene	ug/L	<5.0	<5.0		63	
Styrene	ug/L	<5.0	<5.0		40	N2
Toluene	ug/L	<5.0	<5.0		41	
Xylene (Total)	ug/L	<10.0	<10.0		40	
4-Bromofluorobenzene (S)	%.	98	96			
Dibromofluoromethane (S)	%.	95	96			
Toluene-d8 (S)	%.	97	96			

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: IPP - MAHL
Pace Project No.: 50371395

QC Batch: 787161
QC Batch Method: EPA 624.1
Analysis Method: EPA 624.1
Analysis Description: 624.1 MSV
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50371395005

METHOD BLANK: 3601241 Matrix: Water
Associated Lab Samples: 50371395005

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,2-Dichlorobenzene	ug/L	<5.0	5.0	04/29/24 11:37	
Chloroform	ug/L	<4.8	4.8	04/29/24 11:37	
Ethylbenzene	ug/L	<5.0	5.0	04/29/24 11:37	
Styrene	ug/L	<5.0	5.0	04/29/24 11:37	N2
Toluene	ug/L	<5.0	5.0	04/29/24 11:37	
Xylene (Total)	ug/L	<10.0	10.0	04/29/24 11:37	
4-Bromofluorobenzene (S)	%.	96	85-120	04/29/24 11:37	
Dibromofluoromethane (S)	%.	97	91-114	04/29/24 11:37	1d
Toluene-d8 (S)	%.	96	85-117	04/29/24 11:37	

LABORATORY CONTROL SAMPLE: 3601242

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,2-Dichlorobenzene	ug/L	20	20.9	104	65-135	
Chloroform	ug/L	20	19.4	97	70-135	
Ethylbenzene	ug/L	20	21.3	107	60-140	
Styrene	ug/L	20	22.2	111	77-124	N2
Toluene	ug/L	20	21.0	105	70-130	
Xylene (Total)	ug/L	60	62.7	105	76-119	
4-Bromofluorobenzene (S)	%.			99	85-120	
Dibromofluoromethane (S)	%.			98	91-114	
Toluene-d8 (S)	%.			100	85-117	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: IPP - MAHL
Pace Project No.: 50371395

QC Batch: 787342
QC Batch Method: EPA 608.3
Analysis Method: EPA 608.3
Analysis Description: 608.3 Pesticides
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50371395002, 50371395005

METHOD BLANK: 3601749 Matrix: Water
Associated Lab Samples: 50371395002, 50371395005

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
gamma-BHC (Lindane)	ug/L	<0.050	0.050	04/30/24 19:32	
Decachlorobiphenyl (S)	%.	53	1-133	04/30/24 19:32	

LABORATORY CONTROL SAMPLE: 3601750

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
gamma-BHC (Lindane)	ug/L	0.1	0.098	98	32-140	
Decachlorobiphenyl (S)	%.			69	1-133	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3601751 3601752

Parameter	Units	50371395002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
gamma-BHC (Lindane)	ug/L	<0.048	0.19	0.19	0.19	0.19	98	102	32-140	4	35	
Decachlorobiphenyl (S)	%.						69	76	1-133			

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: IPP - MAHL
Pace Project No.: 50371395

QC Batch: 788241
QC Batch Method: EPA 420.4
Analysis Method: EPA 420.4
Analysis Description: 420.4 Phenolics
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50371395001, 50371395003, 50371395004, 50371395005

METHOD BLANK: 3606145 Matrix: Water
Associated Lab Samples: 50371395001, 50371395003, 50371395004, 50371395005

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Phenolics, Total Recoverable	ug/L	<20.0	20.0	05/07/24 11:28	

LABORATORY CONTROL SAMPLE: 3606146

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Phenolics, Total Recoverable	ug/L	50	52.1	104	90-110	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3606147 3606148

Parameter	Units	50371290005 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Phenolics, Total Recoverable	ug/L	0.0042J mg/L	50	50	53.9	53.5	99	99	90-110	1	20	

MATRIX SPIKE SAMPLE: 3606149

Parameter	Units	50371304001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Phenolics, Total Recoverable	ug/L	<0.010 mg/L	50	49.7	99	90-110	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALIFIERS

Project: IPP - MAHL
Pace Project No.: 50371395

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

1d	Neither matrix spike nor matrix precision data could be provided for this analytical batch due to insufficient sample volume.
D3	Sample was diluted due to the presence of high levels of non-target analytes or other matrix interference.
ED	Due to the extract's physical characteristics, the analysis was performed at dilution.
F1	The sample was analyzed at a dilution due to foaming of the sample in the purge vessel.
N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
S4	Surrogate recovery not evaluated against control limits due to sample dilution.

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: IPP - MAHL
Pace Project No.: 50371395

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50371395002	**CRE**	EPA 608.3	787342	EPA 608.3	787464
50371395005	**ARVCO**	EPA 608.3	787342	EPA 608.3	787464
50371395002	**CRE**	EPA 624.1	786885		
50371395004	**AAR - RT**	EPA 624.1	786885		
50371395005	**ARVCO**	EPA 624.1	787161		
50371395001	**AKWEL 5th St.**	EPA 420.4	788241	EPA 420.4	788475
50371395003	**BORG WARNER**	EPA 420.4	788241	EPA 420.4	788475
50371395004	**AAR - RT**	EPA 420.4	788241	EPA 420.4	788475
50371395005	**ARVCO**	EPA 420.4	788241	EPA 420.4	788475

Pace® Location Requested (City/State):
Pace Analytical Grand Rapids
4171 40th Street SE, Grand Rapids, MI 49512

CHAIN-OF-CUSTODY Analytical Request Document

Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevant fields

LAB USE ONLY- Affix Workorder/Login Label Here
WO#: 50371395
50371395

Company Name: City of Cadillac
Street Address: 1121 Plett Rd, Cadillac, MI 49601
Customer Project #:
Project Name: IPP - MAHL
Site Collection Info/Facility ID (as applicable):

Contact/Report To: Cindy Tomaszewski
Phone #: 231-775-2368
E-Mail: ctomaszewski@cadillac-mi.net
Cc E-Mail:
Invoice To: Cindy Tomaszewski
Invoice E-Mail: ctomaszewski@cadillac-mi.net
Purchase Order # (if applicable):
Quote #:
County / State origin of sample(s): Michigan

Time Zone Collected: [ ] AK [ ] PT [ ] MT [ ] CT [ ] ET
Data Deliverables: [ ] Level II [ ] Level III [ ] Level IV [ ] EQUIS [ ] Other
Regulatory Program (DW, RCRA, etc.) as applicable: Reportable [ ] Yes [ ] No
Rush (Pre-approval required): [ ] Same Day [ ] 1 Day [ ] 2 Day [ ] 3 Day [ ] Other
DW PWSID # or WW Permit # as applicable:
Date Results Requested: STANDARD TURN.
Field Filtered (if applicable): [ ] Yes [X] No
Analysis:

* Matrix Codes (Insert in Matrix box below): Drinking Water (DW), Ground Water (GW), Waste Water (WW), Product (P), Soil/Solid (SS), Oil (OL), Wipe (WP), Tissue (TS), Bioassay (B), Vapor (V), Surface Water (SW),Sediment (SED), Sludge (SL), Caulk (CK), Leachate (LL), Biosolid (BS), Other (OT)

Specify Container Size **: 3 | 1 | 6
Identify Container Preservative Type***: 3 | 1 | 1

**Container Size: (1) 1L, (2) 500mL, (3) 250mL, (4) 125mL, (5) 100mL, (6) 40mL vial, (7) EnCore, (8) TerraCore, (9) 90mL, (10) Other
*** Preservative Types: (1) None, (2) HNO3, (3) H2SO4, (4) HCl, (5) NaOH, (6) Zn Acetate, (7) NaHSO4, (8) Sod. Thiosulfate, (9) Ascorbic Acid, (10) MeOH, (11) Other

Lab Use Only
Proj. Mgr: **Brian Hall**
AcctNum / Client ID:
Table #:
Profile / Template: **10789**
Prelog / Bottle Ord. ID: **EZ 3094262**

Analysis Requested

Customer Sample ID	Matrix *	Comp / Grab	Composite Start Date	Composite Start Time	Collected or Composite End Date	Collected or Composite End Time	# Cont.	Res. Chlorine Results	Res. Chlorine Units	420.4 Phenolics, Total	608.3 Pesticides - Lindane	624.1 Volatile Organics	Sample Comment
AKWEL 5th ST.	WW	G	—		4-23	1:45	1			X			
CRE	WW	G	—		4-23	1255	6				X	X	
BORG WARNER	WW	G	—		4-23	1:25	1			X			
AAR - RT	WW	G	—		4-23	1205	4			X		X	
ARVCO	WW	G	—		4-23	1145	7			X	X	X	

Preservation non-conformance identified for sample.

Additional Instructions from Pace®:
VOCs - Chloroform, 1,2-DCB, Ethylbenzene, Styrene, Toluene, Xylenes

Collected By: (Printed Name) Derek & Patrick
Signature: Derek v Scodi

Customer Remarks / Special Conditions / Possible Hazards:

# Coolers:	Thermometer ID:	Correction Factor (°C):	Obs. Temp. (°C)	Corrected Temp. (°C)	On Ice:

Relinquished by/Company: (Signature)	Date/Time:	Received by/Company: (Signature)	Date/Time:
City of Cadillac	4-23-24	D Sih / Pace	4/24/24 1120

Tracking Number:
Delivered by: [ ] In- Person [ ] Courier [ ] FedEX [ ] UPS [ ] Other

Page: 1 of 1

Submitting a sample via this chain of custody constitutes acknowledgment and acceptance of the Pace® Terms and Conditions found at https://www.pacelabs.com/resource-library/resource/pace-terms-and-conditions/

ENV-FRM-CORQ-0019_v02_110123 ©

Pace

Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 4/24 1430	**Evaluated By:** DAS	**WO#: 50371395** PM: BJH Due Date: 05/08/24 CLIENT: GR-Cadillac	
Client: City of Cadillac	**PM:** BJH		
Lab Notified of Rush or Short Holds: YES NO ~~NA~~			

Project Received Via: FedEx ~~UPS~~ Client Pace Courier Other: ____________

				Comments:
Custody Seal Present and Intact:	YES	NO	~~NA~~	
Received Sample Information Form (SIF): Drinking Waters Only	YES	NO	~~NA~~	
Short Hold Present (≤ 48 Hours):	YES	NO		
Sample Received in Hold:	YES	NO		
Custody Signature Present:	YES	NO		
Collector Signature Present:	YES	NO		
Sample Collected Today and On Ice:	YES	NO	~~N/A~~	
IR Gun #: ~~350~~ 351 **Therm #:** 282 283	**Temp. should be 0°C - 6°C (Initial/Corrected)**			
Ice Type: WET Bagged / ~~WET Loose~~ BLUE NONE	1. Cooler Temp. Upon Receipt: 2.0/1.6 °C			
Ice Location: TOP ~~BOTTOM~~ MIDDLE DISPERSED	2. Cooler Temp. Upon Receipt: ________ °C			
Temp Blank Received:	~~YES~~	NO		
Sample Label Matches COC (ID/Date/Time):	~~YES~~	NO		
Container Intact:	~~YES~~	NO		
Correct Container:	~~YES~~	NO		
Sufficient Volume:	~~YES~~	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with red dot on container lid if unacceptable. **pH Strip Lot #:** HC321337 *Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl*	~~YES~~	NO	N/A	
Residual Chlorine Absent: **Cl_2 Strip Lot #:** HC330789 *Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide*	YES	~~NO~~	N/A	*See Notes/Comments
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.	YES	NO	~~N/A~~	
Trip Blank Received: HCl MeOH Other: ________	YES	~~NO~~	ON HOLD	

Comments:
ARVCO green/sludge like consistency, too thick to visually check for Res. Cl.

3. Cooler Temp. Upon Receipt: ________ °C

4. Cooler Temp. Upon Receipt: ________ °C

Non-Conformance Form Required: ~~YES~~ NO

Pace®

Sample Receiving Non-Conformance Form (NCF)

WO#: 50371395
PM: BJH Due Date: 05/08/24
CLIENT: GR-Cadillac

Affix Workorder/Login Label Here or List Pace Workorder Number Here		**COC Integrity Issues:** Check issues below and add details where appropriate		Che… ate		
Date: 4/24/24		COC does not match samples received (missing, additional, etc.)		Custody seal(s) damaged or missing on coolers, samples, or trip blanks		*Insufficient sample volume received
Evaluated by: DAS		COC sample ID does not match sample label		Cooler or sample container broken or compromised	✓	*Sample contains residual chlorine
Client: City of Cadillac		*COC collection date/time missing or does not match sample label		*Sample past holding time		Improper preservation
***Drinking Water Deficiency:** Samples may be invalid. Analysis must not proceed without client written permission.		*Analyses/ analytes missing or clarification needed		*Temperature not within acceptance criteria (typically 0-6°C)	✓	*Sample contains interferences (multi-phasic, solids, color, odor, etc…)
		*Required signatures are missing		*Sample arrived frozen or partially frozen		Vial(s) received with improper headspace (>6mm)
*No Sample Information Form (SIF) received with sample(s)	✓	*Residual Chlorine presence/ absence not indicated on COC		*Incorrect or improper containers received		Other: See notes below

COC					Sample Label					Sample Notes
Sample ID	Date	Time	Container Type	Quantity	Sample ID	Date	Time	Container Type	Quantity	
CRE	4/23	1255	AG1U	3						2 of the 3 608.3
										contained Res. Cl. Unsure
										why the third did not. All marked.
ARVCO	4/23	1145	AG1U	3						Sample Viscosity too
										thick for strips to test
										for Res. Cl.

General Comments/ Client Instructions:

Green Sludge like substance

Cadillac Castings Inc
Categorical Limit Calculations
Table A18

Regulated under 40 CFR Part 464.36

Processes include:

Dust Collection Scrubbing Operations (c), primarily ductile iron castings

Categorical Limits:

Pollutant	Max Day	Monthly Avg
	lbs/ billion SCF of air scrubbed	
Copper	0.218	0.12
Lead	0.398	0.195
Zinc	0.736	0.278
Total Phenols	0.646	0.225
TTO	2.04	0.664
FOG	22.5	7.51

Dust Scrubber Flow	52,000	scfm, each
Number of scrubbers	2	
Total flow, SCF/day	149,760,000	
Total flow, billion SCF/day	0.150	

Categorical Limits for CCI, based on scrubber flow:

Pollutant	**Max Day**	**Monthly Avg**
	lbs/ day	
Copper	**0.03**	**0.02**
Lead	**0.06**	**0.03**
Zinc	**0.11**	**0.04**
Total Phenols	**0.10**	**0.03**
TTO	**0.31**	**0.10**
FOG	**3.37**	**1.12**

Melting Furnace Scrubbing Operations (f), primarily ductile iron

Categorical Limits:

Pollutant	Max Day	Monthly Avg
	lbs/ billion SCF of air scrubbed	
Copper	1.02	0.561
Lead	1.86	0.911
Zinc	3.44	1.3
Total Phenols	3.01	1.05
TTO	8.34	2.73
FOG	105	35

Melt Furnace Scrubber Flow	30,000
Number of scrubbers	1
Total flow, SCF/day	43,200,000
Total flow, billion SCF/day	0.043

Categorical Limits for CCI, based on scrubber flow:

Pollutant	**Max Day**	**Monthly Avg**
	lbs/ day	
Copper	**0.15**	**0.08**
Lead	**0.28**	**0.14**
Zinc	**0.52**	**0.19**
Total Phenols	**0.45**	**0.16**
TTO	**1.25**	**0.41**
FOG	**15.72**	**5.24**

APPENDIX B

PERMIT NO. MI0020257

STATE OF MICHIGAN
DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

AUTHORIZATION TO DISCHARGE UNDER THE NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM

In compliance with the provisions of the federal Clean Water Act (federal Water Pollution Control Act, 33 U.S.C., Section 1251 *et seq*., as amended); Part 31, Water Resources Protection, of the Natural Resources and Environmental Protection Act, 1994 PA 451, as amended (NREPA); Part 41, Sewerage Systems, of the NREPA; and Michigan Executive Order 2019-06,

City of Cadillac
200 North Lake Street
Cadillac, MI 49601

is authorized to discharge from the **Cadillac Wastewater Treatment Plant** located at

1121 Plett Road
Cadillac, MI 49601

designated as **Cadillac WWTP**

to the receiving water named the Clam River in accordance with effluent limitations, monitoring requirements, and other conditions set forth in this permit.

This permit is based on a complete application submitted on September 21, 2022.

This permit takes effect on January 1, 2024. The provisions of this permit are severable. After notice and opportunity for a hearing, this permit may be modified, suspended, or revoked in whole or in part during its term in accordance with applicable laws and rules. On its effective date, this permit shall supersede National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257 (expiring October 1, 2022).

This permit and the authorization to discharge shall expire at midnight on **October 1, 2027**. In order to receive authorization to discharge beyond the date of expiration, the permittee shall submit an application that contains such information, forms, and fees as are required by the Michigan Department of Environment, Great Lakes, and Energy (Department) by **April 4, 2027**.

Issued: November 29, 2023.

Original signed by Christine Alexander
Christine Alexander, Manager
Permits Section
Water Resources Division

PERMIT FEE REQUIREMENTS

In accordance with Section 324.3120 of the NREPA, the permittee shall make payment of an annual permit fee to the Department for each October 1 the permit is in effect regardless of occurrence of discharge. The permittee shall submit the fee in response to the Department's annual notice. Payment may be made electronically via the Department's MiEnviro Portal system. The MiEnviro Portal website is located at https://mienviro.michigan.gov/ncore/. Payment shall be submitted or postmarked by January 15 for notices mailed by December 1. Payment shall be submitted or postmarked no later than 45 days after receiving the notice for notices mailed after December 1.

Annual Permit Fee Classification: Municipal Major, less than 10 MGD (Individual Permit)

In accordance with Section 324.3132 of the NREPA, the permittee shall make payment of an annual biosolids land application fee to the Department if the permittee land applies biosolids. The permittee shall submit the fee in response to the Department's annual notice. Payment may be made electronically via the Department's MiEnviro Portal system. The MiEnviro Portal website is located at https://mienviro.michigan.gov/ncore/. Payment shall be submitted or postmarked no later than January 31 of each year for notices mailed by December 15. Payment shall be submitted or postmarked no later than 45 days after receiving the notice for notices mailed after December 15.

CONTACT INFORMATION

Unless specified otherwise, all contact with the Department required by this permit shall be made to the Cadillac District Office of the Water Resources Division. The Cadillac District Office is located at 120 West Chapin Street, Cadillac, MI 49601-2158, Telephone: 231-775-3960, Fax: 231-775-1511.

CONTESTED CASE INFORMATION

Any person who is aggrieved by this permit may file a sworn petition with the Michigan Administrative Hearing System within the Michigan Department of Licensing and Regulatory Affairs, c/o the Michigan Department of Environment, Great Lakes, and Energy, setting forth the conditions of the permit which are being challenged and specifying the grounds for the challenge. The Department of Licensing and Regulatory Affairs may reject any petition filed more than 60 days after issuance as being untimely.

PART I

Section A. Limitations and Monitoring Requirements

1. Final Effluent Limitations, Monitoring Point 001A

During the period beginning on the effective date of this permit and lasting until the expiration date of this permit, the permittee is authorized to discharge treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River at Latitude 44.2657, Longitude -85.3984. Such discharge shall be limited and monitored by the permittee as specified below.

	Maximum Limits for Quantity or Loading				Maximum Limits for Quality or Concentration					
Parameter	**Monthly**	**7-Day**	**Daily**	**Units**	**Monthly**	**7-Day**	**Daily**	**Units**	**Monitoring Frequency**	**Sample Type**
Flow	(report)	---	(report)	MGD	---	---	---	---	Daily	Report Total Daily Flow
Carbonaceous Biochemical Oxygen Demand (CBOD5)										
June – September	190	270	(report)	lbs/day	7	---	10	mg/l	5x Weekly	24-Hr Composite
October – November	290	450	(report)	lbs/day	11	---	17	mg/l	5x Weekly	24-Hr Composite
December – March	610	910	(report)	lbs/day	23	---	34	mg/l	5x Weekly	24-Hr Composite
April – May	450	670	(report)	lbs/day	17	---	25	mg/l	5x Weekly	24-Hr Composite
Total Suspended Solids (TSS)										
June – September	530	800	(report)	lbs/day	20	30	(report)	mg/l	5x Weekly	24-Hr Composite
October – May	800	1,200	(report)	lbs/day	30	45	(report)	mg/l	5x Weekly	24-Hr Composite
Ammonia Nitrogen (as N)										
Through September 30, 2026										
June – September	24	53	(report)	lbs/day	0.9	---	2.0	mg/l	5x Weekly	24-Hr Composite
October – November	43	75	(report)	lbs/day	1.6	---	2.8	mg/l	5x Weekly	24-Hr Composite
December – March	99	---	(report)	lbs/day	3.7	---	(report)	mg/l	5x Weekly	24-Hr Composite
April – May	130	---	(report)	lbs/day	4.8	---	(report)	mg/l	5x Weekly	24-Hr Composite
Beginning October 1, 2026										
June – September	24	53	(report)	lbs/day	0.9	---	2.0	mg/l	5x Weekly	24-Hr Composite
October – November	43	75	(report)	lbs/day	1.6	---	2.8	mg/l	5x Weekly	24-Hr Composite
December – March	99	---	(report)	lbs/day	3.7	---	(report)	mg/l	5x Weekly	24-Hr Composite
April – May	80	---	(report)	lbs/day	3.0	---	(report)	mg/l	5x Weekly	24-Hr Composite

PART I

Section A. Limitations and Monitoring Requirements

Parameter	Maximum Limits for Quantity or Loading				Maximum Limits for Quality or Concentration				Monitoring Frequency	Sample Type
	Monthly	7-Day	Daily	Units	Monthly	7-Day	Daily	Units		
Total Phosphorus (as P)										
May – March	13	---	(report)	lbs/day	0.5	---	(report)	mg/l	5x Weekly	24-Hr Composite
April	17	---	(report)	lbs/day	0.65	---	(report)	mg/l	5x Weekly	24-Hr Composite
Total Copper	---	---	---	---	---	---	(report)	ug/l	2x Monthly	24-Hr Composite
Total Arsenic	0.43	---	(report)	lbs/day	16	---	(report)	ug/l	2x Monthly	24-Hr Composite
Total Beryllium	0.11	---	(report)	lbs/day	4.1	---	(report)	ug/l	2x Monthly	24-Hr Composite
Total Cadmium	0.11	0.32	(report)	lbs/day	4.1	---	12	ug/l	2x Monthly	24-Hr Composite
Total Silver	0.037	---	(report)	lbs/day	1.4	---	(report)	ug/l	2x Monthly	24-Hr Composite
Chloride	---	---	---	---	---	---	(report)	mg/l	Monthly	24-Hr Composite
Sulfate	---	---	---	---	---	---	(report)	mg/l	Monthly	24-Hr Composite
Acute Toxicity – Fathead Minnow	---	---	---	---	---	---	(report)	TU_A	Quarterly	24-Hr Composite
Acute Toxicity – *C. dubia*	---	---	---	---	---	---	(report)	TU_A	Quarterly	24-Hr Composite
							Individual Chronic Value			
Chronic Toxicity – Fathead Minnow	---	---	---	---	1.1	---	(report)	TU_C	Quarterly	24-Hr Composite
Chronic Toxicity – *C. dubia*	---	---	---	---	1.1	---	(report)	TU_C	Quarterly	24-Hr Composite
							Maximum Daily			
Fecal Coliform Bacteria	---	---	---	---	200	400	(report)	cts/ 100 ml	5x Weekly	Grab
Perfluorooctanesulfonic Acid (PFOS)	---	---	---	---	---	---	(report)	ng/l	3x Annual	Grab
Available Cyanide	0.16	---	(report)	lbs/day	5.9	---	(report)	ug/l	Monthly	Grab
Total Benzidine	0.0032	---	(report)	lbs/day	0.12	---	(report)	ug/l	2x Monthly	Grab
Total Bromomethane	---	---	---	---	---	---	(report)	ug/l	Quarterly	Grab

PART I

Section A. Limitations and Monitoring Requirements

Parameter	Maximum Limits for Quantity or Loading				Maximum Limits for Quality or Concentration				Monitoring Frequency	Sample Type
	Monthly	7-Day	Daily	Units	Monthly	7-Day	Daily	Units		
Total Mercury										
Corrected	(report)	---	(report)	lbs/day	(report)	---	(report)	ng/l	Quarterly	Calculation
Uncorrected	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab
Field Duplicate	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab
Field Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation
Laboratory Method Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation
	12-Month Rolling Avg				**12-Month Rolling Avg**					
Total Mercury	0.000053	---	---	lbs/day	2.0	---	---	ng/l	Quarterly	Calculation
					Minimum % Monthly		**Minimum % Daily**			
TSS Minimum % Removal										
October – May	---	---	---	---	85	---	(report)	%	Monthly	Calculation
					Minimum Daily		**Maximum Daily**			
pH	---	---	---	---	6.5	---	9.0	S.U.	5x Weekly	Grab
Dissolved Oxygen										
May – November	---	---	---	---	6.0	---	---	mg/l	5x Weekly	Grab
December – April	---	---	---	---	5.0	---	---	mg/l	5x Weekly	Grab

The following design flow was used in determining the above limitations, but is not to be considered a limitation or actual capacity: 3.2 MGD.

a. Narrative Standard
The receiving water shall contain no turbidity, color, oil films, floating solids, foams, settleable solids, or deposits as a result of this discharge in unnatural quantities which are or may become injurious to any designated use.

b. Sampling Locations
Samples for CBOD5, TSS, Ammonia Nitrogen, Total Phosphorus, Total Copper, Total Arsenic, Total Beryllium, Total Cadmium, Total Silver, Chloride, Sulfate, Acute Toxicity, and Chronic Toxicity shall be taken prior to disinfection. Samples for Fecal Coliform Bacteria, PFOS, Available Cyanide, Total Benzidine, Total Bromomethane, Total Mercury, pH, and Dissolved Oxygen shall be taken after disinfection. The Department may approve alternate sampling locations that are demonstrated by the permittee to be representative of the effluent.

c. Quarterly Monitoring and 3x Annual Monitoring
Quarterly samples shall be taken during the months of January, April, July, and October. 3x Annual samples shall be taken during the months of January, May, and September. If the facility does not discharge during these months, the permittee shall sample the next discharge occurring during the period in question. If the facility does not discharge during the period in question, a sample is not required for that period. For any month in which a sample is not taken, the permittee shall enter "*G" on the Discharge Monitoring Report (DMR). (For purposes of reporting on the Daily tab of the DMR, the permittee shall enter “*G” on the first day of the month only).

Section A. Limitations and Monitoring Requirements

d. Ultraviolet Disinfection
It is understood that ultraviolet light will be used to achieve compliance with the fecal coliform limitations. If disinfection other than ultraviolet light will be used, the permittee shall notify the Department in accordance with Part II.C.12. of this permit.

e. Percent Removal Requirements
Monthly percent removal shall be calculated based on the monthly average effluent TSS concentrations and the monthly average influent concentrations for approximately the same period. Daily percent removal shall be calculated based on the daily effluent TSS concentrations and the daily influent concentrations for the same day. Reporting of Daily percent removal is only required on days on which an influent sample is obtained. The calculation shall be made as follows for each parameter: Percent removal = (influent concentration - effluent concentration) / influent concentration x 100.

f. Monitoring Frequency Reduction for Available Cyanide, Total Arsenic, Total Benzidine, Total Beryllium, Total Bromomethane, Total Cadmium, and/or Total Silver
After the submittal of 12 months of data, the permittee may request, in writing, Department approval for a reduction in monitoring frequency for Available Cyanide, Total Arsenic, Total Benzidine, Total Beryllium, Total Bromomethane, Total Cadmium, and/or Total Silver. This request shall contain an explanation as to why the reduced monitoring is appropriate. Upon receipt of written approval and consistent with such approval, the permittee may reduce the monitoring frequency indicated in Part I.A.1. of this permit. The monitoring frequency for Available Cyanide, Total Benzidine, and Total Bromomethane shall not be reduced to less than annually. The monitoring frequency for Total Arsenic, Total Beryllium, Total Cadmium, and Total Silver shall not be reduced to less than quarterly. The Department may revoke the approval for reduced monitoring at any time upon notification to the permittee.

g. Monitoring Frequency Reduction for Perfluorooctanesulfonic Acid (PFOS)
After the submittal of at least 10 equally spaced sample results obtained over a minimum of three (3) months, the permittee may request, in writing, Department approval of a reduction in monitoring frequency for PFOS. This request shall contain an explanation as to why the reduced monitoring is appropriate. Upon receipt of written approval and consistent with such approval, the permittee may reduce the monitoring frequency indicated in Part I.A.1. of this permit. The monitoring frequency for PFOS shall not be reduced to less than annually. The Department may revoke the approval for reduced monitoring at any time upon notification to the permittee.

h. Final Effluent Limitation for Total Mercury
The final limit for total mercury is the Discharge Specific Level Currently Achievable (LCA) based on a multiple discharger variance from the WQBEL of 1.3 ng/l, pursuant to Rule 1103(9) of the Water Quality Standards. Compliance with the LCA shall be determined as a 12-month rolling average, the calculation of which may be done using blank-corrected sample results. The 12-month rolling average shall be determined by adding the present monthly average result to the preceding 11 monthly average results then dividing the sum by 12. For facilities with quarterly monitoring requirements for total mercury, quarterly monitoring shall be equivalent to three (3) months of monitoring in calculating the 12-month rolling average. Facilities that monitor more frequently than monthly for total mercury must determine the monthly average result, which is the sum of the results of all data obtained in a given month divided by the total number of samples taken, in order to calculate the 12-month rolling average. If the 12-month rolling average for any quarter is less than or equal to the LCA, the permittee will be considered to be in compliance for total mercury for that quarter, provided the permittee is also in full compliance with the Pollutant Minimization Program for Total Mercury, set forth in Part I.A.4. of this permit.

PART I

Section A. Limitations and Monitoring Requirements

i. Total Mercury Testing and Additional Reporting Requirements

The analytical protocol for total mercury shall be in accordance with EPA Method 1631, Revision E, "Mercury in Water by Oxidation, Purge and Trap, and Cold Vapor Atomic Fluorescence Spectrometry," EPA-821-R-02-019, August 2002. The quantification level for total mercury shall be 0.5 ng/l, unless a higher level is appropriate because of sample matrix interference. Justification for higher quantification levels shall be submitted to the Department within 30 days of such determination.

The use of clean technique sampling procedures is required unless the permittee can demonstrate to the Department that an alternate sampling procedure is representative of the discharge. Guidance for clean technique sampling is contained in EPA Method 1669, “Sampling Ambient Water for Trace Metals at EPA Water Quality Criteria Levels (Sampling Guidance),” EPA-821-R96-001, July 1996. Information and data documenting the permittee's sampling and analytical protocols and data acceptability shall be submitted to the Department upon request.

In order to demonstrate compliance with EPA Method 1631E and EPA Method 1669, the permittee shall report, on the daily sheet, the analytical results of all field blanks and field duplicates collected in conjunction with each sampling event, as well as laboratory method blanks when used for blank correction. The permittee shall collect at least one (1) field blank and at least one (1) field duplicate per sampling event. If more than 10 samples are collected during a sampling event, the permittee shall collect at least one (1) additional field blank AND field duplicate for every 10 samples collected. A “sampling event” shall be defined herein as all sampling for total mercury conducted on the same day, provided the same sampling team collected all samples using the same sampling methods, procedures, and equipment on that day. Only field blanks or laboratory method blanks may be used to calculate a concentration lower than the actual sample analytical results (i.e., a blank correction). Only one (1) blank (field OR laboratory method) may be used for blank correction of a given sample result, and only if the blank meets the quality control acceptance criteria. If blank correction is not performed on a given sample analytical result, the permittee shall report under "Total Mercury – Corrected" the same value reported under "Total Mercury – Uncorrected." The field duplicate is for quality control purposes only; its analytical result shall not be averaged with the sample result.

PART I

Section A. Limitations and Monitoring Requirements

j. Whole Effluent Toxicity Final Requirements
Test species shall include fathead minnow **and** *Ceriodaphnia dubia*. Testing and reporting procedures shall follow procedures contained in EPA-821-R-02-013, "Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms" (Fourth Edition). When the effluent ammonia nitrogen (as N) concentration is greater than 3 mg/l, the pH of the toxicity test shall be maintained at a pH of 8 Standard Units. The acute toxic unit (TU_A) value and chronic toxic unit (TU_C) value for **each species tested** shall be reported on the DMR. If multiple chronic toxicity tests for the same species are performed during the month, the maximum TU_A value and monthly average TU_C value for the species shall be reported. For **each species not tested**, the permittee shall enter **"*W"** on the DMR. (For purposes of reporting on the Daily tab of the DMR, the permittee shall enter "*W" on the first day of the month only). Completed toxicity test reports for each test conducted shall be retained by the permittee in accordance with the requirements of Part II.B.5. of this permit and shall be available for review by the Department upon request. After 24 months of toxicity testing and upon approval from the Department, the monitoring frequency may be reduced if the test data indicate that the toxicity requirements of R 323.1219 of the Michigan Administrative Code are consistently being met. After one (1) year of toxicity testing and upon approval from the Department, the chronic toxicity tests may be performed using the more sensitive species identified in the chronic toxicity results collected to date. If a more sensitive species cannot be identified, the chronic toxicity tests shall be performed with both species. Toxicity test data acceptability is contingent upon validation of the test method by the testing laboratory. Such validation shall be submitted to the Department upon request.

1) When monitoring shows persistent exceedance of the 1.1 TU_C limit for effluent toxicity or data indicate persistent exceedance of the acute toxicity requirements of R 323.1219, the Department will determine whether the permittee must implement the toxicity control program requirements specified in 2), below.

2) Upon written notification by the Department, the following conditions apply. Within 90 days of the notification, the permittee shall implement a Toxicity Reduction Evaluation (TRE). The objective of the TRE shall be to reduce the toxicity of the final effluent from Monitoring Point 001A to <1.1 TU_C and <1.0 TU_A. The following documents are available as guidance to reduce toxicity to acceptable levels: Phase I, EPA/600/6-91/005F (chronic), EPA/600/6-91/003 (acute); Phase II, EPA/600/R-92/080 (acute and chronic); Phase III, EPA/600/R-92/081 (acute and chronic); and Publicly Owned Treatment Works (POTWs), EPA/833B-99/002. The TRE shall include quarterly chronic toxicity tests of the discharge from Monitoring Point 001A for the duration of the TRE. The tests shall be conducted and reported as specified above. Annual reports shall be submitted to the Department within 30 days of the completion of the last test of each annual cycle.

3) This permit may be modified in accordance with applicable laws and rules to include additional whole effluent toxicity control requirements as necessary.

PART I

Section A. Limitations and Monitoring Requirements

2. Quantification Levels and Analytical Methods for Selected Parameters

Maximum acceptable quantification levels (QLs) are specified for selected parameters in the table below. These QLs apply to all monitoring conducted in compliance with this permit if and when the parameters specified herein are monitored. This includes monitoring conducted to meet the requirements of the application for permit reissuance. These QLs shall be considered the maximum acceptable unless a higher QL is appropriate because of sample matrix interference. Justification for higher QLs shall be submitted to the Department within 30 days of such determination.

Where necessary to help ensure that the QLs specified herein can be achieved, analytical methods may also be specified in the table below. The sampling procedures, preservation and handling, and analytical protocol for all monitoring conducted in compliance with this permit, including monitoring conducted to meet the requirements of the application for permit reissuance, shall be in accordance with the methods specified herein, or in accordance with Part II.B.2. of this permit if no method is specified herein, unless an alternate method is approved by the Department. The Department will consider only alternate methods that meet the requirements of Part II.B.2. and whose QLs are at least as sensitive (i.e., low) as those specified herein. **Not all QLs are expressed in the same units in the table below**. The table is continued on the following page:

Parameter	QL	Units	Analytical Method
1,2-Diphenylhydrazine (as Azobenzene)	3.0	ug/l	
2,4,6-Trichlorophenol	5.0	ug/l	
2,4-Dinitrophenol	19	ug/l	
3,3'-Dichlorobenzidine	1.5	ug/l	
4-Chloro-3-Methylphenol	7.0	ug/l	
4,4'-DDD	0.01	ug/l	
4,4'-DDE	0.01	ug/l	
4,4'-DDT	0.01	ug/l	
Acrylonitrile	1.0	ug/l	
Aldrin	0.01	ug/l	
Alpha-Endosulfan	0.01	ug/l	
Alpha-Hexachlorocyclohexane	0.01	ug/l	
Antimony, Total	1	ug/l	
Arsenic, Total	1	ug/l	
Barium, Total	5	ug/l	
Benzidine	0.1	ug/l	
Beryllium, Total	1	ug/l	
Beta-Endosulfan	0.01	ug/l	
Beta-Hexachlorocyclohexane	0.01	ug/l	
Bis (2-Chloroethyl) Ether	1.0	ug/l	
Bis (2-Ethylhexyl) Phthalate	5.0	ug/l	
Boron, Total	20	ug/l	
Bromomethane, Total	1.0	ug/l	
Cadmium, Total	0.2	ug/l	
Chlordane	0.01	ug/l	
Chloride	1.0	mg/l	
Chromium, Hexavalent	5	ug/l	
Chromium, Total	10	ug/l	
Copper, Total	1	ug/l	

PART I

Section A. Limitations and Monitoring Requirements

Parameter	QL	Units	Analytical Method
Cyanide, Available	2	ug/l	EPA Method OIA 1677
Cyanide, Total	5	ug/l	
Delta-Hexachlorocyclohexane	0.01	ug/l	
Dieldrin	0.01	ug/l	
Di-N-Butyl Phthalate	9.0	ug/l	
Endosulfan Sulfate	0.01	ug/l	
Endrin	0.01	ug/l	
Endrin Aldehyde	0.01	ug/l	
Fluoranthene	1.0	ug/l	
Heptachlor	0.01	ug/l	
Heptachlor Epoxide	0.01	ug/l	
Hexachlorobenzene	0.01	ug/l	
Hexachlorobutadiene	0.01	ug/l	
Hexachlorocyclopentadiene	0.01	ug/l	
Hexachloroethane	5.0	ug/l	
Lead, Total	1	ug/l	
Lindane	0.01	ug/l	
Lithium, Total	10	ug/l	
Mercury, Total	0.5	ng/l	EPA Method 1631E
Nickel, Total	5	ug/l	
PCB-1016	0.1	ug/l	
PCB-1221	0.1	ug/l	
PCB-1232	0.1	ug/l	
PCB-1242	0.1	ug/l	
PCB-1248	0.1	ug/l	
PCB-1254	0.1	ug/l	
PCB-1260	0.1	ug/l	
Pentachlorophenol	1.8	ug/l	
Perfluorooctanesulfonic acid (PFOS)	2.0	ng/l	While EPA Method 1633 remains draft, analyses may be performed using that method, or ASTM D7979, or an isotope dilution method (sometimes referred to as Method 537 modified). Once EPA Method 1633 is promulgated, only that method may be used.
Perfluorooctanoic acid (PFOA)			
Perfluorobutanesulfonic acid (PFBS)			
Phenanthrene	1.0	ug/l	
Selenium, Total	1.0	ug/l	
Silver, Total	0.5	ug/l	
Strontium, Total	1000	ug/l	
Sulfate	2.0	mg/l	
Sulfide, Total	20	ug/l	
Thallium, Total	1	ug/l	
Toxaphene	0.1	ug/l	
Vinyl Chloride	1.0	ug/l	
Zinc, Total	10	ug/l	

PART I

Section A. Limitations and Monitoring Requirements

3. Additional Monitoring Requirements

As a condition of this permit, the permittee shall monitor the discharge from monitoring point 001A for the constituents identified below. This monitoring is an application requirement of 40 CFR 122.21(j), effective December 2, 1999. Testing shall be conducted in August 2024, May 2025, March 2026, and October 2026. Grab samples shall be collected for total phenols and the Perfluoroalkyl and Polyfluoroalkyl Substances (PFAS) and Volatile Organic Compounds identified below. For all other parameters, 24-hour composite samples shall be collected.

The results of such additional monitoring shall be submitted with the application for reissuance (see the cover page of this permit for the application due date). The permittee shall notify the Department within 14 days of completing the monitoring for each month specified above in accordance with Part II.C.5. Additional reporting requirements are specified in Part II.C.11. If, upon review of the analysis, it is determined that additional requirements are needed to protect the receiving waters in accordance with applicable water quality standards, the permit may then be modified by the Department in accordance with applicable laws and rules.

Hardness
calcium carbonate

Perfluoroalkyl and Polyfluoroalkyl Substances
Perfluorooctanoic acid (PFOA) Perfluorobutanesulfonic acid (PFBS)

Metals (Total Recoverable), Cyanide and Total Phenols

antimony	chromium	lead	nickel
selenium	thallium	total phenols	zinc

Volatile Organic Compounds

acrolein	acrylonitrile	benzene	bromoform
carbon tetrachloride	chlorobenzene	chlorodibromomethane	chloroethane
2-chloroethylvinyl ether	chloroform	dichlorobromomethane	1,1-dichloroethane
1,2-dichloroethane	trans-1,2-dichloroethylene	1,1-dichloroethylene	1,2-dichloropropane
1,3-dichloropropylene	ethylbenzene	methyl chloride	
methylene chloride	1,1,2,2-tetrachloroethane	tetrachloroethylene	toluene
1,1,1-trichloroethane	1,1,2-trichloroethane	trichloroethylene	vinyl chloride

Acid-Extractable Compounds

4-chloro-3-methylphenol	2-chlorophenol	2,4-dichlorophenol	2,4-dimethylphenol
4,6-dinitro-o-cresol	2,4-dinitrophenol	2-nitrophenol	4-nitrophenol
pentachlorophenol	phenol	2,4,6-trichlorophenol	

Base/Neutral Compounds

acenaphthene	acenaphthylene	anthracene	benzo(a)anthracene
benzo(a)pyrene	3,4-benzofluoranthene	benzo(ghi)perylene	benzo(k)fluoranthene
bis(2-chloroethoxy)methane	bis(2-chloroethyl)ether	bis(2-chloroisopropyl)ether	
bis(2-ethylhexyl)phthalate	4-bromophenyl phenyl ether	butylbenzyl phthalate	2-chloronaphthalene
4-chlorophenyl phenyl ether	chrysene	di-n-butyl phthalate	di-n-octyl phthalate
dibenzo(a,h)anthracene	1,2-dichlorobenzene	1,3-dichlorobenzene	1,4-dichlorobenzene
3,3'-dichlorobenzidine	diethyl phthalate	dimethyl phthalate	2,4-dinitrotoluene
2,6-dinitrotoluene	1,2-diphenylhydrazine	fluoranthene	fluorene
hexachlorobenzene	hexachlorobutadiene	hexachlorocyclopentadiene	hexachloroethane
indeno(1,2,3-cd)pyrene	isophorone	naphthalene	nitrobenzene
n-nitrosodi-n-propylamine	n-nitrosodimethylamine	n-nitrosodiphenylamine	phenanthrene
pyrene	1,2,4-trichlorobenzene		

PART I

Section A. Limitations and Monitoring Requirements

4. Pollutant Minimization Program for Total Mercury

The goal of the Pollutant Minimization Program is to maintain the effluent concentration of total mercury at or below 1.3 ng/l. The permittee shall continue to implement the Pollutant Minimization Program approved on May 17, 2005, and modifications thereto, to proceed toward the goal. The Pollutant Minimization Program includes the following:

a. an annual review and semi-annual monitoring of potential sources of mercury entering the wastewater collection system;

b. a program for quarterly monitoring of influent and periodic monitoring of sludge for mercury; and

c. implementation of reasonable cost-effective control measures when sources of mercury are discovered. Factors to be considered include significance of sources, economic considerations, and technical and treatability considerations.

On or before March 31 of each year, the permittee shall submit a status report to the Department for the previous calendar year that includes 1) the monitoring results for the previous year, 2) an updated list of potential mercury sources, and 3) a summary of all actions taken to reduce or eliminate identified sources of mercury.

Any information generated as a result of the Pollutant Minimization Program set forth in this permit may be used to support a request to modify the approved program or to demonstrate that the Pollutant Minimization Program requirement has been completed satisfactorily.

A request for modification of the approved program and supporting documentation shall be submitted in writing to the Department for review and approval. The Department may approve modifications to the approved program (approval of a program modification does not require a permit modification), including a reduction in the frequency of the requirements under items a. and b. above.

This permit may be modified in accordance with applicable laws and rules to include additional mercury conditions and/or limitations as necessary.

PART I

Section A. Limitations and Monitoring Requirements

5. Pollutant Minimization and Source Evaluation Program for Perfluorooctanesulfonic Acid (PFOS), Perfluorooctanoic Acid (PFOA), and Perfluorobutanesulfonic Acid (PFBS)

The goal of the Pollutant Minimization and Source Evaluation Program is to identify and address sources of PFOS, PFOA, and PFBS and to reduce and maintain the effluent concentrations of PFOS, PFOA, and PFBS at or below the water quality-based effluent limitations (WQBELs). The WQBELs are 12 ng/l for PFOS, 0.28 ug/l for PFOA, and 1,100 ug/l for PFBS.

a. Within 90 days of written notification from the Department or after the permittee notifies the Department that final effluent concentrations of PFOS, PFOA, and/or PFBS have exceeded the WQBELs, the permittee shall submit to the Department an approvable Pollutant Minimization and Source Evaluation Program for the PFAS analyte(s) in exceedance of the WQBEL(s) to proceed toward the goal. The Pollutant Minimization and Source Evaluation Program shall include the following:

1) identification of and strategies to identify any potential and probable PFOS, PFOA, and PFBS sources, as applicable;

2) monitoring plan for the permitted facility's influent and effluent, as well as effluent from suspected sources;

3) implemented measures thus far to eliminate, reduce, and/or control sources, and an assessment of the degree of success and the strategies used to measure success; and

4) proposed measures and implementation schedules for elimination, control, and/or reduction of the identified sources (prioritizing highest loadings and concentrations), and the strategies that will be used to measure success.

b. Two times annually, on or before February 1 and August 1 of each year following Department approval of the Pollutant Minimization and Source Evaluation Program, the permittee shall submit to the Department a status report for the prior six (6) month period. Status reports shall include:

1) complete list of known or suspected PFOS, PFOA, and PFBS sources, as applicable;

2) summary of influent and effluent monitoring data;

3) summary of monitoring data from known or suspected sources;

4) history and compliance status for sources;

5) implemented measures to eliminate, reduce, or control sources, (prioritizing highest loadings and concentrations), and an assessment of the degree of success and the strategies used to measure success;

6) proposed measures and schedules for elimination, control, or reduction of any newly identified PFOS, PFOA, and PFBS sources (prioritizing highest loadings and concentrations), as applicable, and the strategies that will be used to measure success;

7) barriers to implementation and revisions to the implementation schedule; and

8) any laboratory reports not previously supplied.

Any information generated as a result of the Pollutant Minimization and Source Evaluation Program set forth in this permit may be used to support a request to modify the Pollutant Minimization and Source Evaluation Program or to demonstrate that the requirement has been completed satisfactorily.

Section A. Limitations and Monitoring Requirements

A request for modification of the approved Pollutant Minimization and Source Evaluation Program shall be submitted in writing to the Department along with supporting documentation for review and approval. The Department may approve modifications to the approved Pollutant Minimization and Source Evaluation Program, including a reduction in the frequency of the influent and known or potential source monitoring requirements. Approval of a Pollutant Minimization and Source Evaluation Program modification does not require a permit modification.

This permit may be modified in accordance with applicable laws and rules to include additional PFOS, PFOA, and PFBS conditions and/or limitations as necessary.

6. Untreated or Partially Treated Sewage Discharge Reporting and Testing Requirements

In accordance with Section 324.3112a of the NREPA, if untreated or partially treated sewage is directly or indirectly discharged from a sewer system onto land or into the waters of the state, the permittee shall immediately, but not more than 24 hours after the discharge begins, notify local health departments, a daily newspaper of general circulation in the county in which the permittee is located, and a daily newspaper of general circulation in the county or counties in which the municipalities whose waters may be affected by the discharge are located, that the discharge is occurring. The permittee shall also notify the Department via its MiEnviro Portal system on the form entitled "Report of Discharge (CSO\SSO\RTB)." The MiEnviro Portal website is located at https://mienviro.michigan.gov/ncore/. At the conclusion of the discharge, the permittee shall make all such notifications specified in, and in accordance with, Section 324.3112a of the NREPA, and shall notify the Department via its MiEnviro Portal system on the form entitled "Report of Discharge (CSO\SSO\RTB)."

The permittee shall also annually contact municipalities, including the superintendent of a public drinking water supply with potentially affected intakes, whose waters may be affected by the permittee's discharge of untreated or partially treated sewage, and if those municipalities wish to be notified in the same manner as specified above, the permittee shall provide such notification.

Additionally, in accordance with Section 324.3112a of the NREPA, each time a discharge of untreated or partially treated sewage occurs, the permittee shall test the affected waters for *Escherichia coli* to assess the risk to the public health as a result of the discharge and shall provide the test results to the affected local county health departments and to the Department. The results of this testing shall be submitted to the Department via MiEnviro Portal as part of the notification specified above, or, if the results are not yet available, submitted as soon as they become available. This testing is not required if it has been waived by the local health department, or if the discharge(s) did not affect surface waters. The testing shall be done at locations specified by each affected local county health department but shall not exceed 10 tests for each separate discharge event. The affected local county health department may waive this testing requirement if it determines that such testing is not needed to assess the risk to the public health as a result of the discharge event.

Permittees accepting sanitary or municipal sewage from other sewage collection systems are encouraged to notify the owners of those systems of the above reporting and testing requirements.

Section A. Limitations and Monitoring Requirements

7. Facility Contact

The "Facility Contact" was specified in the application. The permittee may replace the facility contact at any time, and shall notify the Department in writing within 10 days after replacement (including the name, address and telephone number of the new facility contact).

a. The facility contact shall be (or a duly authorized representative of this person):
- for a corporation, a principal executive officer of at least the level of vice president; or a designated representative if the representative is responsible for the overall operation of the facility from which the discharge originates, as described in the permit application or other NPDES form,
- for a partnership, a general partner,
- for a sole proprietorship, the proprietor, or
- for a municipal, state, or other public facility, either a principal executive officer, the mayor, village president, city or village manager or other duly authorized employee.

b. A person is a duly authorized representative only if:
- the authorization is made in writing to the Department by a person described in paragraph a. of this section; and
- the authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the facility (a duly authorized representative may thus be either a named individual or any individual occupying a named position).

Nothing in this section releases the permittee from properly submitting reports and forms as required by law.

8. Monthly Operating Reports

Part 41 of Act 451 of 1994 as amended, specifically Section 324.4106 and associated R 299.2953, requires that the permittee file with the Department, on forms prescribed by the Department, operating reports showing the effectiveness of the treatment facility operation and the quantity and quality of liquid wastes discharged into waters of the state. Applicable forms and guidance are available on the Department's web site at https://www.michigan.gov/egle/about/Organization/Water-Resources/wastewater-construction. The permittee may use alternate forms if they are consistent with the approved treatment facility monitoring program. Unless the Department provides written notification to the permittee that monthly submittal of operating reports is required, operating reports that result from implementation of the approved treatment facility monitoring program shall be maintained on site for a minimum of three (3) years and shall be made available to the Department for review upon request.

Within 30 days of the effective date of this permit, the permittee shall submit to the Department either a revised treatment facility monitoring program to address monitoring requirement changes reflected in this permit, or justification explaining why monitoring requirement changes reflected in this permit do not necessitate revisions to the treatment facility monitoring program. Upon receipt of approval from the Department and consistent with such approval, the permittee shall implement the approved revised treatment facility monitoring program.

Section A. Limitations and Monitoring Requirements

9. Asset Management

The permittee shall at all times properly operate and maintain all facilities (i.e., the sewer system and treatment works as defined in Part 41 of the NREPA), and control systems installed or used by the permittee to operate the sewer system and treatment works and achieve and maintain compliance with the conditions of this permit (also see Part II.D.3 of this permit). The requirements of an Asset Management Program function to achieve the goals of effective performance, adequate funding, and adequate operator staffing and training. Asset management is a planning process for ensuring that optimum value is gained for each asset and that financial resources are available to rehabilitate and replace those assets when necessary. Asset management is centered on a framework of five (5) core elements: the current state of the assets; the required sustainable level of service; the assets critical to sustained performance; the minimum life-cycle costs; and the best long-term funding strategy.

a. Asset Management Program Requirements
On or before May 1, 2014 (received August 12, 2019), the permittee shall submit to the Department an Asset Management Plan for review and approval. An approvable Asset Management Plan shall contain a schedule for the development and implementation of an Asset Management Program that meets the requirements outlined below in 1) – 4). A copy of any Asset Management Program requirements already completed by the permittee should be submitted as part of the Asset Management Plan. Upon approval by the Department the permittee shall implement the Asset Management Plan. (The permittee may choose to include the Operation and Maintenance Manual required under Part II.C.14. of this permit as part of their Asset Management Program).

1) *Maintenance Staff.* The permittee shall provide an adequate staff to carry out the operation, maintenance, repair, and testing functions required to ensure compliance with the terms and conditions of this permit. The level of staffing needed shall be determined by taking into account the work involved in operating the sewer system and treatment works, planning for and conducting maintenance, and complying with this permit.

2) *Collection System Map.* The permittee shall complete a map of the sewer collection system it owns and operates. The map shall be of sufficient detail and at a scale to allow easy interpretation. The collection system information shown on the map shall be based on current conditions and shall be kept up-to-date and available for review by the Department. **Note: Items below referencing combined sewer systems are not applicable to separate sewer systems.** Such map(s) shall include but not be limited to the following:

a) all sanitary sewer lines and related manholes;

b) all combined sewer lines, related manholes, catch basins and CSO regulators;

c) all known or suspected connections between the sanitary sewer or combined sewer and storm drain systems;

d) all outfalls, including the treatment plant outfall(s), combined sewer treatment facility outfalls, untreated CSOs, and any known SSOs;

e) all pump stations and force mains;

f) the wastewater treatment facility(ies), including all treatment processes;

g) all surface waters (labeled);

h) other major appurtenances such as inverted siphons and air release valves;

PART I

Section A. Limitations and Monitoring Requirements

i) a numbering system which uniquely identifies manholes, catch basins, overflow points, regulators and outfalls;

j) the scale and a north arrow;

k) the pipe diameter, date of installation, type of material, distance between manholes, and the direction of flow; and

l) the manhole interior material, rim elevation (optional), and invert elevations.

3) *Inventory and assessment of fixed assets.* The permittee shall complete an inventory and assessment of operations-related fixed assets including portions of the collection system owned and operated by the permittee. Fixed assets are assets that are normally stationary (e.g., pumps, blowers, buildings, manholes, and sewer lines). The inventory and assessment shall be based on current conditions and shall be kept up-to-date and available for review by the Department.

a) The fixed asset inventory shall include the following:

(1) a brief description of the fixed asset, its design capacity (e.g., pump: 120 gallons per minute), its level of redundancy, and its tag number if applicable;

(2) the location of the fixed asset;

(3) the year the fixed asset was installed;

(4) the present condition of the fixed asset (e.g., excellent, good, fair, poor); and

(5) the current fixed asset (replacement) cost in dollars for year specified in accordance with approved schedules;

b) The fixed asset assessment shall include a “Business Risk Evaluation” that combines the probability of failure of the fixed asset and the criticality of the fixed asset, as follows:

(1) Rate the probability of failure of the fixed asset on a scale of 1-5 (low to high) using criteria such as maintenance history, failure history, and remaining percentage of useful life (or years remaining);

(2) Rate the criticality of the fixed asset on a scale of 1-5 (low to high) based on the consequence of failure versus the desired level of service for the facility; and

(3) Compute the Business Risk Factor of the fixed asset by multiplying the failure rating from (1) by the criticality rating from (2).

4) *Operation, Maintenance & Replacement (OM&R) Budget and Rate Sufficiency for the Sewer System and Treatment Works.* The permittee shall complete an assessment of its user rates and replacement fund, including the following:

a) beginning and end dates of fiscal year;

b) name of the department, committee, board, or other organization that sets rates for the operation of the sewer system and treatment works;

Section A. Limitations and Monitoring Requirements

c) amount in the permittee's replacement fund in dollars for year specified in accordance with approved schedules;

d) replacement fund strategy of all assets with a useful life of 20 years or less;

e) expenditures for maintenance, corrective action and capital improvement taken during the fiscal year;

f) OM&R budget for the fiscal year; and

g) rate calculation demonstrating sufficient revenues to cover OM&R expenses. If the rate calculation shows there are insufficient revenues to cover OM&R expenses, the permittee shall document, within three (3) fiscal years after submittal of the Asset Management Plan, that there is at least one rate adjustment that reduces the revenue gap by at least 10 percent. The permittee may prepare and submit an alternate plan, subject to Department approval, for addressing the revenue gap. The ultimate goal of the Asset Management Program is to ensure sufficient revenues to cover OM&R expenses.

b. Annual Reporting
The permittee shall develop a written report that summarizes asset management activities completed during the previous year and planned for the upcoming year. The written report shall be submitted to the Department on or before July 31 of each year. The written report shall include:

1) a description of the staffing levels maintained during the year;

2) a description of inspections and maintenance activities conducted and corrective actions taken during the previous year;

3) expenditures for collection system maintenance activities, treatment works maintenance activities, corrective actions, and capital improvement during the previous year;

4) a summary of assets/areas identified for inspection/action (including capital improvement) in the upcoming year based on the five (5) core elements and the Business Risk Factors computed in accordance with condition a.3)b)(3) above;

5) a maintenance budget and capital improvement budget for the upcoming year that take into account implementation of an effective Asset Management Program that meets the five (5) core elements;

6) an updated asset inventory based on the original submission; and

7) an updated OM&R budget with an updated rate schedule that includes the amount of insufficient revenues, if any.

Section A. Limitations and Monitoring Requirements

10. Discharge Monitoring Report – Quality Assurance Study Program

The permittee shall participate in the Discharge Monitoring Report – Quality Assurance (DMR-QA) Study Program. The purpose of the DMR-QA Study Program is to annually evaluate the proficiency of all in-house and/or contract laboratory(ies) that perform, on behalf of the facility authorized to discharge under this permit, the analytical testing required under this permit. In accordance with Section 308 of the Clean Water Act (33 U.S.C. § 1318); and R 323.2138 and R 323.2154 of Part 21, Wastewater Discharge Permits, promulgated under Part 31 of the NREPA, participation in the DMR-QA Study Program is required for all major facilities, and for minor facilities selected for participation by the Department.

Annually and in accordance with DMR-QA Study Program requirements and submittal due dates, the permittee shall submit to the Michigan DMR-QA Study Program state coordinator all documentation required by the DMR-QA Study. DMR-QA Study Program participation is required only for the analytes required under this permit and only when those analytes are also identified in the DMR-QA Study.

If the permitted facility's status as a major facility should change, participation in the DMR-QA Study Program may be reevaluated. Questions concerning participation in the DMR-QA Study Program should be directed to the Michigan DMR-QA Study Program state coordinator.

All forms and instructions required for participation in the DMR-QA Study Program, including submittal due dates and state coordinator contact information, can be found at https://www.epa.gov/compliance/discharge-monitoring-report-quality-assurance-study-program.

11. Continuous Monitoring

If continuous monitoring equipment is used and becomes temporarily inoperable, the permittee shall manually obtain a minimum of three (3) equally spaced grab samples/readings within each 24-hour period for the affected parameter(s). On such days, in the comment field on the Daily tab of the DMR, the permittee shall indicate "continuous monitoring system inoperable," the date on which the system is expected to become operable again, and the number of samples/readings obtained during each 24-hour period.

12. PFAS Data Reporting Requirements

The permittee shall submit the complete laboratory analysis of each PFAS sample. This may be in addition to required DMR reporting. Submittals shall be made via the MiEnviro Portal using the PFAS POTW Effluent Monitoring Report form. Until EPA Method 1633 is promulgated, the complete laboratory analysis of each PFAS sample shall constitute the first 28 analytes identified on the PFAS POTW Effluent Monitoring Report form. Following promulgation of EPA Method 1633, the complete laboratory analysis of each PFAS sample shall constitute all 40 analytes identified on the PFAS POTW Effluent Monitoring Report form.

PART I

Section B. Storm Water Pollution Prevention

Section B. Storm Water Pollution Prevention is not required for this permit.

PART I

Section C. Industrial Waste Pretreatment Program

1. Michigan Industrial Pretreatment Program

a. The permittee shall implement the Michigan Industrial Pretreatment Program (MIPP) approved on July 23, 1985, and any subsequent modifications approved up to the issuance of this permit.

b. The permittee shall comply with R 323.2301 through R 323.2317 of the Michigan Administrative Code (Part 23 Rules) and the approved MIPP.

c. The permittee shall have the legal authority and necessary interjurisdictional agreements that provide the basis for the implementation and enforcement of the approved MIPP throughout the service area. The legal authority and necessary interjurisdictional agreements shall include, at a minimum, the authority to carry out the activities specified in R 323.2306(a).

d. The permittee shall develop procedures which describe, in sufficient detail, program commitments which enable implementation of the approved MIPP and the Part 23 Rules in accordance with R 323.2306(c).

e. The permittee shall establish an interjurisdictional agreement (or comparable document) with all tributary governmental jurisdictions. Each interjurisdictional agreement shall contain, at a minimum, the following:

1) identification of the agency responsible for the implementation and enforcement of the approved MIPP within the tributary governmental jurisdiction's boundaries; and

2) the provision of the legal authority which provides the basis for the implementation and enforcement of the approved MIPP within the tributary governmental jurisdiction's boundaries

f. The permittee shall prohibit discharges that:

1) cause, in whole or in part, the permittee's failure to comply with any condition of this permit or the NREPA;

2) restrict, in whole or in part, the permittee's management of biosolids;

3) cause, in whole or in part, operational problems at the treatment facility or in its collection system;

4) violate any of the general or specific prohibitions identified in R 323.2303(1) and (2);

5) violate categorical standards identified in R 323.2311; and

6) violate local limits established in accordance with R 323.2303(4).

g. The permittee shall maintain a list of its nondomestic users that meet the criteria of a significant industrial user as identified in R 323.2302(cc).

h. The permittee shall develop an enforcement response plan which describes, in sufficient detail, program commitments which will enable the enforcement of the approved MIPP and the Part 23 Rules in accordance with R 323.2306(g).

i. The Department may require modifications to the approved MIPP which are necessary to ensure compliance with the Part 23 Rules in accordance with R 323.2309.

Section C. Industrial Waste Pretreatment Program

j. The permittee shall not implement changes or modifications to the approved MIPP without notification to the Department.

k. The permittee shall maintain an adequate revenue structure and staffing level for effective implementation of the approved MIPP.

l. The permittee shall develop and maintain, for a minimum of three (3) years, all records and information necessary to determine nondomestic user compliance with the Part 23 Rules and the approved MIPP. This period of retention shall be extended during the course of any unresolved enforcement action or litigation regarding a nondomestic user or when requested by the Department or the United States Environmental Protection Agency. All of the aforementioned records and information shall be made available upon request for inspection and copying by the Department and the United States Environmental Protection Agency.

m. The permittee shall evaluate the approved MIPP for compliance with the Part 23 Rules and the prohibitions set forth in item f. above. Based upon this evaluation, the permittee shall propose to the Department all necessary changes or modifications to the approved MIPP no later than the next Industrial Pretreatment Program Annual Report due date (see item p. below).

n. The permittee shall develop and enforce local limits to implement the prohibitions set forth in item f. above. Local limits shall be based upon data representative of actual conditions demonstrated in a maximum allowable headworks loading analysis. A technical evaluation of whether local limits need to be developed for the following pollutants shall be submitted to the Department by January 1, 2025: Total Arsenic, Total Beryllium, Total Cadmium, Total Silver, Available Cyanide, and Total Benzidine. The submittal shall provide a technical evaluation of the basis upon which this determination was made which includes information regarding the maximum allowable headworks loading, collection system protection criteria, and worker health and safety, based upon data collected since the last local limits review.

o. The permittee is required under this permit and R 323.2303(4) of the Michigan Administrative Code to review and update their local limits when:

1) new pollutants are introduced;

2) new pollutants that were previously unevaluated are identified;

3) new water quality or biosolids standards are established or additional information becomes available about the nature of pollutants, such as removal rates and accumulation in biosolids; or

4) substantial increases of pollutants are proposed as required in the notification of new or increased uses in accordance with the provisions of 40 CFR 122.42.

p. On or before April 1 of each year, the permittee shall submit to the Department, as required by R 323.2310(8), an Industrial Pretreatment Program Annual Report on the status of program implementation and enforcement activities. The reporting period shall begin on January 1 and end on December 31. At a minimum, the Industrial Pretreatment Program Annual Report shall include:

1) the Pretreatment Program Reports data identified in Appendix A to 40 CFR Part 127 – NPDES Electronic Reporting;

Section C. Industrial Waste Pretreatment Program

2) a summary of changes to the approved MIPP that have not been previously reported to the Department;

3) a summary of results of all the sampling and analyses performed of the wastewater treatment plant's influent, effluent, and biosolids conducted in accordance with approved methods during the reporting period. The summary shall include the monthly average, daily maximum, quantification level, and number of samples analyzed for each pollutant. At a minimum, the results of analyses for all locally limited parameters for at least one monitoring event that tests influent, effluent and biosolids during the reporting period shall be submitted with each report, unless otherwise required by the Department. Sample collection shall be at intervals sufficient to provide pollutant removal rates, unless the pollutant is not measurable; and

4) any other relevant information requested by the Department.

q. Survey of Significant Industrial Users
On or before January 1, 2025, the permittee shall submit to the Department for review, the results of a survey which identifies all significant industrial users of the waste collection system. For significant industrial users the following information shall be included:

1) the user's name and mailing address which shall include both the local facility address and the main office address, if different;

2) the principal enterprise(s) of the user; the product(s) produced and raw material(s) processed; the facility's production rate(s); and the Standard Industrial Classification (SIC) code(s);

3) the quantity of process wastewater discharged (in GPD) and an indication of whether the discharge is continuous or batch;

4) a description of any pretreatment provided prior to discharge to the waste collection system; and

5) a description of the wastewater characteristics in terms of pollutant parameters and concentrations.

PART I

Section D. Residuals Management Program

1. Residuals Management Program for Land Application of Biosolids

The permittee is authorized to land-apply bulk biosolids or prepare bulk biosolids for land application in accordance with the permittee's approved Residuals Management Program (RMP) approved on May 6, 2002, and approved modifications thereto, and the requirements established in R 323.2401 through R 323.2418 of the Michigan Administrative Code (Part 24 Rules). The approved RMP, and any approved modifications thereto, are enforceable requirements of this permit. Incineration, landfilling and other residual disposal activities shall be conducted in accordance with Part II.D.7. of this permit. The Part 24 Rules can be obtained via the internet at https://www.michigan.gov/egle/about/organization/water-resources/biosolids/laws-and-rules.

a. Annual Report
On or before October 30 of each year, the permittee shall submit an annual report to the Department for the previous fiscal year of October 1 through September 30. The report shall be submitted electronically via the Department's MiEnviro Portal system at https://mienviro.michigan.gov/ncore/. At a minimum, the report shall contain:

1) a certification that current residuals management practices are in accordance with the approved RMP, or a proposal for modification to the approved RMP; and

2) a completed Annual Report Form for Reporting Biosolids, available at https://mienviro.michigan.gov/ncore/.

b. Modifications to the Approved RMP
Prior to implementation of modifications to the RMP, the permittee shall submit proposed modifications to the Department for approval. The approved modification shall become effective upon the date of approval. Upon written notification, the Department may impose additional requirements and/or limitations to the approved RMP as necessary to protect public health and the environment from any adverse effect of a pollutant in the biosolids.

c. Record Keeping
Records required by the Part 24 Rules shall be kept for a minimum of five (5) years. However, the records documenting cumulative loading for sites subject to cumulative pollutant loading rates shall be kept as long as the site receives biosolids.

d. Contact Information
RMP-related submittals shall be made to the Department.

PART II

Section A. Definitions

Part II may include terms and /or conditions not applicable to discharges covered under this permit.

Acute toxic unit (TUA) means 100/LC50 where the LC50 is determined from a whole effluent toxicity (WET) test which produces a result that is statistically or graphically estimated to be lethal to 50% of the test organisms.

Annual monitoring frequency refers to a calendar year beginning on January 1 and ending on December 31. When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

Authorized public agency means a state, local, or county agency that is designated pursuant to the provisions of Section 9110 of Part 91, Soil and Sedimentation Control, of the NREPA, to implement soil erosion and sedimentation control requirements with regard to construction activities undertaken by that agency.

Best management practices (BMPs) means structural devices or nonstructural practices that are designed to prevent pollutants from entering into storm water, to direct the flow of storm water, or to treat polluted storm water.

Bioaccumulative chemical of concern (BCC) means a chemical which, upon entering the surface waters, by itself or as its toxic transformation product, accumulates in aquatic organisms by a human health bioaccumulation factor of more than 1000 after considering metabolism and other physiochemical properties that might enhance or inhibit bioaccumulation. The human health bioaccumulation factor shall be derived according to R 323.1057(5). Chemicals with half-lives of less than 8 weeks in the water column, sediment, and biota are not BCCs. The minimum bioaccumulation concentration factor (BAF) information needed to define an organic chemical as a BCC is either a field-measured BAF or a BAF derived using the biota-sediment accumulation factor (BSAF) methodology. The minimum BAF information needed to define an inorganic chemical as a BCC, including an organometal, is either a field-measured BAF or a laboratory-measured bioconcentration factor (BCF). The BCCs to which these rules apply are identified in Table 5 of R 323.1057 of the Water Quality Standards.

Biosolids are the solid, semisolid, or liquid residues generated during the treatment of sanitary sewage or domestic sewage in a treatment works. This includes, but is not limited to, scum or solids removed in primary, secondary, or advanced wastewater treatment processes and a derivative of the removed scum or solids.

Bulk biosolids means biosolids that are not sold or given away in a bag or other container for application to a lawn or home garden.

CAFO means concentrated animal feeding operation.

Certificate of Coverage (COC) is a document, issued by the Department, which authorizes a discharge under a general permit.

Chronic toxic unit (TUC) means 100/MATC or 100/IC25, where the maximum acceptable toxicant concentration (MATC) and IC25 are expressed as a percent effluent in the test medium.

Class B biosolids refers to material that has met the Class B pathogen reduction requirements or equivalent treatment by a Process to Significantly Reduce Pathogens (PSRP) in accordance with the Part 24 Rules, Land Application of Biosolids, promulgated under Part 31 of the NREPA. Processes include aerobic digestion, composting, anaerobic digestion, lime stabilization and air drying.

Combined sewer system is a sewer system in which storm water runoff is combined with sanitary wastes.

PART II

Section A. Definitions

Composite sample is a sample collected over time, either by continuous sampling or by mixing discrete samples. A composite sample represents the average wastewater characteristics present during the compositing period. Various methods for compositing are available and are based on either time or flow-proportioning, the choice of which will depend on the permit requirements.

Continuous monitoring refers to sampling/readings that occur at regular and consistent intervals throughout a 24-hour period and at a frequency sufficient to capture data that are representative of the discharge. The maximum acceptable interval between samples/readings shall be one (1) hour.

Daily concentration
FOR PARAMETERS OTHER THAN pH, DISSOLVED OXYGEN, TEMPERATURE, AND CONDUCTIVITY – Daily concentration is the sum of the concentrations of the individual samples of a parameter taken within a calendar day divided by the number of samples taken within that calendar day. The daily concentration will be used to determine compliance with any maximum and minimum daily concentration limitations. For guidance and examples showing how to report and perform calculations using results below quantification levels, see the document entitled "Reporting Results Below Quantification," available at https://www.michigan.gov/-/media/Project/Websites/egle/Documents/Programs/WRD/MiEnviro/results-below-quantification.pdf.

FOR pH, DISSOLVED OXYGEN, TEMPERATURE, AND CONDUCTIVITY – The daily concentration used to determine compliance with maximum daily pH, temperature, and conductivity limitations is the highest pH, temperature, and conductivity readings obtained within a calendar day. The daily concentration used to determine compliance with minimum daily pH and dissolved oxygen limitations is the lowest pH and dissolved oxygen readings obtained within a calendar day.

Daily loading is the total discharge by weight of a parameter discharged during any calendar day. This value is calculated by multiplying the daily concentration by the total daily flow and by the appropriate conversion factor. The daily loading will be used to determine compliance with any maximum daily loading limitations. When required by the permit, report the maximum calculated daily loading for the month in the "MAXIMUM" column under "QUANTITY OR LOADING" on the DMRs.

Daily monitoring frequency refers to a 24-hour day. When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

Department means the Michigan Department of Environment, Great Lakes, and Energy.

Detection level means the lowest concentration or amount of the target analyte that can be determined to be different from zero by a single measurement at a stated level of probability.

Discharge means the addition of any waste, waste effluent, wastewater, pollutant, or any combination thereof to any surface water of the state.

EC_{50} means a statistically or graphically estimated concentration that is expected to cause 1 or more specified effects in 50% of a group of organisms under specified conditions.

Fecal coliform bacteria monthly
FOR WWSLs THAT COLLECT AND STORE WASTEWATER AND ARE AUTHORIZED TO DISCHARGE ONLY IN THE SPRING AND/OR FALL ON AN INTERMITTENT BASIS – Fecal coliform bacteria monthly is the geometric mean of all daily concentrations determined during a discharge event. Days on which no daily concentration is determined shall not be used to determine the calculated monthly value. The calculated monthly value will be used to determine compliance with the maximum monthly fecal coliform bacteria limitations. When required by the permit, report the calculated monthly value in the "AVERAGE" column under "QUALITY OR CONCENTRATION" on the DMR. If the period in which the discharge event occurred was partially in each of two months, the calculated monthly value shall be reported on the DMR of the month in which the last day of discharge occurred.

Section A. Definitions

FOR ALL OTHER DISCHARGES – Fecal coliform bacteria monthly is the geometric mean of all daily concentrations determined during a reporting month. Days on which no daily concentration is determined shall not be used to determine the calculated monthly value. The calculated monthly value will be used to determine compliance with the maximum monthly fecal coliform bacteria limitations. When required by the permit, report the calculated monthly value in the "AVERAGE" column under "QUALITY OR CONCENTRATION" on the DMR.

Fecal coliform bacteria 7-day
FOR WWSLs THAT COLLECT AND STORE WASTEWATER AND ARE AUTHORIZED TO DISCHARGE ONLY IN THE SPRING AND/OR FALL ON AN INTERMITTENT BASIS – Fecal coliform bacteria 7-day is the geometric mean of the daily concentrations determined during any 7 consecutive days of discharge during a discharge event. If the number of daily concentrations determined during the discharge event is less than 7 days, the number of actual daily concentrations determined shall be used for the calculation. Days on which no daily concentration is determined shall not be used to determine the value. The calculated 7-day value will be used to determine compliance with the maximum 7-day fecal coliform bacteria limitations. When required by the permit, report the maximum calculated 7-day geometric mean value for the month in the "MAXIMUM" column under "QUALITY OR CONCENTRATION" on the DMRs. If the 7-day period was partially in each of two months, the value shall be reported on the DMR of the month in which the last day of discharge occurred.

FOR ALL OTHER DISCHARGES – Fecal coliform bacteria 7-day is the geometric mean of the daily concentrations determined during any 7 consecutive days in a reporting month. If the number of daily concentrations determined is less than 7, the actual number of daily concentrations determined shall be used for the calculation. Days on which no daily concentration is determined shall not be used to determine the value. The calculated 7-day value will be used to determine compliance with the maximum 7-day fecal coliform bacteria limitations. When required by the permit, report the maximum calculated 7-day geometric mean for the month in the "MAXIMUM" column under "QUALITY OR CONCENTRATION" on the DMRs. The first calculation shall be made on day 7 of the reporting month, and the last calculation shall be made on the last day of the reporting month.

Flow-proportioned composite sample is a composite sample in which either a) the volume of each portion of the composite is proportional to the effluent flow rate at the time that portion is obtained; or b) a constant sample volume is obtained at varying time intervals proportional to the effluent flow rate.

General permit means an NPDES permit authorizing a category of similar discharges.

Geometric mean is the average of the logarithmic values of a base 10 data set, converted back to a base 10 number.

Grab sample is a single sample taken at neither a set time nor flow.

IC_{25} means the toxicant concentration that would cause a 25% reduction in a nonquantal biological measurement for the test population.

Illicit connection means a physical connection to a municipal separate storm sewer system that primarily conveys non-storm water discharges other than uncontaminated groundwater into the storm sewer; or a physical connection not authorized or permitted by the local authority, where a local authority requires authorization or a permit for physical connections.

Illicit discharge means any discharge to, or seepage into, a municipal separate storm sewer system that is not composed entirely of storm water or uncontaminated groundwater. Illicit discharges include non-storm water discharges through pipes or other physical connections; dumping of motor vehicle fluids, household hazardous wastes, domestic animal wastes, or litter; collection and intentional dumping of grass clippings or leaf litter; or unauthorized discharges of sewage, industrial waste, restaurant wastes, or any other non-storm water waste directly into a separate storm sewer.

PART II

Section A. Definitions

Individual permit means a site-specific NPDES permit.

Inlet means a catch basin, roof drain, conduit, drain tile, retention pond riser pipe, sump pump, or other point where storm water or wastewater enters into a closed conveyance system prior to discharge off site or into waters of the state.

Interference is a discharge which, alone or in conjunction with a discharge or discharges from other sources, both: 1) inhibits or disrupts a POTW, its treatment processes or operations, or its sludge processes, use or disposal; and 2) therefore, is a cause of a violation of any requirement of the POTW's NPDES permit (including an increase in the magnitude or duration of a violation) or, of the prevention of sewage sludge use or disposal in compliance with the following statutory provisions and regulations or permits issued thereunder (or more stringent state or local regulations): Section 405 of the Clean Water Act, the Solid Waste Disposal Act (SWDA) (including Title II, more commonly referred to as the Resource Conservation and Recovery Act (RCRA), and including state regulations contained in any state sludge management plan prepared pursuant to Subtitle D of the SWDA), the Clean Air Act, the Toxic Substances Control Act, and the Marine Protection, Research and Sanctuaries Act. [This definition does not apply to sample matrix interference].

Land application means spraying or spreading biosolids or a biosolids derivative onto the land surface, injecting below the land surface, or incorporating into the soil so that the biosolids or biosolids derivative can either condition the soil or fertilize crops or vegetation grown in the soil.

LC_{50} means a statistically or graphically estimated concentration that is expected to be lethal to 50% of a group of organisms under specified conditions.

Maximum acceptable toxicant concentration (MATC) means the concentration obtained by calculating the geometric mean of the lower and upper chronic limits from a chronic test. A lower chronic limit is the highest tested concentration that did not cause the occurrence of a specific adverse effect. An upper chronic limit is the lowest tested concentration which did cause the occurrence of a specific adverse effect and above which all tested concentrations caused such an occurrence.

Maximum extent practicable means implementation of best management practices by a public body to comply with an approved storm water management program as required by a national permit for a municipal separate storm sewer system, in a manner that is environmentally beneficial, technically feasible, and within the public body's legal authority.

MBTU/hr means million British Thermal Units per hour.

MGD means million gallons per day.

Monthly concentration is the sum of the daily concentrations determined during a reporting period divided by the number of daily concentrations determined. The calculated monthly concentration will be used to determine compliance with any maximum monthly concentration limitations. Days with no discharge shall not be used to determine the value. When required by the permit, report the calculated monthly concentration in the "AVERAGE" column under "QUALITY OR CONCENTRATION" on the DMR.

For minimum percent removal requirements, the monthly influent concentration and the monthly effluent concentration shall be determined. The calculated monthly percent removal, which is equal to 100 times the quantity [1 minus the quantity (monthly effluent concentration divided by the monthly influent concentration)], shall be reported in the "MINIMUM" column under "QUALITY OR CONCENTRATION" on the DMRs.

PART II

Section A. Definitions

Monthly loading is the sum of the daily loadings of a parameter divided by the number of daily loadings determined during a reporting period. The calculated monthly loading will be used to determine compliance with any maximum monthly loading limitations. Days with no discharge shall not be used to determine the value. When required by the permit, report the calculated monthly loading in the “AVERAGE” column under “QUANTITY OR LOADING” on the DMR.

Monthly monitoring frequency refers to a calendar month. When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

Municipal separate storm sewer means a conveyance or system of conveyances designed or used for collecting or conveying storm water which is not a combined sewer and which is not part of a POTW as defined in the Code of Federal Regulations at 40 CFR 122.2.

Municipal separate storm sewer system (MS4) means all separate storm sewers that are owned or operated by the United States, a state, city, village, township, county, district, association, or other public body created by or pursuant to state law, having jurisdiction over disposal of sewage, industrial wastes, storm water, or other wastes, including special districts under state law, such as a sewer district, flood control district, or drainage district, or similar entity, or a designated or approved management agency under Section 208 of the Clean Water Act that discharges to the waters of the state. This term includes systems similar to separate storm sewer systems in municipalities, such as systems at military bases, large hospital or prison complexes, and highways and other thoroughfares. The term does not include separate storm sewers in very discrete areas, such as individual buildings.

National Pretreatment Standards are the regulations promulgated by or to be promulgated by the Federal Environmental Protection Agency pursuant to Section 307(b) and (c) of the Clean Water Act. The standards establish nationwide limits for specific industrial categories for discharge to a POTW.

No observed adverse effect level (NOAEL) means the highest tested dose or concentration of a substance which results in no observed adverse effect in exposed test organisms where higher doses or concentrations result in an adverse effect.

Noncontact cooling water is water used for cooling which does not come into direct contact with any raw material, intermediate product, by-product, waste product or finished product.

Nondomestic user is any discharger to a POTW that discharges wastes other than or in addition to water-carried wastes from toilet, kitchen, laundry, bathing or other facilities used for household purposes.

Nonstructural controls are practices or procedures implemented by employees at a facility to manage storm water or to prevent contamination of storm water.

NPDES means National Pollutant Discharge Elimination System.

Outfall is the location at which a point source discharge first enters a surface water of the state.

Part 91 agency means an agency that is designated by a county board of commissioners pursuant to the provisions of Section 9105 of Part 91 of the NREPA; an agency that is designated by a city, village, or township in accordance with the provisions of Section 9106 of Part 91 of the NREPA; or the Department for soil erosion and sedimentation control activities under Part 615, Supervisor of Wells; Part 631, Reclamation of Mining Lands; or Part 632, Nonferrous Metallic Mineral Mining, of the NREPA, pursuant to the provisions of Section 9115 of Part 91 of the NREPA.

Part 91 permit means a soil erosion and sedimentation control permit issued by a Part 91 agency pursuant to the provisions of Part 91 of the NREPA.

PART II

Section A. Definitions

Partially treated sewage is any sewage, sewage and storm water, or sewage and wastewater, from domestic or industrial sources that is treated to a level less than that required by the permittee's NPDES permit, or that is not treated to national secondary treatment standards for wastewater, including discharges to surface waters from retention treatment facilities.

PFAS means perfluoroalkyl and polyfluoroalkyl substances.

Point of discharge is the location of a point source discharge where storm water is discharged directly into a separate storm sewer system.

Point source discharge means a discharge from any discernible, confined, discrete conveyance, including but not limited to any pipe, ditch, channel, tunnel, conduit, well, discrete fissure, container, or rolling stock. Changing the surface of land or establishing grading patterns on land will result in a point source discharge where the runoff from the site is ultimately discharged to waters of the state.

Polluting material means any material, in solid or liquid form, identified as a polluting material under the Part 5 Rules, Spillage of Oil and Polluting Materials, promulgated under Part 31 of the NREPA (R 324.2001 through R 324.2009 of the Michigan Administrative Code).

POTW is a publicly owned treatment work.

Predevelopment is the last land use prior to the planned new development or redevelopment.

Pretreatment is reducing the amount of pollutants, eliminating pollutants, or altering the nature of pollutant properties to a less harmful state prior to discharge into a public sewer. The reduction or alteration can be by physical, chemical, or biological processes, process changes, or by other means. Dilution is not considered pretreatment unless expressly authorized by an applicable National Pretreatment Standard for a particular industrial category.

Public (as used in the MS4 individual permit) means all persons who potentially could affect the authorized storm water discharges, including, but not limited to, residents, visitors to the area, public employees, businesses, industries, and construction contractors and developers.

Public body means the United States; the state of Michigan; a city, village, township, county, school district, public college or university, or single-purpose governmental agency; or any other body which is created by federal or state statute or law.

Qualified Personnel means an individual who meets qualifications acceptable to the Department and who is authorized by an Industrial Storm Water Certified Operator to collect the storm water sample.

Qualifying storm event means a storm event causing greater than 0.1 inch of rainfall and occurring at least 72 hours after the previous measurable storm event that also caused greater than 0.1 inch of rainfall. Upon request, the Department may approve an alternate definition meeting the condition of a qualifying storm event.

Quantification level means the measurement of the concentration of a contaminant obtained by using a specified laboratory procedure calculated at a specified concentration above the detection level. It is considered the lowest concentration at which a particular contaminant can be quantitatively measured using a specified laboratory procedure for monitoring of the contaminant.

Quarterly monitoring frequency refers to a three-month period, defined as January through March, April through June, July through September, and October through December (or otherwise defined in the permit). When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

PART II

Section A. Definitions

Regional Administrator is the Region 5 Administrator, U.S. EPA, located at R-19J, 77 W. Jackson Blvd., Chicago, Illinois 60604.

Regulated area means the permittee's urbanized area, where urbanized area is defined as a place and its adjacent densely populated territory that together have a minimum population of 50,000 people as defined by the United States Bureau of the Census and as determined by the latest available decennial census.

Secondary containment structure means a unit, other than the primary container, in which significant materials are packaged or held, which is required by state or federal law to prevent the escape of significant materials by gravity into sewers, drains, or otherwise directly or indirectly into any sewer system or to the surface waters or groundwaters of the state.

Separate storm sewer system means a system of drainage, including, but not limited to, roads, catch basins, curbs, gutters, parking lots, ditches, conduits, pumping devices, or man-made channels, which is not a combined sewer where storm water mixes with sanitary wastes, and is not part of a POTW.

Significant industrial user is a nondomestic user that: 1) is subject to Categorical Pretreatment Standards under 40 CFR 403.6 and 40 CFR Chapter I, Subchapter N; or 2) discharges an average of 25,000 gallons per day or more of process wastewater to a POTW (excluding sanitary, noncontact cooling and boiler blowdown wastewater); contributes a process waste stream which makes up five (5) percent or more of the average dry weather hydraulic or organic capacity of the POTW treatment plant; or is designated as such by the permittee as defined in 40 CFR 403.12(a) on the basis that the industrial user has a reasonable potential for adversely affecting the POTW's treatment plant operation or violating any pretreatment standard or requirement (in accordance with 40 CFR 403.8(f)(6)).

Significant materials means any material which could degrade or impair water quality, including but not limited to: raw materials; fuels; solvents, detergents, and plastic pellets; finished materials such as metallic products; hazardous substances designated under Section 101(14) of the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) (see 40 CFR 372.65); any chemical the facility is required to report pursuant to Section 313 of Emergency Planning and Community Right-to-Know Act (EPCRA); polluting materials as identified under the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code); Hazardous Wastes as defined in Part 111, Hazardous Waste Management, of the NREPA; fertilizers; pesticides; and waste products such as ashes, slag, and sludge that have the potential to be released with storm water discharges.

Significant spills and significant leaks means any release of a polluting material reportable under the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code).

Special-use area means storm water discharges for which the Department has determined that additional monitoring is needed from: secondary containment structures required by state or federal law; lands on Michigan's List of Sites of Environmental Contamination pursuant to Part 201, Environmental Remediation, of the NREPA; and/or areas with other activities that may contribute pollutants to the storm water.

Stoichiometric means the quantity of a reagent calculated to be necessary and sufficient for a given chemical reaction.

Storm water means storm water runoff, snowmelt runoff, surface runoff and drainage, and non-storm water included under the conditions of this permit.

Storm water discharge point is the location where the point source discharge of storm water is directed to surface waters of the state or to a separate storm sewer. It includes the location of all point source discharges where storm water exits the facility, including outfalls which discharge directly to surface waters of the state, and points of discharge which discharge directly into separate storm sewer systems.

PART II

Section A. Definitions

Structural controls are physical features or structures used at a facility to manage or treat storm water.

SWPPP means the Storm Water Pollution Prevention Plan prepared in accordance with this permit.

Tier I value means a value for aquatic life, human health or wildlife calculated under R 323.1057 of the Water Quality Standards using a tier I toxicity database.

Tier II value means a value for aquatic life, human health or wildlife calculated under R 323.1057 of the Water Quality Standards using a tier II toxicity database.

Total maximum daily loads (TMDLs) are required by the Clean Water Act for waterbodies that do not meet water quality standards. TMDLs represent the maximum daily load of a pollutant that a waterbody can assimilate and meet water quality standards, and an allocation of that load among point sources, nonpoint sources, and a margin of safety.

Toxicity reduction evaluation (TRE) means a site-specific study conducted in a stepwise process designed to identify the causative agents of effluent toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in effluent toxicity.

Water Quality Standards means the Part 4 Water Quality Standards promulgated pursuant to Part 31 of the NREPA, being R 323.1041 through R 323.1117 of the Michigan Administrative Code.

Weekly monitoring frequency refers to a calendar week which begins on Sunday and ends on Saturday. When required by this permit, an analytical result, reading, value, or observation shall be reported for that period if a discharge occurs during that period. If the calendar week begins in one month and ends in the following month, the analytical result, reading, value, or observation shall be reported in the month in which monitoring was conducted.

WWSL is a wastewater stabilization lagoon.

WWSL discharge event is a discrete occurrence during which effluent is discharged to the surface water up to 10 days of a consecutive 14-day period.

3-portion composite sample is a sample consisting of three equal-volume grab samples collected at equal intervals over an 8-hour period.

7-day concentration
FOR WWSLs THAT COLLECT AND STORE WASTEWATER AND ARE AUTHORIZED TO DISCHARGE ONLY IN THE SPRING AND/OR FALL ON AN INTERMITTENT BASIS – The 7-day concentration is the sum of the daily concentrations determined during any 7 consecutive days of discharge during a WWSL discharge event divided by the number of daily concentrations determined. If the number of daily concentrations determined during the WWSL discharge event is less than 7 days, the number of actual daily concentrations determined shall be used for the calculation. The calculated 7-day concentration will be used to determine compliance with any maximum 7-day concentration limitations. When required by the permit, report the maximum calculated 7-day concentration for the WWSL discharge event in the “MAXIMUM” column under “QUALITY OR CONCENTRATION” on the DMR. If the WWSL discharge event was partially in each of two months, the value shall be reported on the DMR of the month in which the last day of discharge occurred.

FOR ALL OTHER DISCHARGES – The 7-day concentration is the sum of the daily concentrations determined during any 7 consecutive days in a reporting month divided by the number of daily concentrations determined. If the number of daily concentrations determined is less than 7, the actual number of daily concentrations determined shall be used for the calculation. The calculated 7-day concentration will be used to determine compliance with any maximum 7-day concentration limitations in the reporting month. When required by the permit, report the maximum calculated 7-day concentration for the month in the “MAXIMUM” column under

Section A. Definitions

"QUALITY OR CONCENTRATION" on the DMR. The first 7-day calculation shall be made on day 7 of the reporting month, and the last calculation shall be made on the last day of the reporting month.

7-day loading
FOR WWSLs THAT COLLECT AND STORE WASTEWATER AND ARE AUTHORIZED TO DISCHARGE ONLY IN THE SPRING AND/OR FALL ON AN INTERMITTENT BASIS – The 7-day loading is the sum of the daily loadings determined during any 7 consecutive days of discharge during a WWSL discharge event divided by the number of daily loadings determined. If the number of daily loadings determined during the WWSL discharge event is less than 7 days, the number of actual daily loadings determined shall be used for the calculation. The calculated 7-day loading will be used to determine compliance with any maximum 7-day loading limitations. When required by the permit, report the maximum calculated 7-day loading for the WWSL discharge event in the "MAXIMUM" column under "QUANTITY OR LOADING" on the DMR. If the WWSL discharge event was partially in each of two months, the value shall be reported on the DMR of the month in which the last day of discharge occurred.

FOR ALL OTHER DISCHARGES – The 7-day loading is the sum of the daily loadings determined during any 7 consecutive days in a reporting month divided by the number of daily loadings determined. If the number of daily loadings determined is less than 7, the actual number of daily loadings determined shall be used for the calculation. The calculated 7-day loading will be used to determine compliance with any maximum 7-day loading limitations in the reporting month. When required by the permit, report the maximum calculated 7-day loading for the month in the "MAXIMUM" column under "QUANTITY OR LOADING" on the DMR. The first 7-day calculation shall be made on day 7 of the reporting month, and the last calculation shall be made on the last day of the reporting month.

24-hour composite sample is a flow-proportioned composite sample consisting of hourly or more frequent portions that are taken over a 24-hour period and in which the volume of each portion is proportional to the discharge flow rate at the time that portion is taken. A time-proportioned composite sample may be used upon approval from the Department if the permittee demonstrates it is representative of the discharge.

PART II

Section B. Monitoring Procedures

1. Representative Samples
Samples and measurements taken as required herein shall be representative of the volume and nature of the monitored discharge.

2. Test Procedures
Test procedures for the analysis of pollutants shall conform to regulations promulgated pursuant to Section 304(h) of the Clean Water Act (40 CFR Part 136 – Guidelines Establishing Test Procedures for the Analysis of Pollutants), unless specified otherwise in this permit. **Test procedures used shall be sufficiently sensitive to determine compliance with applicable effluent limitations**. For lists of approved test methods, go to https://www.epa.gov/cwa-methods. Requests to use test procedures not promulgated under 40 CFR Part 136 for pollutant monitoring required by this permit shall be made in accordance with the Alternate Test Procedures regulations specified in 40 CFR 136.4. These requests shall be submitted to the Manager of the Permits Section, Water Resources Division, Michigan Department of Environment, Great Lakes, and Energy, P.O. Box 30458, Lansing, Michigan, 48909-7958. The permittee may use such procedures upon approval.

The permittee shall periodically calibrate and perform maintenance procedures on all analytical instrumentation at intervals to ensure accuracy of measurements. The calibration and maintenance shall be performed as part of the permittee's laboratory Quality Assurance/Quality Control program.

3. Instrumentation
The permittee shall periodically calibrate and perform maintenance procedures on all monitoring instrumentation at intervals to ensure accuracy of measurements.

4. Recording Results
For each measurement or sample taken pursuant to the requirements of this permit, the permittee shall record the following information: 1) the exact place, date, and time of measurement or sampling; 2) the person(s) who performed the measurement or sample collection; 3) the dates the analyses were performed; 4) the person(s) who performed the analyses; 5) the analytical techniques or methods used; 6) the date of and person responsible for equipment calibration; and 7) the results of all required analyses.

5. Records Retention
All records and information resulting from the monitoring activities required by this permit, including all records of analyses performed, calibration and maintenance of instrumentation, and recordings from continuous monitoring instrumentation, shall be retained for a minimum of three (3) years, or longer if requested by the Regional Administrator or the Department.

PART II

Section C. Reporting Requirements

1. Start-Up Notification

The permittee shall notify the Department of start-up if one of the following conditions applies and in accordance with the applicable condition:

a. Non-CAFOs

1) **If this is an individual permit** and the permittee will not discharge during the first 60 days following the effective date of this permit, the permittee shall notify the Department via MiEnviro Portal within 14 days following the effective date of this permit, and then again 60 days prior to commencement of the discharge.

2) **If this is a general permit** and the permittee will not discharge during the first 60 days following the effective date of the Certificate of Coverage (COC) issued under this general permit, the permittee shall notify the Department via MiEnviro Portal within 14 days following the effective date of the COC, and then again 60 days prior to commencement of the discharge.

b. CAFOs

1) **If this is an individual permit** and the permittee will not populate with animals during the first 60 days following the effective date of this permit, the permittee shall notify the Department via MiEnviro Portal within 14 days following the effective date of this permit, and then again 60 days prior to populating with animals.

2) **If this is a general permit** and the permittee will not populate with animals during 60 days following the effective date of the Certificate of Coverage (COC) issued under this general permit, the permittee shall notify the Department via MiEnviro Portal within 14 days following the effective date of the COC, and then again 60 days prior to populating with animals.

2. Submittal Requirements for Self-Monitoring Data

Part 31 of the NREPA (specifically Section 324.3110(7)); and R 323.2155(2) of Part 21, Wastewater Discharge Permits, promulgated under Part 31 of the NREPA, allow the Department to specify the forms to be utilized for reporting the required self-monitoring data. Unless instructed on the effluent limitations page to conduct "Retained Self-Monitoring," the permittee shall submit self-monitoring data via the Department's MiEnviro Portal system.

The permittee shall utilize the information provided on the MiEnviro Portal website, located at https://mienviro.michigan.gov/ncore/, to access and submit the electronic forms. Both monthly summary and daily data shall be submitted to the Department no later than the 20th day of the month following each month of the authorized discharge period(s). The permittee may be allowed to submit the electronic forms after this date if the Department has granted an extension to the submittal date.

3. Retained Self-Monitoring Requirements

If instructed on the effluent limits page (or otherwise authorized by the Department in accordance with the provisions of this permit) to conduct retained self-monitoring, the permittee shall maintain a year-to-date log of retained self-monitoring results and, upon request, provide such log for inspection to the staff of the Department. Retained self-monitoring results are public information and shall be promptly provided to the public upon request.

The permittee shall certify, in writing, to the Department, on or before January 10 (April 1 for animal feeding operation facilities) of each year, that: 1) all retained self-monitoring requirements have been complied with and a year-to-date log has been maintained; and 2) the application on which this permit is based still accurately describes the discharge. With this annual certification, the permittee shall submit a summary of the previous

PART II

Section C. Reporting Requirements

year's monitoring data. The summary shall include maximum values for samples to be reported as daily maximums and/or monthly maximums and minimum values for any daily minimum samples.

Retained self-monitoring may be denied to a permittee by notification in writing from the Department. In such cases, the permittee shall submit self-monitoring data in accordance with Part II.C.2., above. Such a denial may be rescinded by the Department upon written notification to the permittee. Reissuance or modification of this permit or reissuance or modification of an individual permittee's authorization to discharge shall not affect previous approval or denial for retained self-monitoring unless the Department provides notification in writing to the permittee.

4. Additional Monitoring by Permittee

If the permittee monitors any pollutant at the location(s) designated herein more frequently than required by this permit, using approved analytical methods as specified above, the results of such monitoring shall be included in the calculation and reporting of the values required in the Discharge Monitoring Report. Such increased frequency shall also be indicated.

Monitoring required pursuant to Part 41 of the NREPA or Rule 35 of the Mobile Home Park Commission Act, 1987 PA 96, as amended, for assurance of proper facility operation, shall be submitted as required by the Department.

5. Compliance Dates Notification

Within 14 days of every compliance date specified in this permit, the permittee shall submit a written notification to the Department via MiEnviro Portal (https://mienviro.michigan.gov/ncore/) indicating whether or not the particular requirement was accomplished. If the requirement was not accomplished, the notification shall include an explanation of the failure to accomplish the requirement, actions taken or planned by the permittee to correct the situation, and an estimate of when the requirement will be accomplished. If a written report is required to be submitted by a specified date and the permittee accomplishes this, a separate written notification is not required.

6. Noncompliance Notification

Compliance with all applicable requirements set forth in the Clean Water Act, Parts 31 and 41 of the NREPA, and related regulations and rules is required. All instances of noncompliance shall be reported as follows:

a. 24-Hour Reporting
Any noncompliance which may endanger health or the environment (including maximum and/or minimum daily concentration discharge limitation exceedances) shall be reported, verbally, within 24 hours from the time the permittee becomes aware of the noncompliance by calling the Department at the number indicated on the second page of this permit (or, if this is a general permit, on the COC). A written submission shall also be provided via MiEnviro Portal (https://mienviro.michigan.gov/ncore/) within five (5) days.

b. Other Reporting
The permittee shall report, in writing via MiEnviro Portal (https://mienviro.michigan.gov/ncore/), all other instances of noncompliance not described in a. above at the time monitoring reports are submitted; or, in the case of retained self-monitoring, within five (5) days from the time the permittee becomes aware of the noncompliance.

Reporting shall include: 1) a description of the discharge and cause of noncompliance; and 2) the period of noncompliance, including exact dates and times, or, if not yet corrected, the anticipated time the noncompliance is expected to continue, and the steps taken to reduce, eliminate and prevent recurrence of the noncomplying discharge.

PART II

Section C. Reporting Requirements

7. Spill Notification

The permittee shall immediately report any release of any polluting material which occurs to the surface waters or groundwaters of the state, unless the permittee has determined that the release is not in excess of the threshold reporting quantities specified in the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code), by calling the Department at the number indicated on the second page of this permit (or, if this is a general permit, on the COC); or, if the notice is provided after regular working hours, by calling the Department's 24-hour Pollution Emergency Alerting System telephone number, 1-800-292-4706.

Within 10 days of the release, the permittee shall submit to the Department via MiEnviro Portal (https://mienviro.michigan.gov/ncore/) a full written explanation as to the cause of the release, the discovery of the release, response measures (clean-up and/or recovery) taken, and preventive measures taken or a schedule for completion of measures to be taken to prevent reoccurrence of similar releases.

8. Upset Noncompliance Notification

If a process "upset" (defined as an exceptional incident in which there is unintentional and temporary noncompliance with technology-based permit effluent limitations because of factors beyond the reasonable control of the permittee) has occurred, the permittee who wishes to establish the affirmative defense of upset shall notify the Department by telephone within 24 hours of becoming aware of such conditions; and within five (5) days, provide in writing, the following information:

a. that an upset occurred and that the permittee can identify the specific cause(s) of the upset;

b. that the permitted wastewater treatment facility was, at the time, being properly operated and maintained (note that an upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation); and

c. that the permittee has specified and taken action on all responsible steps to minimize or correct any adverse impact in the environment resulting from noncompliance with this permit.

No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review.

In any enforcement proceedings, the permittee, seeking to establish the occurrence of an upset, has the burden of proof.

9. Bypass Prohibition and Notification

a. Bypass Prohibition
Bypass is prohibited, and the Department may take an enforcement action, unless:

1) bypass was unavoidable to prevent loss of life, personal injury, or severe property damage;

2) there were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate backup equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass; and

3) the permittee submitted notices as required under 9.b. or 9.c. below.

PART II

Section C. Reporting Requirements

b. Notice of Anticipated Bypass
If the permittee knows in advance of the need for a bypass, the permittee shall submit written notification to the Department before the anticipated date of the bypass. This notification shall be submitted at least 10 days before the date of the bypass; however, the Department will accept fewer than 10 days advance notice if adequate explanation for this is provided. The notification shall provide information about the anticipated bypass as required by the Department. The Department may approve an anticipated bypass, after considering its adverse effects, if it will meet the three (3) conditions specified in a. above.

c. Notice of Unanticipated Bypass
As soon as possible but no later than 24 hours from the time the permittee becomes aware of the unanticipated bypass, the permittee shall notify the Department by calling the number indicated on the second page of this permit (or, if this is a general permit, on the COC); or, if notification is provided after regular working hours, call the Department's 24-hour Pollution Emergency Alerting System telephone number, 1-800-292-4706.

d. Written Report of Bypass
A written submission shall be provided within five (5) working days of commencing any bypass to the Department, and at additional times as directed by the Department. The written submission shall contain a description of the bypass and its cause; the period of bypass, including exact dates and times, and if the bypass has not been corrected, the anticipated time it is expected to continue; steps taken or planned to reduce, eliminate, and prevent reoccurrence of the bypass; and other information as required by the Department.

e. Bypass Not Exceeding Limitations
The permittee may allow any bypass to occur which does not cause effluent limitations to be exceeded, but only if it also is for essential maintenance to ensure efficient operation. These bypasses are not subject to the provisions of 9.a., 9.b., 9.c., and 9.d., above. This provision does not relieve the permittee of any notification responsibilities under Part II.C.11. of this permit.

f. Definitions

1) Bypass means the intentional diversion of waste streams from any portion of a treatment facility.

2) Severe property damage means substantial physical damage to property, damage to the treatment facilities which causes them to become inoperable, or substantial and permanent loss of natural resources which can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production.

10. Bioaccumulative Chemicals of Concern (BCC)

Consistent with the requirements of R 323.1098 and R 323.1215 of the Michigan Administrative Code, the permittee is prohibited from undertaking any action that would result in a lowering of water quality from an increased loading of a BCC unless an increased use request and antidegradation demonstration have been submitted and approved by the Department.

11. Notification of Changes in Discharge

The permittee shall notify the Department via MiEnviro Portal (https://mienviro.michigan.gov/ncore/), as soon as possible but within no more than 10 days of knowing, or having reason to believe, that any activity or change has occurred or will occur which would result in the discharge of: 1) detectable levels of chemicals on the current Michigan Critical Materials Register, priority pollutants or hazardous substances set forth in 40 CFR 122.21, Appendix D, or the Pollutants of Initial Focus in the Great Lakes Water Quality Initiative specified in 40 CFR 132.6, Table 6, which were not acknowledged in the application or listed in the application at less than

PART II

Section C. Reporting Requirements

detectable levels; 2) detectable levels of any other chemical not listed in the application or listed at less than detection, for which the application specifically requested information; or 3) any chemical at levels greater than five times the average level reported in the complete application (see the first page of this permit, for the date(s) the complete application was submitted). Any other monitoring results obtained as a requirement of this permit shall be reported in accordance with the compliance schedules.

12. Changes in Facility Operations

Any anticipated action or activity, including but not limited to facility expansion, production increases, or process modification, which will result in new or increased loadings of pollutants to the receiving waters must be reported to the Department by a) submission of an increased use request (application) and all information required under R 323.1098 (Antidegradation) of the Water Quality Standards or b) by written notice if the following conditions are met: 1) the action or activity will not result in a change in the types of wastewater discharged or result in a greater quantity of wastewater than currently authorized by this permit; 2) the action or activity will not result in violations of the effluent limitations specified in this permit; 3) the action or activity is not prohibited by the requirements of Part II.C.10.; and 4) the action or activity will not require notification pursuant to Part II.C.11. Following such written notice, the permit or, if applicable, the facility's COC, may be modified according to applicable laws and rules to specify and limit any pollutant not previously limited.

13. Transfer of Ownership or Control

In the event of any change in ownership or control of facilities from which the authorized discharge emanates, the following requirements apply: Not less than 30 days prior to the actual transfer of ownership or control – for non-CAFOs, or within 30 days of the actual transfer of ownership or control – for CAFOs, the permittee shall submit to the Department via MiEnviro Portal (https://mienviro.michigan.gov/ncore/) a written agreement between the current permittee and the new permittee containing: 1) the legal name and address of the new owner; 2) a specific date for the effective transfer of permit responsibility, coverage and liability; and 3) a certification of the continuity of or any changes in operations, wastewater discharge, or wastewater treatment.

If the new permittee is proposing changes in operations, wastewater discharge, or wastewater treatment, the Department may propose modification of this permit in accordance with applicable laws and rules.

14. Operations and Maintenance Manual

For wastewater treatment facilities that serve the public (and are thus subject to Part 41 of the NREPA), Section 4104 of Part 41 and associated Rule 2957 of the Michigan Administrative Code allow the Department to require an Operations and Maintenance (O&M) Manual from the facility. An up-to-date copy of the O&M Manual shall be kept at the facility and shall be provided to the Department upon request. The Department may review the O&M Manual in whole or in part at its discretion and require modifications to it if portions are determined to be inadequate.

At a minimum, the O&M Manual shall include the following information: permit standards; descriptions and operation information for all equipment; staffing information; laboratory requirements; record keeping requirements; a maintenance plan for equipment; an emergency operating plan; safety program information; and copies of all pertinent forms, as-built plans, and manufacturer's manuals.

Certification of the existence and accuracy of the O&M Manual shall be submitted to the Department at least sixty days prior to start-up of a new wastewater treatment facility. Recertification shall be submitted sixty days prior to start-up of any substantial improvements or modifications made to an existing wastewater treatment facility.

PART II

Section C. Reporting Requirements

15. Signatory Requirements

All applications, reports, or information submitted to the Department in accordance with the conditions of this permit and that require a signature shall be signed and certified as described in the Clean Water Act and the NREPA.

The Clean Water Act provides that any person who knowingly makes any false statement, representation, or certification in any record or other document submitted or required to be maintained under this permit, including monitoring reports or reports of compliance or noncompliance, shall, upon conviction, be punished by a fine of not more than $10,000 per violation, or by imprisonment for not more than 6 months per violation, or by both.

The NREPA (Section 3115(2)) provides that a person who at the time of the violation knew or should have known that he or she discharged a substance contrary to this part, or contrary to a permit, COC, or order issued or rule promulgated under this part, or who intentionally makes a false statement, representation, or certification in an application for or form pertaining to a permit or COC or in a notice or report required by the terms and conditions of an issued permit or COC, or who intentionally renders inaccurate a monitoring device or record required to be maintained by the Department, is guilty of a felony and shall be fined not less than $2,500.00 or more than $25,000.00 for each violation. The court may impose an additional fine of not more than $25,000.00 for each day during which the unlawful discharge occurred. If the conviction is for a violation committed after a first conviction of the person under this subsection, the court shall impose a fine of not less than $25,000.00 per day and not more than $50,000.00 per day of violation. Upon conviction, in addition to a fine, the court in its discretion may sentence the defendant to imprisonment for not more than 2 years or impose probation upon a person for a violation of this part. With the exception of the issuance of criminal complaints, issuance of warrants, and the holding of an arraignment, the circuit court for the county in which the violation occurred has exclusive jurisdiction. However, the person shall not be subject to the penalties of this subsection if the discharge of the effluent is in conformance with and obedient to a rule, order, permit, or COC of the Department. In addition to a fine, the attorney general may file a civil suit in a court of competent jurisdiction to recover the full value of the injuries done to the natural resources of the state and the costs of surveillance and enforcement by the state resulting from the violation.

16. Electronic Reporting

Upon notice by the Department that electronic reporting tools are available for specific reports or notifications, the permittee shall submit electronically via MiEnviro Portal (https://mienviro.michigan.gov/ncore/) all such reports or notifications as required by this permit, on forms provided by the Department.

PART II

Section D. Management Responsibilities

1. Duty to Comply
All discharges authorized herein shall be consistent with the terms and conditions of this permit. The discharge of any pollutant identified in this permit, more frequently than, or at a level in excess of, that authorized, shall constitute a violation of the permit.

It is the duty of the permittee to comply with all the terms and conditions of this permit. Any noncompliance with the Effluent Limitations, Special Conditions, or terms of this permit constitutes a violation of the NREPA and/or the Clean Water Act and constitutes grounds for enforcement action; for permit or COC termination, revocation and reissuance, or modification; or denial of an application for permit or COC renewal.

It shall not be a defense for a permittee in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this permit.

2. Operator Certification
The permittee shall have the waste treatment facilities under direct supervision of an operator certified at the appropriate level for the facility certification by the Department, as required by Sections 3110 and 4104 of the NREPA. Permittees authorized to discharge storm water shall have the storm water treatment and/or control measures under direct supervision of a storm water operator certified by the Department, as required by Section 3110 of the NREPA.

3. Facilities Operation
The permittee shall, at all times, properly operate and maintain all treatment or control facilities or systems installed or used by the permittee to achieve compliance with the terms and conditions of this permit. Proper operation and maintenance includes adequate laboratory controls and appropriate quality assurance procedures.

4. Power Failures
In order to maintain compliance with the effluent limitations of this permit and prevent unauthorized discharges, the permittee shall either:

a. provide an alternative power source sufficient to operate facilities utilized by the permittee to maintain compliance with the effluent limitations and conditions of this permit; or

b. upon the reduction, loss, or failure of one or more of the primary sources of power to facilities utilized by the permittee to maintain compliance with the effluent limitations and conditions of this permit, the permittee shall halt, reduce or otherwise control production and/or all discharge in order to maintain compliance with the effluent limitations and conditions of this permit.

5. Adverse Impact
The permittee shall take all reasonable steps to minimize or prevent any adverse impact to the surface waters or groundwaters of the state resulting from noncompliance with any effluent limitation specified in this permit including, but not limited to, such accelerated or additional monitoring as necessary to determine the nature and impact of the discharge in noncompliance.

6. Containment Facilities
The permittee shall provide facilities for containment of any accidental losses of polluting materials in accordance with the requirements of the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code). For a POTW, these facilities shall be approved under Part 41 of the NREPA.

PART II

Section D. Management Responsibilities

7. Waste Treatment Residues
Residuals (i.e., solids, sludges, biosolids, filter backwash, scrubber water, ash, grit, or other pollutants or wastes) removed from or resulting from treatment or control of wastewaters, including those that are generated during treatment or left over after treatment or control has ceased, shall be disposed of in an environmentally compatible manner and according to applicable laws and rules. These laws may include, but are not limited to, the NREPA, Part 31 for protection of water resources, Part 55 for air pollution control, Part 111 for hazardous waste management, Part 115 for solid waste management, Part 121 for liquid industrial wastes, Part 301 for protection of inland lakes and streams, and Part 303 for wetlands protection. Such disposal shall not result in any unlawful pollution of the air, surface waters or groundwaters of the state.

8. Right of Entry
The permittee shall allow the Department, any agent appointed by the Department, or the Regional Administrator, upon the presentation of credentials and, for animal feeding operation facilities, following appropriate biosecurity protocols:

a. to enter upon the permittee's premises where an effluent source is located or any place in which records are required to be kept under the terms and conditions of this permit; and

b. at reasonable times to have access to and copy any records required to be kept under the terms and conditions of this permit; to inspect process facilities, treatment works, monitoring methods and equipment regulated or required under this permit; and to sample any discharge of pollutants.

9. Availability of Reports
Except for data determined to be confidential under Section 308 of the Clean Water Act and Rule 2128 (R 323.2128 of the Michigan Administrative Code), all reports prepared in accordance with the terms of this permit and required to be submitted to the Department shall be available for public inspection via MiEnviro Portal (https://mienviro.michigan.gov/ncore/). As required by the Clean Water Act, effluent data shall not be considered confidential. Knowingly making any false statement on any such report may result in the imposition of criminal penalties as provided for in Section 309 of the Clean Water Act and Sections 3112, 3115, 4106 and 4110 of the NREPA.

10. Duty to Provide Information
The permittee shall furnish to the Department via MiEnviro Portal (https://mienviro.michigan.gov/ncore/), within a reasonable time, any information which the Department may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this permit or the facility's COC, or to determine compliance with this permit. The permittee shall also furnish to the Department, upon request, copies of records required to be kept by this permit.

Where the permittee becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Department, it shall promptly submit such facts or information.

PART II

Section E. Activities Not Authorized by This Permit

1. Discharge to the Groundwaters

This permit does not authorize any discharge to the groundwaters. Such discharge may be authorized by a groundwater discharge permit issued pursuant to the NREPA.

2. POTW Construction

This permit does not authorize or approve the construction or modification of any physical structures or facilities at a POTW. Approval for the construction or modification of any physical structures or facilities at a POTW shall be by permit issued under Part 41 of the NREPA.

3. Civil and Criminal Liability

Except as provided in permit conditions on "Bypass" (Part II.C.9. pursuant to 40 CFR 122.41(m)), nothing in this permit shall be construed to relieve the permittee from civil or criminal penalties for noncompliance, whether or not such noncompliance is due to factors beyond the permittee’s control, such as accidents, equipment breakdowns, or labor disputes.

4. Oil and Hazardous Substance Liability

Nothing in this permit shall be construed to preclude the institution of any legal action or relieve the permittee from any responsibilities, liabilities, or penalties to which the permittee may be subject under Section 311 of the Clean Water Act except as are exempted by federal regulations.

5. State Laws

Nothing in this permit shall be construed to preclude the institution of any legal action or relieve the permittee from any responsibilities, liabilities, or penalties established pursuant to any applicable state law or regulation under authority preserved by Section 510 of the Clean Water Act.

6. Property Rights

The issuance of this permit does not convey any property rights in either real or personal property, or any exclusive privileges, nor does it authorize violation of any federal, state or local laws or regulations, nor does it obviate the necessity of obtaining such permits, including any other Department of Environment, Great Lakes, and Energy permits, or approvals from other units of government as may be required by law.

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

INTEROFFICE COMMUNICATION

TO: Claire Lowe, Cadillac District Office, Water Resources Division

FROM: Glen Schmitt, Permits Section, Water Resources Division

DATE: April 19, 2022

SUBJECT: Local Limits Review
MiWaters Submission Number: HPG-XCC7-NCG09
Cadillac Wastewater Treatment Plant (WWTP)
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257

Pursuant to your request, we have reviewed the Industrial Pretreatment Program Local Limits request dated April 13, 2022 for the Cadillac WWTP. The facility is currently authorized to continuously discharge treated municipal wastewater from Monitoring Point 001A through Outfall 001 to the Clam River at Latitude 44.2657, Longitude -85.3985. A design flow of 3.2 million gallons per day (MGD) was used in determining the final effluent limitations and monitoring requirements in the current NPDES permit, but this flow is not considered a limitation or actual capacity.

The Clam River has a lowest 95 percent exceedance low flow, harmonic mean low flow, and 90dQ10 low flow of 2.6 cubic feet per second (cfs), 13 cfs, and 3.4 cfs (Low Flow File Number 10194). An ambient total hardness value for the Clam River of 70 milligrams per liter (mg/l) was used for the parameters that are total hardness dependent and was collected by Permits Section staff on August 19, 2021 from the Clam River at Latitude 44.263376, Longitude -85.4011.

The theoretical water quality-based effluent limits (WQBELs) are presented in micrograms per liter (ug/l) and pounds per day (lbs./day) for the parameters requested.

Parameter	CAS Number	Daily Maximum WQBEL (µg/L)	Daily Maximum Load (lbs./day)	Monthly Average WQBEL (µg/l)	Monthly Average Load (lbs./day)
Arsenic #	7440382	680	18	16	0.42
Cadmium	7440439	12	0.32	4.1	0.11
Chromium	7440473	1,300	34	94	2.5
Copper	7440508	29	0.77	11	0.29
Cyanide	57125	44	1.2	5.9	0.16
Lead	7439921	820	22	94	2.5
Mercury @	7439976	---	---	0.0013	3.5×10^{-5}
Molybdenum	7439987	58,000	1,500	3,600	97
Nickel	7440020	760	20	47	1.3
Selenium	7782492	120	3.2	5.7	0.15
Silver	7440224	22	0.59	1.4	0.036
Zinc	7440666	360	9.7	210	5.5
Phenol	108952	6,800	180	510	14

Parameter	CAS Number	Daily Maximum WQBEL (µg/L)	Daily Maximum Load (lbs./day)	Monthly Average WQBEL (µg/l)	Monthly Average Load (lbs./day)
1,2-Dichlorobenzene	95501	240	6.4	15	0.39
Chloroform #	67663	11,000	290	710	19
Ethylbenzene #	100414	370	9.9	24	0.63
Lindane # @	58899	---	---	0.026	0.00069
Polychlorinated Biphenyls (PCBs) # @	1336363	---	---	0.000026	6.9×10^{-7}
Styrene #	100425	2,900	77	130	3.5
Toluene	108883	2,600	69	290	7.8
Xylene	1330207	890	24	55	1.5

- Carcinogen
@ - bioaccumulative chemical of concern (BCC)

1) The current NPDES permit includes a monthly average total phosphorus limit of 0.5 milligrams per liter (mg/l) and a corresponding load limitation of 13 pounds per day (lbs./day) from May through March and a monthly average total phosphorus limit of 0.65 mg/l and a corresponding load limit of 17 lbs./day during the month of April. These limits are to protect the Clam River and downstream surface waters from excessive nutrients.

2) The recommendations contained herein are based on the Part 4 Water Quality Standards and Part 8 Water Quality-Based Effluent Limit Development for Toxic Substances. Treatment methods or economic concerns have not been addressed. We understand that other considerations may be used in deciding permit limits and requirements.

Please contact me at (517) 290-6424 or schmittg1@michigan.gov if you have any questions regarding the recommendations above.

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

INTEROFFICE COMMUNICATION

TO: Claire Lowe, Cadillac District Office, Water Resources Division

FROM: Phillip DePetro, Permits Section, Water Resources Division

DATE: April 19, 2022

SUBJECT: Local Limits Review (MiWaters Submission HPG-XCC7-NCG09)
Cadillac WWTP
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257

Permits Section has evaluated your request for a local limits review for the Cadillac WWTP. This memo summarizes the calculated effluent limits for five-day carbonaceous biochemical oxygen demand ($CBOD_5$), ammonia (NH3-N), minimum effluent dissolved oxygen (DO), and total suspended solids (TSS). The Cadillac WWTP discharges treated municipal wastewater to the Clam River at 44.2657°N, -85.3985°E in Wexford County. The facility has an annual average design flow of 3.2 million gallons per day (MGD)

Note that the calculated limits provided in this memo are based on water quality and not the current permit effluent limits. In most cases the effluent limits set in the current permit are based on anti-degredation. In the 1980s, WQBELs were developed based on a trout stream designation (coldwater protection) for the Clam River between Lake Cadillac and the Wexford-Missaukee County line in the non-summer seasons. The Michigan DNR Fisheries order in 1997 and the most current fisheries order list the Clam River as being protected for coldwater fish downstream of the Wexford-Missaukee County line, and as warmwater protected between the lake and the county line. The difference in stream designation between the most recent modeling and the modeling done in the 1980's is the primary driver for the difference in limits.

The Clam River from Lake Cadillac to the Wexford-Missaukee county line in protected for warmwater fish species, other indigenous aquatic life and wildlife, agriculture, navigation, industrial water supply, public water supply at the point of intake, partial body contact recreation, total body contact recreation from May 1 to October 31, and fish consumption. The Clam River downstream of the county line is protected for the same designated uses in addition to being protected for coldwater fish.

The monthly exceedance flows (in cfs) for the Clam River at the Cadillac WWTP outfall are as follows based on Low Flow Discharge Request 10314:

	JAN.	**FEB.**	**MAR.**	**APR.**	**MAY**	**JUNE**
50%	17	17	36	78	35	13
95%	5.6	5.1	9.0	14	6.9	4.1

	JULY	**AUG.**	**SEP.**	**OCT.**	**NOV.**	**DEC.**
50%	7.4	6.3	6.3	10	18	19
95%	3.0	3.0	3.1	3.7	5.5	5.5

The lowest monthly 95% exceedance flows in up to four seasons are used as the background flow of the receiving stream for the purposes of developing WQBELs. The following seasons are used, consistent with the current permit structure: June-September, October-November, December-March, and April-May.

A multiple reach Streeter-Phelps model was used to model the impact of the Cadillac WWTP on the Clam River at critical low-flow conditions. The Haring Twp WWTP (NPDES Permit No. MI0059076) was included in this model given its proximity downstream. Effluent limit recommendations are determined that are always protective of the applicable DO standards. A diurnal variation of 0.75 mg/l per day between the daily average and daily minimum DO concentration was assumed based on a 2006 field study for the warmwater-protected stream reaches. A diurnal variation of 0.51 mg/l was assumed for the coldwater-protected reaches based on the same study. Kinetic rates were also set based on department procedure defaults for $CBOD_5$. A 1983 field study determined that nitrification was not occurring in the Clam River due to a lack of substrate for nitrifying bacteria. Site-specific time of passage and reaeration studies completed in 1983 were used to determine stream velocity and reaeration. Stream slopes were updated using LiDAR data. The recommended local limits based on existing stream designations and water quality for the Cadillac WWTP are presented in Table 1.

As previously stated, the effluent limit recommendations of the Cadillac WWTP in the summer season are less restrictive than current permit effluent limits due to the change in stream designation. The current permit effluent limits are based on anti-degradation in the basis of decision memo associated with the last permit reissuance. There is no daily maximum effluent limit recommendation for NH3-N based on the assumption that nitrification is not occurring in the Clam River.

Effluent limit recommendations based on federal secondary treatment standards (STS) are sufficient to protect the DO standard in the Clam River in the remaining three seasons. A minimum effluent DO concentration of 3.0 mg/l is recommended to protect against acute toxicity in the effluent-receiving water mixing zone. The STS-based effluent limits are less restrictive than current permit effluent limits for the Cadillac WWTP.

The Cadillac WWTP discharge was also evaluated to determine 30-day average NH3-N effluent limits that are protective of the chronic NH3-N toxicity criterion. A new chronic toxicity criterion was adopted by the department in 2019 and is being implemented for permits being issued in the current fiscal year. An effluent pH of 7.3 SU was assumed for the Cadillac WWTP based on two years of discharge monitoring data. The recommended effluent limits to protect against chronic toxicity are more restrictive than current permit effluent limits at the Cadillac WWTP in April. The limits for the remaining months at the Cadillac WWTP are equivalent to or less restrictive than current permit effluent limits.

There is no applicable numeric water quality standard for TSS in the Part 4 Water Quality Standards. The narrative standard set in Rule 50 [R323.1050] would apply for TSS.

These effluent limit recommendations are based on water quality standards. We have not addressed treatment practicality or cost effectiveness. Our recommendations do not imply that other considerations, such as the existing permit effluent limits, should not be taken into account when deciding on permit limits.

Table 2. Cadillac WWTP (MI0020257) Local Limits Review Recommendations

Parameter	Months	Concentration (mg/l)	Basis	Rationale
$CBOD_5$	June-September	17	Daily Maximum	DO Standard
	October-May	25	7-Day Average	Secondary Treatment
		40	30-Day Average	Secondary Treatment
NH3-N	June-September	0.9	30-Day Average	Chronic Toxicity
	October-November	3.1	30-Day Average	Chronic Toxicity
	December-March	5.4	30-Day Average	Chronic Toxicity
	April-May	3.0	30-Day Average	Chronic Toxicity
D.O.	June-September	6.0	Minimum	DO Standard
	October-May	3.0	Minimum	Acute DO Toxicity

Design Flow = 3.2 MGD

filter. The vacuum filter removes water from the sludge and procudes a sludge cake which may be disposed of at a landfill. Water removed from the sludge is returned to the wastewater at the quick-mix tank.

B. PLANT FLOW PATTERN

Figure I shows a simplified flow diagram of main process flow at the Cadillac Wastewater Treatment Plant.

C. PRINCIPLE DESIGN CRITERIA

1. Design population (year of 1990) - 15,000 capita

2. Influent characteristics

   a. BOD5(five day biological oxygen demand) @ 0.22 pounds/day - 3,300 lbs./day or 200 mg/l.

   b. SS (suspended solids) @ 0.28 pounds/day -4200 lbs/day or 252 mg/l.

   c. NH_3-N (ammonia-nitrogen found in the ammonia form) - 20 mg/l.

   d. PO_4-P (phosphate-phosphorous or phosphorous found in the phosphate form) - 40 mg/l.

4. Design Flows

Average day flow	2.0 MGD
Minimum day flow	0.8 MGD
Maximum day flow	4.5 MGD

D. EXPECTED EFFICIENCY OF TREATMENT

Overall plant efficiency is expected to be 97 percent for BOD, 96 percent for suspended solids, 92.5 percent for ammonia and 80 percent for phosphates. Individually, the primary, secondary and tertiary phases of treatment are expected to perform as follows:

PERCENT REMOVAL

	BOD	SS	NH_3-N	PO_4-P
Primary	45	65	---	60
Secondary	82	65	---	---
Tertiary	70	70	92.5	50
Total	97	96	92.5	80

E. OPERATION AND MANAGERIAL RESPONSIBILITY

Personnel for the Treatment Plant should consist of a superintendent, a laboratory technician, and six to eight assistant operators. The superintendent is responsible for the management and total operation of the Plant. In general, his duties include supervision and training of all subordinate personnel and responsibility for daily operations, laboratory procedures, equipment maintenance and the general care and upkeep of the facility. He is expected to acquire a thorough knowledge of the Plant, the wastewater it treats, the effects of the effluent on the receiving stream and applicable regulations or laws governing his operations.

A more detailed discussion of plant personnel and their responsibilities is included in Chapter X.

4.6.3.3 Aeration Equipment

As indicated earlier, three blowers are used to supply air to the aeration tanks. Since continuous aeration is required for the operation of the activated sludge process, the blowers should be kept in good mechanical condition to avoid any shut down of the process. Normally, one large 125-hp blower is in service and the other large unit remains on standby. The small 60-hp blower is used to supplement air to the aeration tanks at times of high demand when the large blower capacity is not sufficient for the demand.

4.6.3.4 Final Clarifiers

Flows exit the three aeration tanks by passing over effluent weirs into effluent boxes, joining and exiting the main building pipe gallery via a 24 inch pipe leading to the final clarifiers' flow distribution box. Here the flow is again divided and passes through the four final clarifiers. These are circular clarifiers, each 45 foot in diameter with side water depth of nine feet. The surface overflow rate for the final clarifiers is 375 and 540 gd/ft^2 for the average and maximum flows, respectively, with the four clarifiers in operation.

Wastewater enters each of the clarifiers by way of the final tanks' distribution chamber, in which the mixed liquor from the aeration tanks is divided into four streams and directed to each clarifier via 16 inch lines. Clarifiers 1 and 2 were installed when the plant was first built. Clarifiers 3 and 4 were added later as part of plant expansions in 1989 and 1995, respectively.

These lines feed the wastewater to influent wells in the center of each clarifier. The influent wells consist essentially of seven foot diameter; four foot long baffles which serve to evenly distribute the influent flow. The mixed liquor flows from the influent wells radially to the perimeter of the clarifiers. Here, the flow proceeds beneath a scum baffle, over effluent weirs and into effluent troughs which direct the effluent from each clarifier to a point between them. The flows are here joined and flow through a 30 inch line to the effluent box of the final clarifier's distribution chamber.

In traveling from the influent wells to the effluent weirs, the solids initially in the clarifier influent settle out of the wastewater and deposit in the bottoms of the clarifiers as activated sludge. Like the primary clarifiers, the final clarifiers are equipped with sludge collection mechanisms. Instead of plowing the sludge to a central well, however, the sludge collectors in the final clarifiers have hollow collector arms with 1-3/8 inch suction orifices which remove the sludge from the tank floor as the collector arm revolves over the tank floor surface. The sludge flows into the arm through the suction orifices and to an eight inch cast iron pipe in the center of each clarifier. These pipes lead the sludge from the final clarifiers to the suction line of the activated sludge return pumps in the main building. Scum is collected in the final clarifiers in much the same way as it is collected in the primary clarifiers. Scum baffles separate the scum from the clarifiers' effluent and the scum is directed from the water surface to a scum trough and from

♦ Original O & M Manual prepared by ALNM 1975, was used in the preparation of this revision.

there to a four inch scum line which leads the scum to the main building waste well. From here it is pumped along with the waste activated sludge back to a point ahead of the primary clarifiers.

The flow into the final clarifiers is regulated with the help of the stop plates in the flow dividing chamber located near the clarifiers. Under normal conditions, all clarifiers are used. Provision is made to bypass the clarifiers for maintenance and repair work. Stop plates are used to achieve the desired flow pattern. Sludge removal from the final clarifiers has a very significant effect on the quality of the plant effluent and offers an important tool for activated sludge process control.

The WEF Manual of Practice No. 11 suggests:

"It is good practice to operate final settling tanks with as little depth of sludge on the bottom as possible. This can be accomplished by withdrawing the sludge at a rate such that its concentration of solids is the same as that of the sludge obtained form a sample of the mixed liquor after settling 30 minutes in a graduated cylinder (standard test).

Thus, if a 30-minute sludge volume is 200 ml in a cylinder settles to 20 percent, the quantity of return sludge drawn from the tank should be 20 percent of the total flow of aeration liquor into the tank. It is sound practice to take measurements for sludge depth in the final clarifiers occasionally. When excessive depths are noted, the rate of sludge withdrawal should be increased."

4.6.4 Methods of Control

Controls for the clarifiers consist of on-off selector switches for the sludge-scum collection mechanism. The collection mechanism should be continuously operated so that the sludge is kept in constant motion toward the sludge hopper. The sludge is removed from the clarifiers as it accumulates by means of telescoping sludge valves located in the sludge well of the main building.

Scum is directed to the scum collection trough by a rotating arm. Under normal conditions sufficient wastewater is directed to the trough along the scum line to flush the scum from the trough to the waste well in the main building. The scum collection trough should be checked daily to ensure that scum is not building up within the trough due to plugging in the pipe between the trough and the sludge well. Any blockage within the pipe must be removed by flushing or pushing a heavy cable or flexible rod through the pipe. Under maximum plant flow conditions it may be necessary to reduce the amount of re circulated activated sludge in order to prevent overloading the clarifiers. Some of the most commonly encountered operating troubles in the clarifier operation are listed in the WEF Manual of Practice No. 11, which should serve as a guide for efficient operation of these units.

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